Survey of Advances in Guidance, Navigation, and Control of Unmanned Rotorcraft Systems

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Recently, there has been growing interest in developing unmanned aircraft systems (UAS) with advanced onboard autonomous capabilities. This paper describes the current state of the art in autonomous rotorcraft UAS (RUAS) and provides a detailed literature review of the last two decades of active research on RUAS. Three functional technology areas are identified as the core components of an autonomous RUAS. Guidance, navigation, and control (GNC) have received much attention from the research community, and have dominated the UAS literature from the nineties until now. This paper first presents the main research groups involved in the development of GNC systems for RUAS. Then it describes the development of a framework that provides standard definitions and metrics characterizing and measuring the autonomy level of a RUAS using GNC aspects. This framework is intended to facilitate the understanding and the organization of this survey paper, but it can also serve as a common reference for the UAS community. The main objective of this paper is to present a comprehensive survey of RUAS research that captures all seminal works and milestones in each GNC area, with a particular focus on practical methods and technologies that have been demonstrated in flight tests. These algorithms and systems have been classified into different categories and classes based on the autonomy level they provide and the algorithmic approach used. Finally, the paper discusses the RUAS literature in general and highlights challenges that need to be addressed in developing autonomous systems for unmanned rotorcraft.
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1. INTRODUCTION

Unmanned aircraft systems (UAS) are encountered in an increasing number of applications, mostly military but also civilian. They offer major advantages when used for aerial surveillance, reconnaissance, and inspection in complex and dangerous environments. Lower downside risk and higher confidence in mission success are two strong motivators for continued expansion of UAS. Many other technological, economical, and political factors have also contributed to the development and deployment of UAS. Currently, the main UAS applications are defense-related and the main investments are driven by future military scenarios. However, a civil UAS market is predicted to emerge over the next decade, starting with government organizations requiring surveillance and inspection systems, such as coast guards, border patrol organizations, emergency services, and police. Several market studies (Dickerson, 2007; Visiongain, 2009) predict that the worldwide UAS market will expand significantly in the next decade. The reader can refer to Valavanis (2007, Chapter 1) and to the UAS roadmap (Roadmap, 2010) for more details on current and predicted UAS market and expenditures.

1.1. Rotorcraft Unmanned Aircraft Systems

The rotorcraft segment is rapidly evolving and dynamic sector of the overall UAS market. Rotorcraft UAS (RUAS[1]) are better suited for many civilian applications than manned aircraft and fixed-wing UAS. Potential civil applications of UAS include infrastructure inspection, law enforcement, surveillance of coast borders and road traffic, disaster and crisis management, and agriculture and forestry. Most of these applications require low-altitude flights with hovering and vertical take-off and landing (VTOL) capabilities, and small RUAS are more appropriate for these applications than fixed-wing UAS. Furthermore, most academic and governmental research groups use RUAS as experimental platforms to validate their GNC algorithms. RUAS can be categorized into five classes based on some attributes such as size and payload (Figure 1).

- **Category I**: Full-scale unmanned helicopters or optionally piloted autonomous helicopters. The main feature of this category of RUAS is their significant payload, endurance, and range, but also important the possibility of

[1]See Section 3.1 for the definition of "RUAS."



Figure 1. Categories of unmanned rotorcraft systems.

carrying a safety pilot onboard the vehicle, thereby providing an excellent test bed for complex and risky flight tests. A good example of this category is the Boeing Unmanned Little Bird (ULB) helicopter (the ULB has been used in the Piasecki/CMU CMUAV-CBRNE project).

- **Category II**: Medium-scale UAS helicopters that are available as autonomous or semiautonomous platforms and have significant payload (more than 10 kg), with a total weight of more than 30 kg. Some examples are the Yamaha RMAX (Japan), Shiebel S-100 (Austria), Rotomotion SR500 (USA), Hirobo SkySurveyor (Japan), Zaval (Russia), ServoHeli-120 (China), and RemoH-C100 (South Korea). The main advantage of these platforms is their significant payload which allows them to carry different heavy and high-quality navigation and mission sensors. These platforms are generally well-engineered with some level of dependability.
- **Category III**: Small-scale RUAS that are based on RC helicopters with optionally integrated autopilot. They have a payload of several kilograms (2 to 10 kg) and a total weight of less than 30 kg. Helicopters such as the Vario Benzin Trainer, Bergen Industrial Twin, Rotomotion SR100, and Hirobo and others, fall into this category. Platforms of this class have less payload than those of category II, but they can still carry most of the navigation and mission sensors. They feature low cost and availability, but they require considerable maintenance and engineering work to convert them from RC helicopters to dependable UAS.
- **Category IV**: Mini RUAS that are man-portable and can fly outdoors as well as in confined and indoor environments. Their payload is generally less than 2 kg and their total weight can range from hundreds of grams to a few kilograms. Most of them are electrically powered, with a flight time that ranges from 5 min to 1 h, depending on the payload. Most multirotors fall into this category, with some other designs. Their limited payload still allows them to carry small conventional avionics and lightweight perception sensors. Their low cost, easy maintenance, and safe operation make them excellent test beds for research.
- **Category V**: Micro air vehicles (MAVs) with less than 100 g payload, such as the Epson micro flying robot. Standard navigation sensors and avionics are difficult to carry on these machines. Research challenges include novel sensing and navigation solutions, such as those based on bio-mimetic principles. These RUAS are mainly designed for indoor applications and can be launched and recovered by hand.

RUAS of categories I, II, and III are generally suitable for outdoor applications and require a suitable area

for launch and recovery. These three categories are dominated by conventional helicopter configurations with a single main rotor and a tail rotor. On the other hand, most mini and micro UAS of categories IV and V are multirotor platforms (e.g., quadrotors and coaxial vehicles) that can fly outdoors as well as indoors and can be launched by hand or from small and narrow spaces.

1.2. Motivation for and Scope of This Survey

Nonmilitary research in RUAS only began in the early 1990s, and they have now become a popular research area. Over the past 20 years, an enormous amount of research has gone into guidance, navigation, and control (GNC) for RUAS, resulting in veried techniques and a large number of published papers. Although some survey papers have tried to review small subsets of methods in a particular area—Ollero and Merino (2004) for flight controllers, Chao, Cao, and Chen (2010) for autopilots, Goerzen, Kong, and Mettler (2010) for path planning algorithms, and Valavanis (2007) for UAS in general[2]—there is a real need for a comprehensive survey to report and organize the large variety of GNC methods, providing a context for viewing and comparing autonomy technologies developed for RUAS. This work was mainly motivated by the fact that we have not found a single survey specializing on GNC systems for UAS in general and RUAS in particular, despite the need for a such work after two decades of active research in these areas. This paper provides an overview of GNC systems developed to date to increase the autonomous capabilities of unmanned rotorcraft systems. The approaches that have been reported are organized into three main categories: control, navigation, and guidance. For each category, methods are grouped at the highest level based on the autonomy level they provide, and then according to the algorithmic approach used, which in most cases is closely associated with the type of sensors used. The central objective of this survey paper is to serve the UAS research community by providing an overview of the state of the art, major milestones, and unsolved problems in the areas of GNC for RUAS. This will help researchers to reduce reinvention and enable them to better identify the key critical gaps that prevent advance in the field.

The rest of the paper is organized as follows: The main research groups involved in research and development of autonomous RUAS are presented in Section 2. Section 3 introduces autonomy aspects onboard RUAS, including autonomy definition from the UAS perspective, autonomy levels and metrics, and the main components of a typical autonomous system. Sections 4, 5, and 6 provide a comprehensive survey of major works focusing on flight control, autonomous navigation, and guidance, respectively. Finally, discussions and conclusions about published papers and developed systems are given in Section 7.

[2]The book (Valavanis, 2007), published in 2007, also provided some overview of recent advances in UAS, but it is more a concatenation of contributed chapters from different groups than a survey.

2. RESEARCH GROUPS INVOLVED IN RUAS RESEARCH AND DEVELOPMENT

A number of research groups are working on the development of autonomy technologies for RUAS. Figures 2 lists some 27 research groups that are involved in research and development of autonomous RUAS. The list is not exhaustive and excludes military and industrial research groups.

3. AUTONOMY LEVELS FOR UNMANNED ROTORCRAFT SYSTEMS

Before review of recent advances in research and development of autonomous RUAS, it is important to develop a framework that provides standard definitions and metrics characterizing and measuring the autonomy level of a RUAS. In this section, the autonomy levels for unmanned rotorcraft systems (ALFURS) framework is proposed, which is based on the generic NIST[3] ALFUS framework[4] (Huang, Messina, & Albus, 2007) with some modifications and extensions to make it specific to RUAS and research-oriented. Another of the main objectives of this section is to identify the key components that constitute the autonomy architecture onboard a RUAS. This section is intended to facilitate the understanding of this survey paper, but it can also serve as a common reference for the UAS community.

3.1. Terminology and Key Definitions

In this section, definitions of terms that are most relevant to RUAS autonomy are proposed. As such, consistency with the NIST ALFUS (Huang, 2008) definitions is assumed. However, for certain terms, the NIST generic definitions have been modified to better suit RUAS. For some other terms, *new* definitions are proposed based on information from various sources (ICAO, FAA, UAS Roadmap 2010–2035, NIST, etc.).

Definition 1. *Rotorcraft: A heavier-than-air aircrafts*[5] *that is supported in flight by the dynamic reaction of the air against its power-driven rotors on a substantially vertical axis.*

Definition 2. *Rotorcraft Unmanned Aerial Vehicle (RUAV): A powered rotorcraft that does not require an onboard crew, can*

[3]National Institute of Standards and Technology.
[4]The Autonomy Levels For Unmanned Systems (ALFUS) Ad Hoc Workgroup is a NIST sponsored effort that aims at formulating a logical framework for characterizing the autonomy of unmanned systems in general, covering issues of definitions, metrics, levels of autonomy, etc.
[5]See ICAO and FAA definitions of aircraft.

Name of the Group and Institution	Rotorcraft Platforms	Current Research Areas & projects	Milestones and Major Achievements
Field Robotics Centre, CMU, CMU-FRC-US *www.frc.ri.cmu.edu/projects/landingdemo/* *www.frc.ri.cmu.edu/cademo*	**Class I:** Eurocopter EC135 and the Boeing unmanned Little Bird (ULB) helicopter **Class II:** Yamaha RMAX and R-50. **Class IV**: Otto Quadrotor	- Obstacles detection and avoidance using a 3D LIDAR. - Landing zone detection and safe landing using sweeping 2D LIDAR. - Visual localization and perception.	- Successful obstacle detection and avoidance with RMAX helicopter. - Safe landing area detection and automatic landing with full-scale helicopter .
UAV Research Facility, Georgia Institute of Technology, UAVRF-GTech-US *http://controls.ae.gatech.edu/wiki/uavrf*	**Class II**: Yamaha RMAX and SungWoo Eng. Remo-H helicopters. **Class IV**: Getpsy ducted-fan), GTQ (quadrotor) and GTLama (co-axial).	- Rotorcraft control using adaptive techniques and neural networks. - Vision-based navigation.	- Adaptive autopilot for 3D trajectory tracking and flight control. - Vision-based target tracking. - Vision-based formation flight .
The NASA/Army Autonomous Rotorcraft Project. NASA-ARP-US *http://ti.arc.nasa.gov/projects/apex/projectARP.php*	**Class II:** Yamaha RMAX helicopter.	- 3D navigation in urban environments using vision and LIDAR. - Safe landing area detection (PALACE project). - Mission and path planning.	- Mapping and path planning using a spinning LIDAR. - Vision-based state estimation. - Successful SLAD tests using stereo and LIDAR. - Mission planning for surveillance.
BEAR Group, Berkeley University, BEAR-US robotics.eecs.berkeley.edu/bear/	**Class II**: Yamaha RMAX and R-50 helicopters. **Class III**: Electric Maxi-Joker helicopters.	No current research activities on UAS (to our knowlwdge).	- Flight control using MPC. - Formation flight. - Vision-based landing. - LIDAR-based obstacle avoiadnce.
UASTech Lab., Linkopings University; UASTech-SE *www.ida.liu.se/divisions/aiics/aiicssite/uastech/*	**Class II**: Yamaha RMAX helicopter. **Class IV**: Quadrotors.	- Vision-based landing and localization. - Mission and path planning. - Artificial Intelligence	- Vision-based landing and localization - Path planning. - Mission planning and execustion monitoring.
The French Aerospace Labs (ONERA); ONERA-FR *http://action.onera.fr/accueil*	**Class II**: Yamaha RMAX helicopters.	- Vision-based navigation (target tracking and safe landing). - Mission planning and decisional autonomy (*ReSSAC project*). - UAS cooperation.	- Ground target tracking using vision. - Safe landing area detection using stereo vision. - Mission management.
USL, Univ. of South Florida; USL-USF-US *www.cse.usf.edu/USL*	**Class II:** Yamaha RMAX, **Class III**: Bergen, Raptor 90, Maxi Joker 2	- Flight control. - Vision-based navigation - Fault detection and isolation.	- Automatic flight. - Trafic data collection and analysis.
Kenzo Lab., Chiba University; CHIBA-U-JP *mec2.tm.chiba-u.jp/~nonami* *mec2.tm.chiba-u.jp/uav/main/*	**Class II**: Sky Surveyor, QTW, **Class III**: Hirobo SST-Eagle. **Class IV**: Astech quadrotors. **Class V**: Epson micro Flying Robot (uFR).	- Autopilots design. - Flight control. - Formation flight control. - Vision-based state estimation.	- Automatic flight of different rotorcraft platforms. - Vision-based flight. - Formation flight of two helicopters.
USW@ADFA, Australia *http://seit.unsw.adfa.edu.au/staff/sites/hrp/research/UAV/uav.html* *http://seit.unsw.adfa.edu.au/acme/researchconsult/flight/index.html*	**Class II:** Yamaha RMAX heli. **Class III**: Hirobo Eagle helicopter.	- Flight control. - Optic flow-based navigation. - Automatic landing on a ship.	- Optic flow-based terrain following using the Yamaha RMAX helicopter.
Shenyang Institute of Automation (SIA), SIA-CN *http://uav.sia.cn/en/index.php*	**Class II**: ServoHeli-120 (120 kg), ServoHeli-40 (40 kg).	- Avionics development/integration. - Flight control. - Path planning and UAS cooperation.	- Waypoint navigation with automatic take-off and landing. - Vision-based ground target tracking.
German Aerospace Centre (DLR); DLR-DE *http://www.dlr.de/ft/en/Desktopdefault.aspx/tabid--1377/1905_read-3350/*	**Class III**: ARTIS helicopter (25 kg)	- Vision-based navigation - Stereo vision-based mapping and obstacle avoidance. - Mission management.	- 3D mapping and path planning using stereo vision. - Vision-based flight through obstacle gates.
ARCAA (CSIRO-QUT), ARCAA-AU http://www.arcaa.aero/ http://research.ict.csiro.au/research/labs/autonomous--systems/field-robotics/field-robotics	**Class III**: Vario helicopters. **Class IV**: Quadrotors and Octocopters.	- Robust control of rotorcraft. - Dependable autonomous UAS. - Obstacle detection and path planning using vision and LIDAR.	- Beyond visual range (BVR) flights. - BVR infrastructure inspection. - Mapping and obstacle avoidance using LIDAR and/or stereo vision .
Aerospace Systems and Control Lab., KAIST, ASCL-KR *http://ascl.kaist.ac.kr/*	**Class III**: Voyager GSR 260 helicopter. **Class IV**: Electric T-REX 600 helicopter.	- UAS flight control. - Vision-based navigation. - Obstacle detection and path planning.	- Trajectory tracking and waypoint navigation. - Vision-based flight.
- National University of Singapore, NUS-SG *http://vlab.ee.nus.edu.sg/*	**Class III**: AF25B from Copterworks Inc; Raptor 90 **Class IV**: Trex-450 heli; coaxial rotorcraft.	- Nonlinear control of RUAS. - 3D indoor navigation. - Vision-based navigation.	- Automatic flight control. - Formation flight of a RUAS with a virtual leader. - Vision-based indoor flight.

Figure 2. Research groups working on RUAS.

Name of the Group and Institution	Rotorcraft Platforms	Current Research Areas & projects	Milestones and Major Achievements
- Computer Vision Group, Universidad Politécnica de Madrid, CVG-ES *http://www.vision4uav.com/*	**Class III**: Bergen Industrial Twin and Rotomotion SR20 heli. **Class IV**: Octocopter from Mikrokopter; quadrotor.	- Vision-based pose estimation. - Object tracking. - 3D mapping using vision. - Vision-based flight.	- Vision-based state estimation and ground target tracking. - Visual 3D SLAM. - Visual servoing.
- Robotics, Vision and Control Group , University of Seville, GRVC-ES *http://grvc.us.es/en/*	**Class III**: Heliv and HERO helicopters.	- Control of UAS. - Vision-based navigation and forest fire detection. - Cooperative perception and control of multiple UAS.	- Cooperative detection of forest fire using two RUAS. -Visual odometry and SLAM. - Cooperative faults detection.
- The Center for Advanced Aerospace Technologies (CATEC), CATEC-ES *http://www.catec.aero/*	fleet of heterogeneous UASs (about 6 fixed-wing UASs, 6 unmanned helicopters, and 10 quadrotors).	- Automatic flight control. - Navigation and sense & avoid. - Multi-vehicle coordination. - Avionics and onboard systems. - Automation and robotics.	- European COMETS and AWARE projects.
- Laboratory for Autonomous Flying Robots, TU Berlin, LFAFR-DE *http://pdv.cs.tu-berlin.de/lfafr/*	**Class III**: Aero-Tec CB-5000 heli. (12-16 kg). **Class IV**: Mikado Logo14, self-made quadrotor (5 kg).	- Control of VTOL UAS. - Collision detection & avoidance. - Distributed control of multiple RUAS.	- Load transportation using 3 helicopters. - Sensor nodes deployment by three autonomous helicopters.
- Autonomous Vehicles Group, Aalborg University, AVG-DK *http://www.es.aau.dk/projects/uav*	**Class III**: Bergen Industrial Twin, Vario XLC, Maxi Joker 2, T-Rex 700 Nitro, Thunder Tiger E550. **Class IV**: LMH-120 Corona, T-Rex 450, X-3D-BL quadrotor.	- Flight control with slung load. - Urban UAS navigation. - Task allocation and coordinated flight of AV	- Control of a helicopter slung load system. - Robust control. - Collision-free path generation.
- Autonomous Helicopter project, *http://heli.stanford.edu/* - STARMAC project, *http://hybrid.stanford.edu/~starmac/project.htm*	**Class III**: 90-size XCell Tempest, 90-size Synergy N9. **Class IV:** STARMAC platform with quadrotors.	- Aerobatic and aggressive flight control. - Learning-based control. - Multi-agent control.	- Autorotation-based landing. - Learning-based aerobatic flight. - Bakflip control for a quadrotor. - collision avoidance flight.
- ACFR, University of Sysdney, ACFR-AU *http://www.acfr.usyd.edu.au/research/aerospace.shtml*	**Class III**: UAV vision G18 helicopter.	- Weed monitoring and animals tracking. - Multi-UAV active SLAM. - Cooperative UAS Systems	- Vision-based mapping and classification (fixed-wing). - Aquatic weed surveillance and management (helicopter).
-Aerospace Controls Lab, MIT, ACL-US. *http://acl.mit.edu/*	**Class IV**: Dragonfly quadrotors, AscTech quadrotors.	- UAS modeling and control. - Multi-UAS task assignement. - Multi-vehicle health management.	- Indoor flight using VICON. - Persistent mission planning. - Health management of UAS.
- Robust Robotics Group, MIT, RRG-US *http://groups.csail.mit.edu/rrg/*	**Class IV**: AscTech quadrotors.	- Non-GPS indoor navigation. - Planning under uncertainties. - SLAM-based navigation and flight control.	- waypoint guidance and geolocation of ground targets. - Autonomous flight in unknown indoor environments. - Planning for target tracking.
-Autonomous Systems Lab., ETH, ASL-CH http://www.asl.ethz.ch/research/asl	**Class IV**: home-designed quadrotors and coaxial rotorcraft **Class V**: under-development for sFly project.	- AIRobots project (Inspection Rotorcrafts). - sFly project (Swarm of Micro Flying Robots).	- Design of miniature platforms. - Indoor hovering.
- Heudiasyc Lab, University of Technology of Compiegne, HDS-FR *http://www.hds.utc.fr*	**Class IV**: quadrotors and other home-designed multi-rotor configurations.	- Design of multi-rotor configurations. - Modeling and control of RUAS. - Vision-based navigation.	- Automatic hovering. - Vision-based hovering.
GRASP Lab., University of Pennsylvania,GRASP-US *http://alliance.seas.upenn.edu/~kumar/wiki/*	**Class IV:** AscTech quadrotors.	- Autonomous navigation and exploration. - Trajectory generation and control for 3D dynamic environments.	- Aggressive and accurate control of quadrotors using VICON system. - Cooperative manipulation and transportation with aerial robots.
CEA-USI-ANU Collaborative research between France (CEA and USI) and Australia (ANU)	**Class IV:** Home-built quadrotor.	- Nonlinear flight control. - Optic flow-based terrain following. - Visual servoing.	- Automatic flight. - vision-based hovering. - Terrain following in a structured indoor environment.

Figure 2. *Continued*

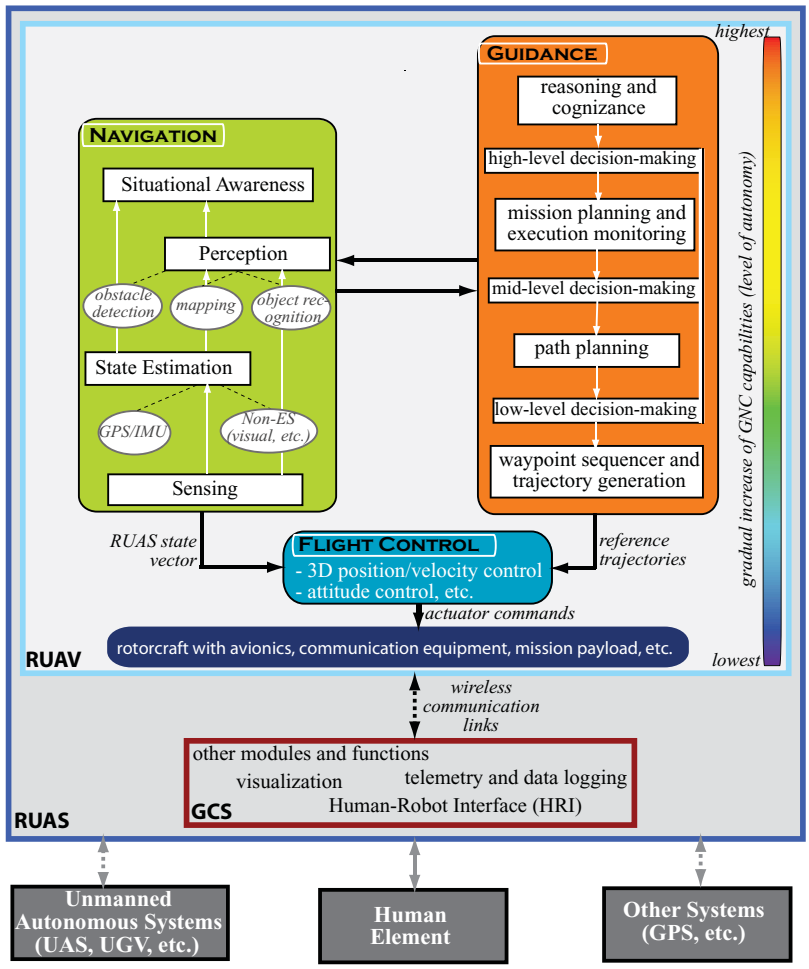


Figure 3. The overall structure of guidance, navigation, and control systems onboard a RUAS.

operate with some degree of autonomy, and can be expendable or reusable. Most RUAVs include integrated equipment such as avionics, data links, payload, and various algorithms needed for flight.

Definition 3. *Rotorcraft Unmanned Aerial or Aircraft System (RUAS): A RUAS*[6] *is a physical system that includes a RUAV, communication architecture, and a ground control station with no human element*[7] *aboard any component. The RUAS acts on the physical world for the purpose of achieving an assigned mission. Contrary to the UAS definition proposed in the US DoD UAS Roadmap 2010–2035 (Roadmap, 2010), here the human element is not part of the RUAS but rather an external system that interacts with the RUAS; see Figure 3.*

Definition 4. *Autonomy: The condition or quality of being self-governing. When applied to RUAS, autonomy can be defined as RUAS's own*[8] *abilities of integrated sensing, perceiving, analyzing, communicating, planning, decision-making, and acting/executing, to achieve its goals as assigned by its human operator(s) through a designed human–robot interface (HRI) or by another system that the RUAS communicates with.*

Definition 5. *Autonomous RUAS: A RUAS is defined to be autonomous relative to a given mission (relational notion) when it accomplishes its assigned mission successfully, within a defined scope, with or without further interaction with human or other external systems. A RUAS is fully autonomous if it accomplishes its assigned mission successfully without any intervention from a*

[6]The plural of RUAS will be also denoted RUAS.
[7]See the UAS Roadmap (Roadmap, 2010) for the definition of human element.

[8]"Own" implies independence from human or any other external systems.

human or any other external system while adapting to operational and environmental conditions.

Definition 6. *Autonomy Level (AL): The term "autonomy level" is used in different contexts in the research community. In Huang (2008) and Huang, Messina, & Albus (2007) for example, AL is equivalent to human independence (HI). In this paper, AL is defined as a set of progressive indices, typically numbers and/or names, identifying a RUAS capability for performing autonomously assigned missions. A RUAS's AL can be characterized by the missions that the RUAS is capable of performing (mission complexity or MC), the environments within which the missions are performed (environment complexity or EC), and independence from any external system including any human element (external system independence or ESI). Note that this AL definition is similar to the contextual autonomous capability (CAC) definition in the NIST ALFUS framework (Huang, 2008), except for HI, which is replaced here by ESI.*

As in Clough (2002) and Merz (2004), we make a distinction between automatic, autonomous, and intelligent systems. An automatic system will do exactly as programmed because it has no capabilities of reasoning, decision-making, or planning. An autonomous system has the capability to make decisions and to plan its tasks and path in order to achieve its assigned mission. An intelligent system has the capabilities of an autonomous system plus the ability to generate its own goals from inside by motivations and without any instruction or influence from outside. In this paper, we are interested in automatic and autonomous systems; intelligent systems are out of the scope of this paper because such systems do not yet exist for UAS. Generally, we do not want an intelligent system, but an autonomous system that does the job assigned.

3.2. Autonomy Levels and Metrics

From reviewing the RUAS literature, it became evident that there is an overall need for a comprehensive framework that allows RUAS practitioners, particularly researchers, to evaluate and characterize the autonomous capabilities of RUAS. The next paragraph gives a brief overview of autonomy-related works, which are also some of the sources that were consulted for developing the ALFURS framework.

Many of the autonomy articles use Sheridan's work (Sheridan, 1992) as a reference for initial understanding of autonomy and human-computer interaction. In his book (Sheridan, 1992), Sheridan proposed a 10-level scale of degrees of autonomy based on who makes the decision (machine or human) and on how those decisions are executed. In 2000, Parasuraman, Sheridan, and Wickens (2000) introduced a revised model of autonomy levels based on four classes of functions: information acquisition, information analysis, decision and action selection, and action implementation. Other relevant concepts and results have been developed by academia, especially from the human–machine interaction and artificial intelligence areas (Castelfranchi & Falcone, 2003; Zeigler, 1990), as well as by NASA and the military,[9] using mainly the OODA (Observe, Orient, Decide, and Act) loop. A more fully developed framework for defining autonomy levels for unmanned systems (ALFUS) has been proposed by an NIST-sponsored ad hoc workgroup (Huang, Messina, & Albus, 2007). In the ALFUS framework, the autonomy level, later renamed contextual autonomous capability (CAC), is measured by weighting the score of various metrics for three aspects, or axes, which are human independence (HI), mission complexity (MC), and environmental complexity (EC). In 2002, the U.S. Air Force Research Laboratory (AFRL) presented the results of a research study on how to measure the autonomy level of a UAV (Clough, 2002). The result of this study is the autonomous control levels (ACL) chart, where 11 autonomy levels have been identified and described. The autonomy level is determined using the OODA concept, namely, perception/situationalawareness (observe), analysis/coordination (orient), decision-making (decide), and capability (act). A few other papers also briefly discussed the UAS autonomy (Fabiani, Fuertes, Piquereau, Mampey, & Teichteil-Knigsbuch, 2007; Lacroix, Alami, Lemaire, Hattenberger, & Gancet, 2007; Merz, 2004; Mettler et al., 2003; Roadmap, 2010), but to our knowledge there have been no other papers published about metrics and autonomy levels for UAS in general and RUAS in particular.

Although the NIST ALFUS and AFRL ACL frameworks provide significant insight and progress in the field of unmanned system autonomy characterization and evaluation, they are difficult to apply directly to RUAS, especially from the research perspective. Indeed, the AFRL ACL chart is most useful and applicable to relatively large UAS operating at high altitudes in obstacle-free environments. Furthermore, the used metrics are military scenario-oriented and are based on the OODA loop, originally developed by the military to illustrate how to take advantage of an enemy. On the other hand, ALFUS is a generic framework covering all unmanned systems, and its application to RUAS is not straightforward. It is also important to note that autonomy metrics and taxonomies have evolved and expanded in theory and practice since then. In this section, we attempt to address the challenge of RUAS autonomy characterization by proposing the autonomy levels for unmanned rotorcraft systems (ALFURS) framework. Based on research, including NIST ALFUS and AFRL ACL studies, and the desire to have a research-oriented autonomy framework that better suits RUAS operating at low altitudes and in cluttered environments, the ALFURS framework was

[9] Future combat systems (FCS) program, autonomous collaborative operations (ACO) program, etc.

developed. ALFURS is based on the RUAS onboard functions that enable its autonomy. These autonomy enabling functions (AEF) can be regrouped into three main categories: guidance, navigation, and control (GNC). Before elaborating this concept, let us first define GNC systems and their relevant components or AEF when related to RUAS, as well as their interaction in a typical RUAS autonomy software implementation; see Figure 3. Indeed, we found in our literature review that GNC terms are commonly used in UAS research, but they are rarely defined and are sometimes mistakenly used.

Definition 7. *Automatic Flight Control System (AFCS): Automatic control*[10] *can be defined as the process of manipulating the inputs to a dynamical system to obtain a desired effect on its outputs without a human in the control loop. For RUAS, the design of flight controllers consists of synthesizing algorithms or control laws that compute inputs for vehicle actuators (rotors, aileron, elevator, etc.) to produce torques and forces that act on the vehicle in controlling its 3D motion (position, orientation, and their time derivatives). AFCS, called also autopilot, is thus the integrated software and hardware that serve the control function as defined.*

Definition 8. *Navigation System (NS): In the broad sense, navigation is the process of monitoring and controlling the movement of a craft or vehicle from one place to another. For RUAS, navigation can be defined as the process of data acquisition, data analysis, and extraction and inference of information about the vehicle's states and its surrounding environment with the objective of accomplishing assigned missions successfully and safely. This information can be metric, such as distances, topological, such as landmarks, or any other attributes that are useful for mission achievement. The main autonomy-enabling functions of a navigation system, from lower to higher level, are as follows:*

- *Sensing: A sensing system involves one or a group of devices (sensors) that respond to a specific physical phenomenon or stimulus and generate signals that reflect some features of or information about an object or a physical phenomenon. Sensors such as gyroscopes, accelerometers, magnetometers, static and dynamic pressure sensors, cameras, and LIDARs are commonly used onboard UAS to provide raw measurements for state estimation and perception algorithms.*
- *State Estimation: This concerns mainly the processing of raw sensor measurements to estimate variables that are related to the vehicle's state, particularly those related to its pose and motion, such as attitude, position, and velocity. These estimates can be absolute or relative. Localization is a particular case of state estimation that is limited to position estimation relative to some map or other locations.*
- *Perception: RUAS perception is the ability to use inputs from sensors to build an internal model of the environment within which the vehicle is operating, and to assign entities, events, and situations perceived in the environment to classes. The classification (or recognition) process involves comparing what is observed with the RUAS's a priori knowledge (Huang, 2008). Perception can be further divided into various functions on different levels such as mapping, obstacle and target detection, and object recognition.*
- *Situational Awareness (SA): The notion of SA is commonly used in aviation systems, and numerous definitions of SA have been proposed. In this paper, we adopt Endsley's definition (Endsley, 1999) of SA as "the perception of elements in the environment within a desirable volume of time and space, the comprehension of their meaning, and the projection of their status in the near future."*[11] *SA therefore is higher than perception because it requires the comprehension of the situation and then the extrapolation or projection of this information forward in time to determine how it will affect future states of the operational environment.*

Definition 9. *Guidance System (GS): A guidance system can be defined as the "driver" of a RUAS that exercises planning and decision-making functions to achieve assigned missions or goals. The role of a guidance system for RUAS is to replace the cognitive processes of a human pilot and operator. It takes inputs from the navigation system and uses targeting information (mission goals) to make appropriate decisions at its high level and to generate reference trajectories and commands for the AFCS at its low level. GS decisions can also spark requests to the navigation system for new information. A guidance system comprises various autonomy-enabling functions including trajectory generation, path planning, mission planning, and reasoning and high-level decision making.*

- *Trajectory Generation: A trajectory generator has the role of computing different motion functions (reference position, reference heading, etc.) that are physically possible, satisfy RUAS dynamics and constraints, and can be directly used as reference trajectories for the flight controller. Reference trajectories can be preprogrammed, uploaded, or generated in real-time onboard the RUAS (dynamic trajectory generation) according to the outputs of higher-level guidance modules.*
- *Path Planning: The process of using accumulated navigation data and a priori information to allow the RUAS to find the best and safest way to reach a goal position/configuration or to accomplish a specific task. Dynamic path planning refers to onboard, real-time path planning.*
- *Mission Planning: The process of generating tactical goals, a route (general or specific), a commanding structure, coordination, and timing for a RUAS or a team of unmanned systems (Huang, 2008). The mission plans can be generated either in*

[10] In the remainder of the paper, the term control refers to automatic control.

[11] This definition is also similar to the one used in the ALFUS framework (Huang, 2008).

advance or in real time. They can be generated by operators or by onboard software systems in either centralized or distributed ways. The term "dynamic mission planning" can also be used to refer to onboard, real-time mission planning.

- *Decision Making: The RUAS's ability to select a course of actions and choices among several alternative scenarios based on available analysis and information. The decisions reached are relevant to achieving assigned missions efficiently and safely. Decision-making processes can differ in type and complexity, ranging from low (e.g., fly home if the communication link is lost) to high-level decision making. Trajectory generation, path planning, and mission planning also involve some decision-making processes.*
- *Reasoning and Cognizance: The RUAS's ability to analyze and reason using contextual associations between different entities. These are the highest level AEF that a RUAS can perform, with varying levels of augmentation or replacement of human cognitive process. Reasoning and cognizance occur prior to the point of decision making. Note that transition from high-level navigation (situational awareness) to high-level guidance (reasoning and cognizance) is of course quite blurry.*

For better understanding of these GNC-related definitions, the reader is encouraged to read the NIST document (Huang, 2008) for more details about the meaning of key terms such as "mission," "goal," "operator," and "environment." Figure 3 shows a simple block diagram of key GNC functions and their interaction in a typical autonomous RUAS. Traditionally, GNC has always been the bottom blobs of Figure 3, i.e., flight control, state estimation, and trajectory generation/waypoint navigation. However, many of the research programs today are geared toward realizing all of the GNC functions in Figure 3 onboard the RUAS.

ALFURS levels are determined based on degrees of RUAS involvement and efforts in performing AEF or GNC functions. The general trend may be that RUAS autonomy level increases when the levels of GNC functions increase. In other words, autonomy level is higher when the GNC systems include high-level AEF functions, and they are performed by the RUAS to a greater extent. Because the main focus of this paper is not on autonomy characterization, this concept[12] will not be elaborated in detail. However, it is important to note that there is a direct correspondence between GNC functions or systems and the MC, EC, and ESI metrics used in the ALFUS project. Therefore, it is possible to establish GNC metrics by mapping ALFUS metrics to the ALFURS framework. Indeed, to achieve a complex mission in a complex environment without any interaction with an external system, the RUAS needs higher levels of GNC. One of the primary motivations for using GNC as aspects or axes for characterizing the autonomy level of RUAS is to use terms and concepts that are familiar to the UAS research community. Indeed, we are interested in a framework that describes the autonomy levels in a simple but meaningful way, so that it can easily be understood and used by other researchers. The intent of this GNC-based ALFURS framework is also to help categorize the RUAS literature and research presented in Sections 4, 5, and 6. Differentiating among consecutive autonomy levels is not trivial and may even be subjective. On the other hand, autonomy levels need to be distinguished to be useful for evaluation and comparison, and to be easily usable by the research community. Therefore, an 11-level[13] scale of autonomy, shown in Figure 4, was proposed, based on *gradual increase* (autonomy as a gradual property) of GNC functions and capabilities. Main or key GNC functions that enable each autonomy level are verbally described, along with their correspondences with MC, EC, and ESI metrics (illustrated by color gradient). The GNC category aspect of this scale is advantageous because it helps RUAS developers to easily and correctly determine the autonomy level of an existing algorithm or system, but also to identify the AEF needed to achieve a certain autonomy level during the design of a new system.

Observation 1. Although the ALFURS framework was proposed for RUAS, it can be used for UAS in general.

In addition to autonomy characterization and evaluation, two other aspects are important for comparing and evaluating RUAS or autonomy technologies: performance and dependability. Autonomy is related to what the RUAS can do (MC, EC, ESI), performance is related to how well the RUAS meets mission requirements (accuracy, time, etc.), and dependability is related to how often the RUAS accomplishes the mission without problems (success rate, failure rate, etc.). In fact, algorithms or GNC systems that are developed to serve the same autonomy level can still be compared and evaluated based on performance and dependability metrics. This is out of the scope of this paper, but such a work will benefit the research community and UAS practitioners in general.

As a direct application of the ALFURS framework, RUAS-related works, reviewed in Sections 4, 5, and 6, are classified based on the GNC aspects and the level of autonomy they are addressing, starting from low-level AEF such as automatic control to high-level functions such as cooperative mission planning.

[12]Metrics for measuring the level of GNC functions, and process for determining the RUAS's level of autonomy using the scores of these various metrics.

[13]Eleven scales, to be consistent with AFRL ACL chart and NIST ALFUS levels.

Level	Level Descriptor	Guidance	Navigation	Control	ESI	EC	MC
10	Fully Autonomous	Human-level decision-making, accomplishment of most missions without any intervention from ES (100% ESI), cognizant of all within the operation range.	Human-like navigation capabilities for most missions, fast SA that outperforms human SA in extremely complex environments and situations.	Same or better control performance as for a piloted aircraft in the same situation and conditions.	*approaching 100% ESI*	*extreme environment*	*highest complexity, all missions*
9	Swarm Cognizance and Group Decision Making	Distributed strategic group planning, selection of strategic goals, mission execution with no supervisory assistance, negotiating with team members and ES.	Long track awareness of very complex environments and situations, inference and anticipation of other agents intents and strategies, high-level team SA.	Ability to choose the appropriate control architecture based on the understanding of the current situation/context and future consequences.	*high level ESI*	*difficult environment*	*collaborative, high complexity missions*
8	Situational Awareness and Cognizance	Reasoning and higher level strategic decision-making, strategic mission planning, most of supervision by RUAS, choose strategic goals, cognizance.	Conscious knowledge of complex environments and situations, inference of self/others intent, anticipation of near-future events and consequences (high fidelity SA).	Ability to change or switch between different control strategies based on the understanding of the current situation/context and future consequences.			
7	RT Collaborative Mission Planning	Collaborative mission planning and execution, evaluation and optimization of multi-vehicle mission performance, allocation of tactical tasks to each agent.	Combination of capabilities in levels 5 and 6 in highly complex, adversarial and uncertain environment, collaborative mid fidelity SA.	same as in previous levels (no-additional control capabilities are required)	*mid level ESI*	*moderate environment*	*mid complexity, multi-functional missions*
6	Dynamic Mission Planning	Reasoning, high-level decision making, mission driven decisions, high adaptation to mission changes, tactical task allocation, execution monitoring.	Higher-level of perception to recognize and classify detected objects/events and to infere some of their attributes, mid fidelity SA.	same as in previous levels (no-additional control capabilities are required)			
5	RT Cooperative Navigation and Path Planning	Collision avoidance, cooperative path planning and execution to meet common goals, swarm or group optimization.	Relative navigation between RUAS, cooperative perception, data sharing, collision detection, shared low fidelity SA.	Distributed or centralised flight control architectures, coordinated maneuvers.			
4	RT Obstacle/Event Detection and Path Planning	Hazard avoidance, RT path planning and re-planning, event driven decisions, robust response to mission changes.	Perception capabilities for obstacle, risks, target and environment changes detection, RT mapping (*optional*), low fidelity SA.	Accurate and robust 3D trajectory tracking capability is desired.			
3	Fault/Event Adaptive RUAS	Health diagnosis, limited adaptation, onboard conservative and low-level decisions, execution of pre-programmed tasks.	Most health and status sensing by the RUAS, detection of hardware and software faults.	Robust flight controller, reconfigurable or adaptive control to compensate for most failures, mission and environment changes.	*low level ESI*	*simple environment*	*low level tasks*
2	ESI Navigation (*e.g., Non-GPS*)	Same as in Level 1	All sensing and state estimation by the RUAS (no ES such as GPS), all perception and situation awareness by the human operator.	Same as in Level 1			
1	Automatic Flight Control	Pre-programmed or uploaded flight plans (waypoints, reference trajectories, etc.), all analyzing, planning and decision-making by ES.	Most sensing and state estimation by the RUAS, all perception and situational awareness by the human operator.	Control commands are computed by the flight control system (automatic control of the RUAS 3D pose).			
0	Remote Control	All guidance functions are performed by external systems (mainly human pilot or operator).	Sensing may be performed by the RUAS, all data is processed and analyzed by an external system (mainly human).	Control commands are given by a remote ES (mainly human pilot).	*0% ESI*	*lowest EC*	*lowest MC*

Figure 4. Illustration of ALFURS autonomy levels as a gradual increase of GNC capabilities and corresponding MC, EC, and ESI. Acronyms: ESI (external system independence), EC (environment complexity), MC (mission complexity), ES (external system), SA (situational awareness), RT (real time).

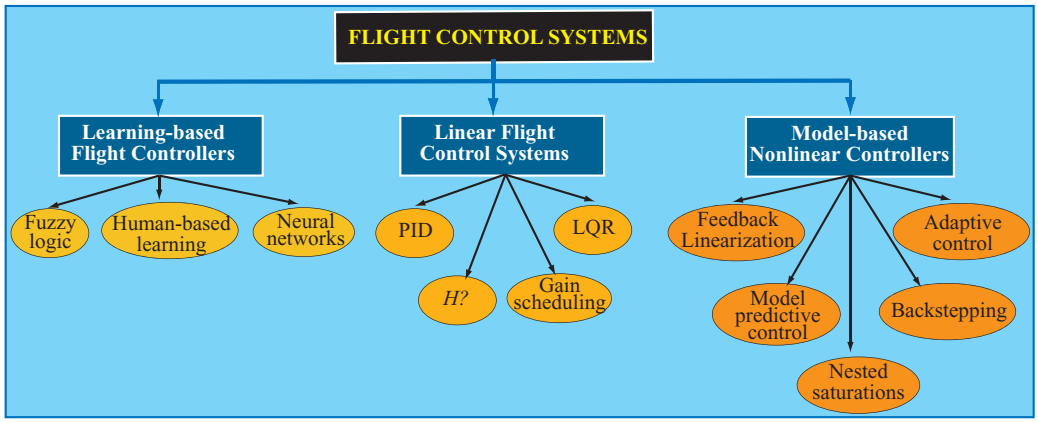


Figure 5. Classification of flight control systems developed for RUAS.

4. FLIGHT CONTROL SYSTEMS

Automatic flight control is a key autonomy-enabling function that increases the RUAS autonomy level from "level 0" to "level 1," as shown in Figure 4. Different control architectures and algorithms have been developed for full-scale manned helicopters and unmanned rotorcraft. Traditional control systems for manned helicopters have been primarily stability augmentation systems (SAS), which are concerned with attitude or altitude control. In addition to SAS, flight control systems developed for RUAS also include velocity/position control, heading control, 3D trajectory tracking, etc. Although RUAS control systems are often based on the control methodologies used for manned aerial vehicles, various other control techniques have been developed in academia in the last decade. Indeed, the RUAS control problem has attracted the attention of many researchers from both the control and robotics communities, because it presents interesting control challenges and an excellent opportunity for developing and testing new control design methodologies. Existing RUAS flight controllers can be classified into three main categories, as in Figure 5: (1) learning-based control methods, (2) linear flight control systems, and (3) model-based nonlinear controllers. To better understand the differences between these control techniques, we first briefly describe the RUAS modeling problem and related works in this area.

4.1. RUAS Dynamics Modeling and Identification

RUAS belong to the class of underactuated mechanical systems, which have fewer control inputs than state variables. Control design for RUAS is generally based on their dynamical model, which features high nonlinearities and strong couplings between different subsystems. Modeling is thus a crucial stage in designing flight controllers for RUAS. Furthermore, it is important to have high-fidelity models for simulation purposes. A model of RUAS dynamics is a mathematical representation that links its inputs to its outputs, as shown in Figure 6. It can be divided into four different subsystems, rigid-body dynamics, force and torque generation mechanism, rotor aerodynamics and dynamics, and actuator dynamics. The development of an accurate model that includes the flexibility and flapping of rotors and fuselage aerodynamics is very complex (Prouty, 1995). For control design purposes, the RUAS is generally considered as a rigid body evolving in 3D space with a mechanism for generating force and torque vectors. The rigid-body dynamics is generally described by the Newton–Euler equations of motion, or energy-oriented approaches such as the Lagrange formulation. The rigid-body equations of motion can be expressed in the body frame or in the inertial frame, and can have different model structures and parameterizations. The force and moment generation mechanism, shown in Figure 6, is not a dynamical system but a process for computing the resultant force and torque vectors experienced by the rigid body. These force and moment vectors (F, M) depend mainly on the thrust T and torque Q generated by each rotor, geometrical parameters, and the orientation of each produced thrust. The orientation of thrust T is controlled by servomotors for tilting the blades (swashplate mechanism) or the rotor itself. Most published papers on modeling and control of RUAS, especially for small platforms such as quadrotors, use this simple model for control design (Bisgaard, Cour-Harbo, & Bendtsen, 2010; Castillo, Dzul, & Lozano, 2004; Frazzoli, Dahleh, & Feron, 2000; Guenard, Hamel, & Mahony, 2008; Johnson and Kannan, 2005; Kendoul, Yu, & Nonami, 2010; Penga et al., 2009; Qi, Song, Dai, Han, & Wang, 2010). For more accurate models, the rigid-body model is essentially extended or augmented with simplified rotor dynamics and aerodynamics, using a combination of momentum and blade element theory. Indeed, detailed modeling of rotor dynamics and aerodynamics can be extremely complex, because the rotor itself is a multibody system. Furthermore,

the produced aerodynamic forces and torques depend on operating conditions and vehicle motion (coupling). Examples of rotor modeling can be found in Prouty's book (Prouty, 1995) for full-scale helicopters and in Mettler's book (Mettler, 2003) for small unmanned rotorcraft. For small-scale RUAS, other dynamics can have major effects on the vehicle response. Indeed, it is necessary to model the dynamics of actuators (swashplate servomotors, gasoline engines, electrical motors, etc.) and stabilizing bars (for helicopters) to better improve model fidelity, especially for simulation purposes.

There is considerable published research on modeling of unmanned rotorcraft. The most popular techniques used for RUAS modeling are the first-principles technique, the system-identification technique, or a combination of both approaches. As described in Mettler (2003), the first-principles modeling technique is a physical approach that involves deriving the mathematical equations of motion using the fundamental laws of mechanics and aerodynamics. It requires considerable knowledge of all the phenomena involved in rotorcraft flight. The resulting models are typically nonlinear and coupled and describe the rotorcraft dynamics in a large portion of its flight envelope. However, they contain dozens of unknown physical parameters such as geometrical data and aerodynamics coefficients. In our literature survey, we found that researchers have employed a variety of dynamic models with different degrees of fidelity and accuracy (nonlinear fully coupled, nonlinear semicoupled, nonlinear decoupled, linear coupled, linear decoupled, etc.). These models are obtained by using a first-principles approach and then by applying model simplification techniques such as Jacobian linearization (Taylor series), feedback linearization (also called dynamic inversion), or decoupling. As mentioned in the previous paragraph, rigid-body models are most often used for control design without comprehensive validation against flight data (Bisgaard et al., 2010; Castillo et al., 2004; Frazzoli et al., 2000; Guenard et al., 2008; Johnson & Kannan, 2005; Kendoul et al., 2010; Penga et al., 2009; Qi et al., 2010). Among the few works that have addressed the problem of physical parameter estimation and experimental validation of the obtained models are Gavrilets, Mettler, & Feron (2001), Bhandari, Colgren, Lederbogen, & Kowalchuk (2005), and Nonami, Kendoul, Suzuki, & Wang (2010). Indeed, Gavrilets et al. (2001) developed a 17-state nonlinear dynamic model of a small aerobatic helicopter and documented the experimental methods used for the estimation of the model's parameters. Experimental validation of the proposed model showed that it is valid up to high speeds for a variety of flight conditions. Another interesting method for modeling the dynamics of a small unmanned helicopter (Thunder Tiger Raptor 50) was proposed by Bhandari et al. (2005). They developed a 6-DoF linear-parameter-varying (LPV) dynamic model based on stability and control derivatives derived from rigid-body equations of motion. It employs the state-space representation of the equations of motion and accounts for the coupling between longitudinal and lateral dynamics of the helicopter. The developed model was evaluated against flight data and the CIFER model and the obtained results were encouraging. At Chiba University, Japan, different linear and nonlinear models have been developed for a variety of RUAS using first-principles or system-identification techniques. Chapter 5 of Nonami et al. (2010) describes the derivation of an analytical linear model for unmanned helicopters, which includes rigid-body dynamics, main rotor dynamics, and aerodynamics, fuselage dynamics and stabilizer bar dynamics. By collaborating with the helicopter's manufacturer, Hirobo Limited, the physical parameters of three unmanned helicopters, SST-Eagle, SF40, and Sky Surveyor, were estimated from platform specifications and refined manually. The derived models have been validated experimentally for the three platforms, and simulation data showed a good match with flight data.

On the other hand, the system-identification technique is essentially a data-fitting process that uses experimental input-output data to produce a mathematical linear model (typically based on transfer functions) of the RUAS dynamics. Shim, Kim, and Sastry (2000) used the time-domain analysis tool from the Matlab System Identification Toolbox to identify a linear model of the Yamaha R-50 helicopter from experimental data. However, effective and popular rotorcraft identification methods are those based on the frequency-domain technique, such as the CIFER tool. CIFER[14] (Tischler and Cauffman, 1992), developed by army/NASA Rotorcraft Division, is one of today's standard tools for identifying linear models of rotorcraft. Some researchers from academia have applied CIFER to identify models of different RUAS configurations (Nonami et al., 2010). A more rigorous and well-documented work about the application of CIFER tool for small RUAS modeling is probably Mettler's work (Mettler, 2003; Mettler, Tischler, & Kanade, 2002). Mettler developed and identified parameterized linear state-space models for hover and cruise flight conditions for the Yamaha R-50 and MIT's X-Cell 60 unmanned helicopters (Mettler, 2003). Key dynamics of both flight conditions is accurately captured with a minimum level of complexity. Although these linear models have been successful for simulation and control around hover and around forward flight, they are valid only within a certain range of the nominal operating point. Furthermore, they naturally lack the expressiveness to comprehensively capture nonlinear aspects of rotorcraft dynamics. To overcome this problem, (LaCivita, Messner, & Kanade, 2002) developed a modeling technique called modeling for flight simulation and control analysis (MOSCA), which integrates first-principles and system-identification

[14]Comprehensive identification from frequency responses.

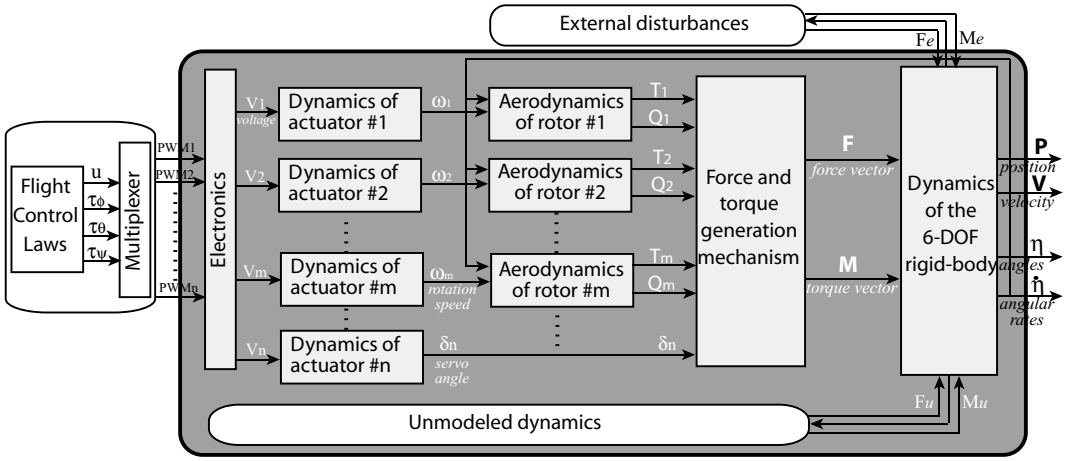


Figure 6. Main components of a typical RUAS dynamical model.

techniques through the use of global optimization methods in the frequency domain. Unknown or uncertain physical parameters of the nonlinear model are automatically tuned to match frequency responses from flight data and from multiple operating points. When applied to the CMU Yamaha R-50 helicopter, the MOSCA technique yields real-time 30-state accurate linear and nonlinear models, which have been used to design an H_∞ controller (La Civita, Papageorgiou, Messner, & Kanade, 2006). Another interesting work that combined first-principles and system-identification approaches for modeling the dynamics of a small aerobatic helicopter (XCell Tempest) was presented in Abbeel, Coates, and Ng (2010). Unknown parameters of a simple nonlinear model of the rigid-body dynamics were identified from flight data by optimizing the prediction accuracy in the time domain. This modeling approach provided good simulation accuracy in flight regimes around level flight, but it still exhibited large prediction errors during simulation of aggressive aerobatic maneuvers. To compensate for this problem, the baseline nonlinear model was refined by learning (identifying) local corrections to the model parameters using time-aligned trajectories of the desired maneuver. Experimental results showed that the resulting models are sufficiently accurate to develop controllers for highly aggressive aerobatic maneuvers; see Section 4.2.2.

4.2. Learning-Based Flight Control Systems

The main characteristic of this control scheme is that the rotorcraft dynamics model is not used, but several trials and flight data are needed to train the system. Among the used methods, fuzzy logic, human-based learning, and neural networks are the most popular.

4.2.1. Fuzzy-Logic-Based Controllers

Fuzzy logic has been successfully applied to control different rotorcraft configurations. The idea is to translate the information and knowledge used by human pilots into rules that can be used by a fuzzy control system. The pioneering work by Sugeno, Griffin, and Bastian (1993); Sugeno, Howard, Isao, and Satoru (1995) applied model-free fuzzy control to a Yamaha R-50 unmanned helicopter. The fuzzy controller was organized hierarchically, with modules for primitive control inputs in a lower layer that could be activated by basic flight mode modules in an upper layer. A combination of expert knowledge and training data was used to generate and adjust the fuzzy rules base. The experimental results showed that the fuzzy-controlled helicopter was able to execute basic flight behavior such as hovering, forward flight, and climbing turns. A similar approach was developed by Montgomery and Bekey (1998) for control of the USC AVATAR helicopter. A hierarchical behavior-based control architecture was used, with each behavior implemented as a hybrid fuzzy logic controller (FLC) and general regression neural network controller (GRNNC). Fuzzy control rules were generated automatically from training data using a model-free "teaching by showing" methodology. The GRNNCs were incrementally built and modified whenever the controller did not meet performance criteria, enhancing the control of the FLCs. This hybrid flight controller was validated in simulations, but it turned out that it was inadequate when flight-tested on an actual robotic helicopter. There are also some works on applying fuzzy controllers for full-scale helicopters. For example, the work presented in Phillips, Karr, and Walker (1996) demonstrated the feasibility of replacing the human pilot on a UH-1H helicopter with a fuzzy flight control system for some standard maneuvers such as hovering, forward flight, and coordinated turning. Effective rules for the fuzzy controller were computed using a genetic algorithm and a

numerical model of the UH-1H helicopter. In a recent paper, Garcia and Valavanis (2009) presented an implementation of a robotic helicopter test bed based on an RC Maxi-Joker II helicopter. The developed flight control system includes four fuzzy controllers for pitch–longitudinal motion, roll–lateral motion, yaw, and collective-vertical control. These decoupled fuzzy controllers were developed in Matlab using Sugeno constant fuzzy logic and a weighted average defuzzification method. Fuzzy rules were designed based on general RC helicopter flight. The developed fuzzy flight controller was validated in outdoor experiments over 300 autonomous flights including hovering, takeoff and landing, forward flight, and waypoint flight.

4.2.2. Human Based Learning Techniques

A different learning-based approach, based on analysis of the pilot's execution of aggressive maneuvers from flight test data, was developed by MIT researchers for acrobatic maneuvering of a small robotic helicopter. Gavrilets, Frazzoli, Mettler, Piedmonte, and Feron (2001) presented an analysis of a series of input–output sequences for an aileron roll and a hammerhead maneuver, collected from an instrumented small-scale helicopter performing acrobatic flight. The objective was to understand and extract the input sequences and feedback mechanisms that a human pilot uses to execute aggressive maneuvers. The insight gained in this study has been used to develop and implement an intuitive control logic for autonomous aggressive flight (Gavrilets, Mettler, & Feron, 2004). Conventional multivariable trim trajectory controllers were used before and on exit from the maneuvers. This flight control system was flight-tested with split-S, hammerhead, and 360° axial roll maneuvers, as well as a split-S-hammerhead maneuver sequence, using a small unmanned helicopter. Abeel et al. (2010) from Stanford University investigated the application of reinforcement (or apprenticeship) learning to aerobatic helicopter flight. The proposed approach is characterized by the following (1) an apprenticeship learning algorithm (expectation maximization) is used to extract the intended maneuver trajectory from multiple suboptimal expert demonstrations; (2) the helicopter dynamic model is obtained by first building a baseline dynamic model using a mix of first-principles modeling and fitting to flight data, and then learning a high-accuracy dynamics model (local corrections) from the flight data of the desired maneuver; (3) the learning of the cost-to-go (reward) function from collected data and the use of an iterative linear–quadratic regulator (iLQR) and differential dynamic programming (DDP) as a flight controller. The flight control system has been demonstrated on an instrumented 90-size XCell Tempest helicopter for which state estimation and control algorithms have been implemented on a ground computer. Experimental results included the autonomous execution of a wide range of maneuvers, including in-place flips, in-place rolls, loops and hurricanes, autorotation landings, chaos and tic–tocs, and complete air shows. These results illustrate that complex control theory is not needed to perform aerobatic and sophisticated-looking maneuvers such as barrel rolls. However, precise, repeatable trajectory tracking may not be achievable with such learning-based techniques.

4.2.3. Neural-Network-Based Controllers

Another interesting method for learning-based control is artificial neural networks (ANN). There are several works that use ANN to identify rotorcraft dynamic models offline or online, as in Dierks and Jagannathan (2010), but most of these works present simulation results only. ANN are generally used to identify some unknowns and then combined with standard control techniques, as in Johnson and Kannan (2005). In Buskey, Wyeth, and Roberts (2001), a ANN-based controller was developed for helicopter hovering. It uses direct mapping of inertial data to actuator control via a feedforward network using the back-propagation training regime. Partial hovering for several seconds has been achieved with a small helicopter.

Observation 2. Learning-based approaches are promising for helicopter control and have been already demonstrated successfully for achieving several flight maneuvers. Their main advantage is their flexibility for implementation on different platforms, because they are generally model-free. Moreover, some learning methods such as ANN-based controllers allow direct mapping between navigation data and actuator deflections, which may be good for fast reactive behavior and vehicle survivability. However, stability and robustness of these approaches are difficult to analyze. Furthermore, no extensive experimental evaluation over a wide range of environments and scenarios has been performed yet, compared to standard linear methods such as proportional-integral-derivative (PID) control.

4.3. Linear Flight Controllers

Conventional approaches to flight control and most initial attempts to achieve autonomous helicopter flight have been based on linear controllers such as PID, LQR and H_∞. Indeed, in the late 1960s and early 1970s, the CH-53A full-scale helicopter achieved autonomous waypoint navigation using classical linear control techniques.

4.3.1. Proportional Integral Derivative

One of the most successful and widely used linear controllers is the PID controller. Generally, a hierarchical two-loop architecture is used, assuming a time-scale separation between the inner- and outer-loop subsystems. The inner loop controls the attitude using single-input-single output (SISO) PID for each axis, and the outer loop is responsible for translation motion control using decoupled PID

controllers also. These PID-type flight controllers were implemented and are still in use in most robotic helicopter projects that are listed in Figure 2. In this control scheme, the rotorcraft model may not be used, and controller gains can be tuned empirically by trial and error. This is a time-consuming process that can be improved by identifying the rotorcraft dynamic model from flight data and then tuning PID controllers in simulation using the identified model, as described in Section 4.1.

4.3.2. Linear–Quadratic Regulator/Gaussian

The Linear–quadratic regulator (LQR) or linear–quadratic gaussian (LQG) is also a popular optimal control technique that has been successfully applied to control several RUAS configurations. In How et al. (2008), the LQR was used for accurate orientation and position control of MIT's RAVEN quadrotors. At Chiba University, RUAS of different weights (2 to 48 kg) achieved stable hovering and accurate trajectory tracking using LQG-based cascaded controllers designed from identified linear models (Nonami et al., 2010; Shin, Fujiwara, Nonami, & Hazawa, 2005). In Bergerman, Amidi, Miller, Vallidis, and Dudek (2007), an LQR controller was used as an inner loop to stabilize the unstable poles of the identified linear model of the RMAX helicopter. This controller was then combined with a feedback linearization controller that decoupled the linear dynamics of the lateral, longitudinal, vertical, and heading axes and enabled trajectory tracking. Experimental results using the RMAX helicopter were presented for an 8-shaped trajectory tracking with speed 3 m/s. Some of the learning-based controllers presented in the previous section also used the LQR technique (Abbeel et al., 2010; Gavrilets et al., 2004), for designing the baseline controller.

4.3.3. H_∞

The H_∞ control approach belongs to the family of model-based robust control design methods that can cope with the problem of parametric uncertainty and unmodeled dynamics. It has already been used for the control of a full-scale helicopter (Smerlas et al., 1998), and it is also used in some commercial autopilots such as the "wePilot" family from the Swiss company "weControl"[15]. One of the bedocumented H_∞-based controllers to be implemented and flown on an unmanned RUAS is probably the H_∞ loop-shaping controller developed by La Civita et al. (2006). The CMU Yamaha R-50 helicopter linear and nonlinear models were obtained by combining first-principles and system identification techniques. The controller design employed a linear model extracted at hover, whereas the simulation and evaluation used the nonlinear model. The implemented control structure is based on a cascaded architecture, where a multiple-input multiple-output (MIMO) inner loop stabilizes the helicopter's attitude, and four independent SISO controllers are responsible for trajectory tracking. All loop designs used an H_∞ loop-shaping technique. This controller was implemented onboard the Yamaha R-50 helicopter and demonstrated good tracking performance during a set of maneuvers. H_∞ has also been applied for attitude and altitude control of the ETH coaxial micro helicopter (Schafroth, Bermes, Bouabdallah, & Siegwart, 2010). There are also a number of recent theoretical works (without experimental results) (Gadewadikar, Lewis, Subbarao, & Chen, 2008; Gadewadikar, Lewis, Subbarao, Peng, & Chen, 2009; He & Han, 2010) on using H_∞ for helicopter control.

[15]http://www.wecontrol.ch/.

4.3.4. Switched Dynamics and Gain Scheduling

To extend the capabilities of linear flight controllers, nonlinear dynamics of RUAS can be modeled as a collection of simplified linear models, with each model representing a particular operating regime. This approach is commonly used to design flight controllers for aerospace systems. Gain scheduling control is the most used technique. An interesting model-based linear controller that used this principle was developed by the Army/NASA rotorcraft division and implemented on the Yamaha RMAX helicopter (Takahashi, Schulein, & Whalley, 2008). Control design is based on two linear models of the rotorcraft that are developed by collecting frequency sweeps in hover and forward flights and using the CIFER software. PID-like feedback blocks were constructed to have high gain at lower frequencies and low gain at higher frequencies. Gain scheduling was used to handle hover-forward transitions. This flight control system was extensively evaluated in simulations and flight-tested on the RMAX robotic helicopter to achieve hovering, automatic takeoff and landing, trajectory tracking, and waypoint navigation. The system is being used in a variety of research projects, including obstacle field navigation. An alternate approach that allowed a small quadrotor UAS to perform aerobatic maneuvers was proposed by Gillula, Hoffmann, Huang, Vitus, and Tomlin (2011). In this method, the behavior of the system is approximated as a discrete set of simpler hybrid modes representing the dynamics in specific portions of the state space. Linear control tools and reachable sets are then used to design the control laws and to construct maneuvers that safely transition through a sequence of modes. This strategy has been implemented on a STARMAC quadrotor and successfully demonstrated for a backflip maneuver.

Observation 3. Designing and implementing linear flight controllers is straightforward, and there are many available tools to tune their gains and to analyze their performance and robustness. Moreover, they have been successfully used in aerospace systems and RUAS to achieve wide

range of tasks and maneuvers. However, it is well known that these linear controllers suffer from performance degradation when the helicopter leaves the nominal conditions or performs aggressive maneuvers. From the theoretical point of view, it is also difficult to prove the asymptotic stability of the complete closed-loop system. Despite these limitations, PID, LQR, H_∞, and gain scheduling are the most widely accepted methods of flight control.

4.4. Model-Based Nonlinear Flight Controllers

To overcome some of the limitations of linear approaches, a variety of nonlinear flight controllers have been developed and applied to rotorcraft control. These nonlinear flight controllers are generally based on the nonlinear model of the rotorcraft dynamics, obtained using the first-principles technique with parameter identification in some cases. Among these, feedback linearization or dynamic inversion, adaptive control, and model predictive control have received much of the attention and have been successfully applied to helicopter control. Other control techniques such as backstepping and nested saturations have also been researched for the control of small and mini RUAS such as quadrotors.

4.4.1. Feedback Linearization

Feedback linearization is a broad class of techniques that are commonly used to control nonlinear systems. It involves the use of nonlinear transformation techniques to transform the state variables of the system into a new coordinate system where the dynamics are linear (diffeomorphism). Linear tools can then be applied and subsequently converted back into the original coordinates via an inverse transformation. Dynamic inversion is a specific case of feedback linearization control that has been investigated at great length for application to manned and unmanned aircraft. Koo and Sastry (1998), investigated the use of exact and approximate input–output feedback linearization for nonlinear control of an unmanned helicopter. The study showed that exact input–output linearization results in unstable zero dynamics because of the coupling in control inputs. A tracking controller was then synthesized using approximate linearization. However, this controller was verified in simulations only. Researchers from Singapore National University have proposed a hierarchical nonlinear flight controller that combines "composite" nonlinear feedback control for the inner loop and dynamic inversion for the outer loop (Penga et al., 2009). The inner loop consists of a linear feedback control law and a nonlinear feedback control law for velocity, attitude, heave velocity, and heading control, using a linear model of a small helicopter, identified from flight data. Reference velocities for the inner loop are computed by the outer loop position controller using a dynamic inversion method. A Raptor 90 helicopter, equipped with the developed autopilot, was able to achieve automatic takeoff and landing, hovering, slithering, pirouetting, vertical turning, spiral turning, etc. In Kendoul, Yu, and Nonami (2010), the design of a nonlinear flight control system and its implementation onboard a 700-g quadrotor are presented. The controller was designed by deriving a mathematical model of the quadrotor dynamics and exploiting its structural properties to transform it into two cascaded subsystems (attitude and translation) coupled by a nonlinear interconnection term. Partial passivation design and inverse dynamics techniques were used to synthesize control laws for each subsystem, thereby resulting in a hierarchical and nonlinear inner- and outer-loop controller. The asymptotic stability of the entire connected nonlinear system was proven by exploiting the theories of systems in cascade. This flight controller was validated through several flight tests including accurate attitude tracking, automatic takeoff and landing, long-distance flight, waypoint navigation, and spiral trajectory tracking. The same platform equipped with this nonlinear flight controller has been also used for vision-based flight research (Kendoul, Fantoni, & Nonami, 2009; Kendoul, Nonami, Fantoni, & Lozano, 2009; Nonami et al., 2010). Nonlinear controllers based on the dynamic inversion technique can also be used for aerobatic and aggressive maneuvers control, as demonstrated in Mellinger and Kumar (2011). The developed algorithm generates optimal trajectories through a sequence of 3D positions and yaw angles. These trajectories are then accurately tracked using a nonlinear inner- and outer-loop controller. Impressive experimental results for a small quadrotor[16] flying through static and thrown circular hoops are presented.

4.4.2. Adaptive Control

Dynamic inversion can be vulnerable to modeling errors and uncertainties. Therefore, some robust nonlinear control schemes have been proposed and implemented on unmanned helicopters. Adaptive control is such a robust control technique, which can handle unmodeled dynamics and parametric uncertainties. The UAS group at Georgia Tech. is very active in researching adaptive techniques for flight control. They have developed different variants of nonlinear flight controllers and implemented them on various fixed-wing as well as rotary-wing UAS. Their research has focused on the use of a direct neural-network-based adaptive control architecture that compensates for unknown plant nonlinearities in a feedback linearizing control framework. The attitude and translational dynamics are controlled separately, using approximate dynamic inversion and linear controllers for the linearized dynamics. This generic inner–outer loop controller has been augmented by a multi–layer neural network that parameterizes the unknown model error and provides online adaptation to

[16] VICON motion capture system was used for navigation.

minimize the effect of nonlinear parametric uncertainty arising because of approximate inversion. A pseudo-control hedging (PCH) method was also used to protect the adaptation process from actuator limits and dynamics. Johnson and Kannan (2005) and Kannan (2005) described the theoretical basis of the developed adaptive controller and presented experimental results for trajectory tracking by the Yamaha RMAX helicopter. This adaptive flight controller has also been flown on different RUAS at Georgia Tech., and it is still used as the main autopilot in different research projects, including formation flight and vision-based navigation research. It is available today as a commercial autopilot (Christophersen et al., 2006) and it is also integrated into commercially available autonomous helicopters.[17] Several extensions of the standard adaptive flight controller have been published. In Yavrucuk, Prasad, and Unnikrishnan (2009) for example, the baseline adaptive flight controller has been augmented by an envelope protection system (EPS) that prevents large structural loads on the RMAX helicopter by limiting maximum values of the load factor. The effectiveness of this system was demonstrated by using the RMAX helicopter to perform aggressive maneuvers such as hover-to-hover acceleration–deceleration and E-turn maneuvers at high forward speeds.

There also exist some works on developing and implementing indirect adaptive controllers on small unmanned rotorcraft. Kendoul, Nonami, et al. (2009b) proposed a nonlinear adaptive controller for vision-based flight of a quadrotor UAS. The nonlinear hierarchical controller, described in Kendoul et al. (2010), has been augmented by an adaptive mechanism (observer) that estimates the visual unknown scale factor online by fusing optic flow and inertial measurements. The developed adaptive vision-based autopilot was implemented on a quadrotor UAS and validated in outdoor and indoor autonomous flights. An integrated observer–controller scheme was also used in Bisgaard et al. (2010) to design an adaptive control system for autonomous helicopter slung load operations. The observer uses vision to estimate the length of the suspension system and the position of the slung load, thereby allowing the model to be adapted online. Then a combined feedforward (input shaping) and feedback (delayed feedback) scheme for simultaneous avoidance of swing excitation and active swing damping is used. The performance of this control scheme was evaluated through simulations and laboratory flight tests, and a significant reduction in slung load swing was observed.

4.4.3. Model Predictive Control

Another approach to the control of nonlinear systems, which does not rely on dynamic inversion, is model predictive control (MPC), also referred to as receding horizon control (RHC). MPC employs an explicit model of the plant to predict the future output behavior, and then tracking error over a future horizon is minimized, preferably online, by solving optimal control problems. The UCB Aerobot (BEAR) team has developed a nonlinear model predictive controller for tracking control of unmanned helicopters (Kim, Shim, & Sastry, 2002). The tracking control problem was formulated as a cost minimization problem in the presence of input and state constraints. The minimization problem was then solved with a gradient-descent method, which is computationally light and fast. The proposed flight control system has been successfully implemented on a number of small helicopters and validated in various applications. Shim, Kim, and Sastry (2003b) presented experimental results from waypoint navigation, a probabilistic pursuit–evasion game, and vision-based target tracking using the UCB Yamaha R-50 helicopter. The nonlinear MPC has also been extended to perform LIDAR-based obstacle avoidance (Shim, Chung, & Sastry, 2006), collision avoidance using centralized (Shim & Sastry, 2007) and decentralized architectures (Shim & Sastry, 2006), and formation flight of two unmanned helicopters together with six simulated helicopters (Shaw, Chung, Hedrick, & Sastry, 2007). Formation flight of two unmanned helicopters was also achieved at Chiba University in Japan, using a leader–follower configuration and model predictive control (Nonami et al., 2010, Chapter 9). The Shenyang Institute of Automation (China) has also implemented modified generalized predictive control on the ServoHeli-40 unmanned helicopter (Qi et al., 2010). They have tested and compared the following three flight controllers: generalized predictive control (GPC), stationary increment predictive control (SIPC), and active model-based stationary increment predictive control (AMSIPC). The proposed AMSIPC is a kind of adaptive controller that is composed of two main components: (1) a linear model predictive controller that is designed using a reference model obtained by linearizing the nonlinear dynamics of a helicopter at one flying mode and (2) an active modeling algorithm that uses the adaptive set membership filter to estimate on line the error between the reference model and the actual dynamics of the rotorcraft. Experimental results showed that when the helicopter increases its longitudinal velocity and changes flight mode from hovering to cruising, the AMSIPC controller results in better tracking performance.

4.4.4. Backstepping Methodology

Backstepping is a well-known recursive methodology for the control of linear and nonlinear underactuated systems. The backstepping technique was originally introduced in adaptive control theory to recursively construct the feedback control law and the associated Lyapunov function for a class of nonlinear systems satisfying certain structural properties. The structural properties of the

[17] http://www.adaptiveflight.com/index.html.

rotorcraft dynamics model allowed the application of the backstepping technique for designing nonlinear flight controllers for these machines. Indeed, several papers about backstepping-based control of RUAS have been published. However, most of the proposed controllers have been validated in simulations only. Early theoretical work on using backstepping techniques to design nonlinear controllers for RUAS can be found in Frazzoli et al. (2000), Mahony, Hamel, and Dzul (1999) and Olfati-Saber (2001). The control design is generally based on an approximate mathematical nonlinear model of the rotorcraft, and the stability of the closed-loop system is analyzed using Lyapunov theory. Bouabdallah and Siegwart (2005) applied the backstepping and sliding-mode techniques to the control of a mini indoor quadrotor. The proposed strategies have been evaluated in simulations as well as in flight tests for the attitude stabilization of a quadrotor on a test bench. Successful implementations of backstepping-based nonlinear flight controllers on actual RUAS have been reported in Pflimlin, Soures, & Hamel (2006) for a ducted-fan UAS and Guenard et al. (2008) for a mini quadrotor. The hierarchical controller proposed in Pflimlin et al. (2006) exploits the cascade structure of the system to design a position controller that computes the required thrust vector and an attitude controller that tracks the desired angles. The stability of the connected system has been proven and its performance has been evaluated in a waypoint navigation flight test using the Bertin Technologies Inc. HoverEye ducted-fan UAS. An extension of this flight controller has been implemented on the CEA (French Atomic Energy Commission) quadrotor for image-based visual servoing of the vehicle (Guenard et al., 2008). Based on the kinematics of an image centroid under spherical projection, a visual error signal was computed and used together with IMU data by a backstepping-based nonlinear controller to stabilize the vehicle. A similar approach has been also used in Herisse, Hamel, Mahony, and Russotto (2010) to perform optic-flow-based terrain following.

4.4.5. Nested Saturation Technique

Control system design for RUAS with actuator saturation is an important practical design problem that many previous approaches did not consider. This problem is more significant in the case of small and mini RUAS, where actuator saturation occurs frequently because of aggressive maneuvers (to avoid obstacles) or external disturbances (wind). Actuator saturation limits the operational envelope of the vehicle and may induce instability in the controlled system. Different strategies have been proposed in the literature to handle the problem of RUAS control with saturated inputs, but very few of them have been implemented on actual rotorcraft and demonstrated in real time. Among them, the Georgia Tech. adaptive flight controller, previously described, has been augmented by pseudocontrol hedging (Johnson & Kannan, 2005) to cope with actuator limits and dynamics. The Heudiasyc Laboratory at the University of Technology of Compiegne (France) has been very active in designing flight controllers for mini quadrotors with actuator saturations. Their approach, described in Castillo et al. (2004), consists of feedback linearizing the attitude dynamics and applying a nested saturation technique to stabilize the longitudinal, lateral, heave, and yaw dynamics. The resulting four control laws are nonlinear and prevent actuator saturation by sequentially bounding state variables, from angular rates to positions. Nested saturations are a low-gain technique with low performance but have global stability results in the presence of control input saturations. This control strategy was used to stabilize a mini quadrotor UAS. Experimental results from indoor flights are reported in Castillo et al. (2004). Different variants of this control strategy have been developed and implemented on different rotorcraft configurations (Kendoul, Lara, Fantoni, & Lozano, 2007; Lozano, 2010; Romero, Salazar, & Lozano, 2009).

Observation 4. Model-based nonlinear controllers are an interesting alternative for advanced flight control with better flight capabilities. Nonlinear techniques such as feedback linearization, adaptive control, MPC, and backstepping control have been employed successfully in various universities and organizations to control various RUAS. The best-documented and flight-tested nonlinear flight controllers are probably the *Georgia Tech* adaptive controller and the UCB MPC controller. Nonlinear controllers outperform linear techniques in terms of robustness to unmodeled dynamics and disturbances, tracking accuracy over a wider flight envelope, etc. However, the reported experimental results have not shown significant progress in flying capabilities compared to standard linear controllers. This may be due to the lack of proper implementation and tuning of these controllers, which are even more complex when the model parameters are unknown. Because the theory of many of these nonlinear flight controllers is well developed, we recommend more experimental work in rigorously implementing and extensively flight-testing developed algorithms.

5. NAVIGATION SYSTEMS

In addition to flight control, designing an autonomous UAS with higher levels of autonomy requires an appropriate navigation system for sensing, state estimation, environment perception, and situational awareness. A definition of navigation and its components from the UAS perspective is provided in Section 3. In this section, we attempt to review the existing navigation systems developed for RUAS and classify them into categories, groups, and classes based on the autonomy level they provide (first level of classification), the sensing technology used (second level of classification), and the algorithmic approach or method used

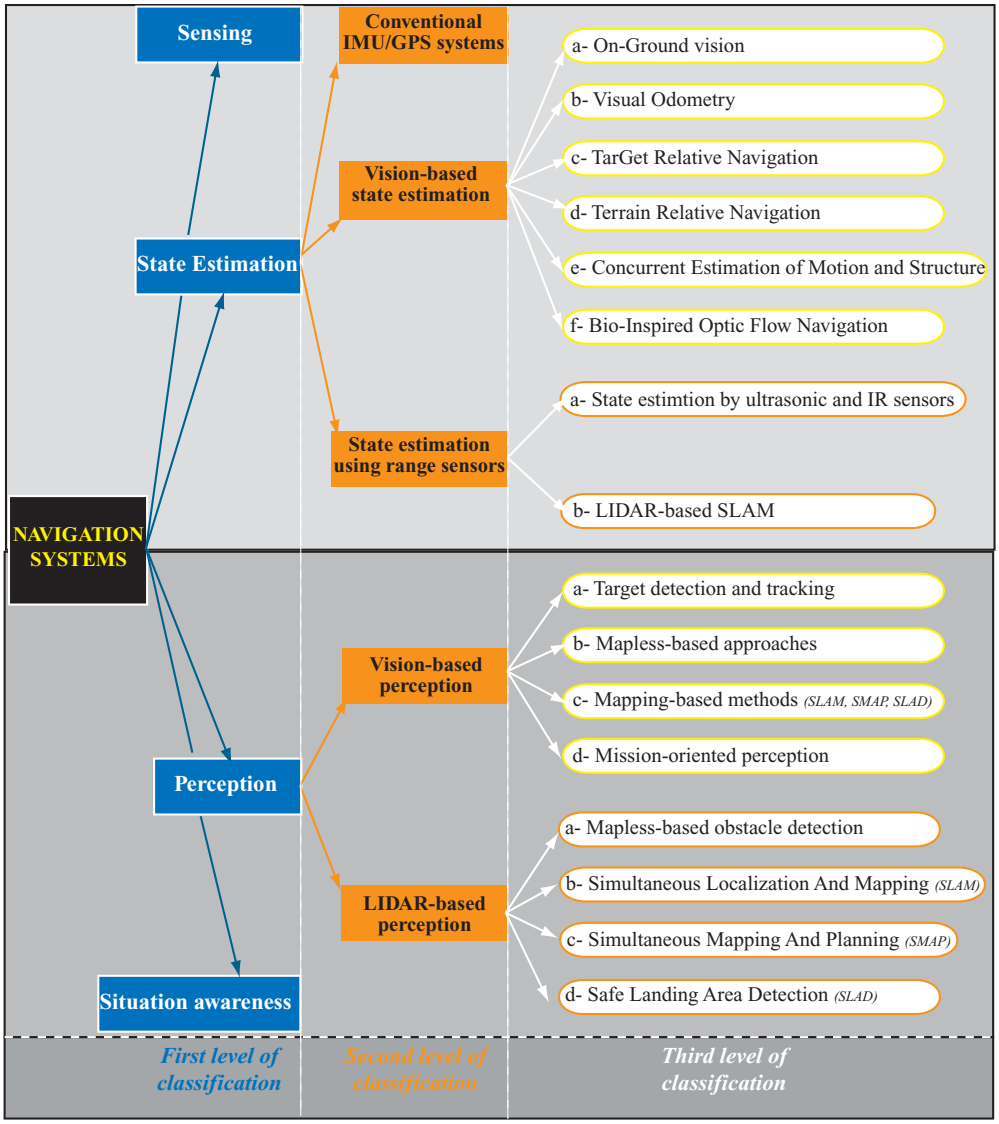


Figure 7. Classification of navigation systems developed for RUAS.

(third level of classification), see Figure 7 for the different levels of classification.

5.1. Sensing Technologies

The first component of a navigation system is sensing, which involves an actual sensory device such as an inertial measurement unit (IMU), a camera, or any other sensor. Sensing capabilities are required for all autonomy levels, a presented in Figure 4. This section is dedicated to a general introduction to sensing technologies commonly used for RUAS navigation. A detailed description of navigation sensors, a comparison of their characteristics and performance, and the state of the art in research and development of sensing technologies for UAS are out of the scope of this survey.

Conventional and basic navigation sensors (minimal sensors suite) for RUAS include state estimation sensors for flight control, such as an IMU (three gyroscopes, three accelerometers, and three magnetometers) for attitude estimation, a global navigation satellite system (GNSS) for position and velocity estimation, and an altimeter (barometric, LIDAR, radar) for enhancing the height estimation. Most current UAS research platforms and commercial autopilots contain these conventional navigation sensors. Progress in electronics miniaturization and

microprocessors has made the integration of these sensors in one small and compact device possible. Indeed, lightweight integrated IMU/GPS/altimeter devices (e.g., Xsens MTi-G 68 g, sbg IG-500N 45 g, Microstrain 3DM-GX3-35 23 g) with acceptable accuracy and affordable prices are available today.

For perception, ranging sensors and cameras are the sensors used most on RUAS. Cameras or electrooptics sensors are a popular approach for environment sensing because they are light, passive, and compact and provide rich information about the vehicle's self-motion and its surrounding environment. Computer vision is a popular solution for applications that require target detection and tracking, landmark recognition, and other tasks. There exist different types of imaging sensors, such as single cameras, stereo cameras, omnidirectional cameras, and optic flow sensors. The drawback of camera-based approaches is their sensitivity to the ambient light and scene texture. Furthermore, the complexity of image processing algorithms makes their real-time implementation on embedded microprocessors challenging.

LIDAR (light detection and ranging) is a suitable device for mapping and obstacle detection, because it directly measures the range by scanning a laser beam in the environment and measuring distance through time of flight or interference. Other common terms for LIDAR are LADAR (laser detection and ranging) and laser radar, which are often used in military contexts. LIDAR does not rely on ambient lighting and scene texture. However, LIDAR units they are heavier than cameras and energy-consuming (active), and most available off-the-shelf systems use a single-line scan (2D LIDAR). For 3D navigation, these 2D LIDAR units have been mounted on sweeping or spinning mechanisms when used on RUAS and ground robots. A few 3D compact LIDAR units exist, but they are not commercially available, such as the one from Fiberteck Inc. (Scherer, Singh, Chamberlain, & Elgersma, 2008) or they are very expensive and heavy, such as the Velodyne 3D LIDAR. Flash LIDAR unit or 3D time-of-flight cameras are a promising emerging 3D sensing technology that will certainly increase the perception capabilities of RUAS. Unlike traditional LIDAR devices that scan a collimated laser beam over the scene, a flash LIDAR unit illuminates the entire scene with diffuse laser light and computes the time of flight for every pixel in an imager, thereby resulting in a dense 3D depth image. Recently, several companies have begun offering flash LIDAR units commercially, such as the SwissRanger SR4000 (510 g) from MESA Imaging AG, Canesta 3D cameras, the Kinect sensor (based on the Canesta 3D camera), the TigerEye 3D Flash LIDAR unit (1.6 kg) from Advanced Scientific Concepts Inc., the Ball Aerospace 5th Generation flash LIDAR.

Radar is the sensor of choice for long-range collision and obstacle detection in larger UAS and manned aircraft. Radar provides nearly all-weather, high-resolution, and broad-area imagery. The problem with radar is that it requires a lot of power, and the localization of the beam requires a large antenna. Radar units are also heavier than previous sensors, which makes their integration into small RUAS difficult. Some lightweight radar systems are available today, but they are expensive, such as the miniature radar altimeter (MRA) Type 1 from Roke Manor Research Ltd., which weighs only 400 g and has a range of 700 m. We are not aware of any work by an academic research group, on using radar onboard RUAS for obstacle and collision avoidance, except the work in Viquerat, Blackhall, Reid, Sukkarieh, & Brooker (2007) but using a fixed-wing UAS. The use of other ranging sensors such as ultrasonic and infrared sensors has been limited to a few indoor flights or ground detection during automatic landing.

5.2. State Estimation

State estimation for RUAS can be defined as the process of tracking the current vehicle 3D pose, which can be absolute or relative to some object or initial location. The design of motion estimation algorithms is necessary for flight control and a crucial step in the development of autonomous flying machines. Typically, state estimation algorithms fuse information from many sources (multiple sensors, preregistered maps, ground segment data, etc.) to estimate the vehicle's attitude, height, velocity, and position, which may be absolute or relative. We classify state estimation algorithms into three main categories based on the sensing technology used: (1) conventional systems, (2) vision-based systems, and (3) systems that rely on active ranging sensors.

5.2.1. Conventional GPS–Inertial Measurement Unit Systems

UAS normally rely on an IMU and a GNSS such as GPS to provide the flight controller with attitude and position information. These measurements are usually sufficient for UAS operating in obstacle-free environments such as at high altitudes. An extended kalman filter (EKF) provides the most common approach to fusing sensor data to estimate the rotorcraft 3D pose. This can be done in one step by using one filter with all the state variables that need to be estimated, as for example in Christophersen et al. (2006) (16-states EKF) and (Bristeau, Dorveaux, Vissiere, & Petit, 2010) (23-states EKF). Another alternative is to use two cascaded EKFs, one for attitude and heading estimation (attitude and heading reference system) using IMU raw data, and the other one for position and velocity estimation using GPS raw measurements and translational accelerations (GPS/INS solution) Kendoul et al. (2010). Height estimation can be enhanced by incorporating altimeter measurements into the second EKF, or estimated separately using another Kalman filter that fuses altimeter data with vertical accelerations. There are also other approaches for sensor data fusion, such as particle filters and complementary

filters (Mahony, Hamel, & Pflimlin, 2008) but they are less popular for UAS state estimation.

5.2.2. Vision-Based State Estimation Systems

GNSS-based navigation depends on the existence of, and access to, signals from satellites. There is a well-recognized need for support and backup systems in case GNSS is not available, in particular to reach the requirements for integrity and accuracy for use in civilian applications. According to Figure 4 a RUAS with non-GNSS state estimation capability has autonomy level 2, which allows it to achieve more complex missions in a wide range of environments. Indeed, many RUAS missions are defined within GNSS-denied environments such as urban and indoor environments. To achieve realistic missions in such environments, a non-GNSS navigation system is necessary. Vision systems are passive and outperform active navigation systems in terms of cost, weight, power consumption, and size, and are therefore an excellent sensing technology for many aerial platforms and various environments. Computer vision can be used as part of the state estimation system for flight control, as a perception system for obstacle and target detection, and as a mission sensor to collect required information. In this section, we are interested in approaches and algorithms that are used for vision-based state estimation. There is no trivial way of classifying these algorithms, but we try here to classify them based on the principles and approach used to estimate the motion from the general perspective of RUAS navigation. We have identified six main classes of vision-based state estimation systems:

- On-ground vision (OGV)
- Visual odometry (VO)
- Target relative navigation (TGRN)
- Terrain/landmark relative navigation (TRN)
- Concurrent estimation of motion and structure (SFM and SLAM)
- Bio-inspired optic flow navigation (BIOFN).

• **On-Ground Vision:** In some research projects, cameras were placed on the ground and used to track the vehicle and to estimate its attitude and/or position in a local frame or image frame. The advantage of this configuration is the absence of constraints related to RUAS limited payload and onboard computational resources, thereby enabling the use of high-quality imaging systems and sophisticated vision algorithms. On the other hand, these setups are not practical for real-world applications and are generally used as experimental test beds to provide a quick navigation solution in order to test and evaluate flight control and guidance algorithms.

VICON systems are commercial digital optical motion capture systems that use several onground cameras to track the 3D position of a specified object. Some research groups have developed indoor flight test beds composed of several quadrotors and a VICON system. For example, the MIT Aerospace Control Lab. (How, Bethke, Frank, Dale, & Vian, 2008) and the GRASP Lab. (Michael, Mellinger, Lindsey, & Kumar, 2010) use this setup in their research on flight control and multivehicle cooperation. Another setup, used in many research works, consists of one or several ground cameras and an aerial platform with colored landmarks that can easily be tracked from images. Altug, Ostrowski, and Taylor (2005) used a pair of pan/tilt ground and onboard cameras and colored blobs attached to the bottom of a quadrotor to estimate the 3D pose of the quadrotor and to stabilize its motion. In Wang, Song, Nonami, Hirata, and Miyazawa (2006), a ground USB-camera was used to track a square marker on the bottom of the Epson 12-g micro flying robot and to provide a hovering controller with attitude and position measurements. This approach has been also used for flight control of the Colibri outdoor helicopter, Figure 8. A trinocular system, composed of three FireWire cameras fixed on the ground, was used to estimate the helicopter position and heading by tracking color landmarks, fixed at the bottom of the helicopter. This visual state estimator was compared against GPS data during manual flights (Martinez, Campoy, Mondragon, & Olivares-Mendez, 2009) and has also been integrated into the flight control system to perform automatic vision-based hovering and landing (Martinez, Mondragon, Olivares-Mendez, & Campoy, 2010).

In some research work, ground cameras were used to directly track a vehicle without using color landmarks on the vehicle. The 3D position of the vehicle was then calculated by triangulation using two or more cameras. At Chiba University, a ground-based Bumblebee stereo vision system was used to estimate the 3D position of a quadrotor

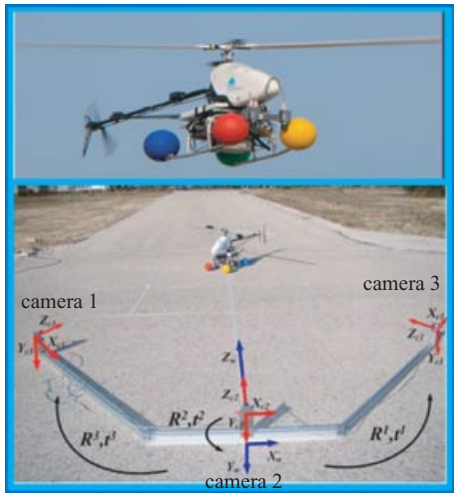


Figure 8. The trinocular ground-based vision system developed by the Universidad Politecnica de Madrid (Martinez, et al., 2010).

and to achieve autonomous hovering and automatic landing outdoors (Pebrianti, Kendoul, Azrad, Wang, & Nonami, 2010). Autonomous aerobatic flights of an instrumented helicopter were achieved at Stanford University (Abbeel et al., 2010) using two (or more) Point Grey Research DragonFly2 cameras with known positions and orientations on the ground. The accuracy of the estimates obtained was about 25 cm at about 40 m distance from the cameras.

• **Visual Odometry:** Traditional odometry techniques use data from the movement of actuators to estimate changes in position and/or orientation over time. VO is an incremental procedure that estimates a vehicle's orientation and/or relative position (traveled distance) by analyzing a sequence of images. In the general case, the RUAS position relative to the initial or a known location is computed by integrating over time the flown distances, obtained from tracking visual features in an unknown environment using an onboard imaging system. A typical VO algorithm includes the following main components: (1) detection of salient features that can be robustly tracked across successive images, (2) feature correspondence or tracking between consecutive images, and (3) motion parameter estimation using feature correspondences.

Most VO algorithms need compute a scale factor to recover the real rotorcraft position and velocity. Another common problem of VO is that position estimates drift over time. IMU/INS data are generally used in the VO algorithm to compensate for rotation effects and to improve the estimation accuracy and robustness. VO approaches reported in the literature vary according to how these challenges are addressed. We distinguish between visual odometers that are based on stereo, monocular, and omnidirectional vision.

Stereo-vision-based odometry: the concept of visual odometry for RUAS was implemented in the CMU autonomous helicopter (Amidi, Kanade, & Fujita, 1999) using template matching on stereo images. The VO locks and tracks arbitrary objects (target template) in consecutive images and uses attitude measurements and range to objects (sensed by triangulation) in order to compute the position of the helicopter relative to its initial location. This visual odometer, aided by a set of inexpensive angular sensors were the only sensors employed to stabilize and maneuver a Yamaha R-50 helicopter (Amidi et al., 1999). Researchers from CSIRO (Buskey et al., 2003; Roberts, Corke, & Buskey, 2003), have also developed a low-cost state estimator for a helicopter using stereo vision and an in-house IMU. The proposed approach relies first on height estimation from stereo disparity, and then fusing optic flow and inertial data using complementary filters to compute the translational velocities. Experimental results for attitude control, velocity, and height tracking were presented using an Xcell 60 helicopter. A stereo visual odometer was also proposed in Kelly, Saripalli, and Sukhatme (2007). The proposed VO is based on feature tracking and helicopter pose estimation using a maximum likelihood estimator. The accuracy of the system was evaluated offline using data collected from manual flights of the AVATAR helicopter. The obtained results demonstrated that VO+IMU can produce a final position estimate within 1% of the measured GPS+IMU position over a flight distance of about 400 m. In Mejias, Campoy, Mondragon, and Doherty (2007), a similar stereo VO system has been implemented and tested on a ground robot and on the COLIBRI helicopter. The implemented stereo VO system tracks features in the ground and computes the helicopter height using the stereo disparity principle. A least-squares algorithm is then used to compute the helicopter rotation and translation between two successive images. The experimental results obtained from a GPS-based flight showed that the mean squared error between GPS and online vision estimates was about 1 m during a 10-m-long flight. Whereas previous stereo VO systems were used for outdoor flights, the stereo VO system presented in Achtelik, Bachrach, Heb, Prentice, and Roy (2009) has been implemented on a small quadrotor and flight-tested in an indoor environment. Like previous systems, it includes feature tracking, 6-DOF motion estimation using the least-squares algorithm, nonlinear motion estimation by bundle adjustment over a window of consecutive frames, and fusion of visual estimates and IMU data using an EKF. A quadrotor UAS, outfitted with a customized stereo camera rig (35-cm baseline) and IMU, was used to compare the position and velocity estimates from the VO and the VICON system (ground truth). The vision algorithm was implemented on a ground control station and the flight control was based on the VO estimates. The obtained results showed that the estimates originating from the VO match the ground-truth values closely in both position and velocity, and the bundle adjustment substantially reduced the total error.

Monocular-vision-based odometry: Monocular vision (single camera) is another alternative for stereo VO. It has been used successfully for VO onboard RUAS, in particular for those of class IV, such as quadrotors, where the payload limitation prevents the use of stereo cameras. The challenge with monocular-based VO is scale factor ambiguity due to the fact that the depth cannot be recovered from images only. In Caballero, Merino, Ferruz, and Ollero (2005) and Caballero, Merino, Ferruz, and Ollero (2009), a homography-based VO technique is proposed. It basically consists of a point-feature tracker that obtains matches between images and a combination of least median of squares and M-estimator for outlier rejection and homography estimation from these matches. Motion parameters (orientation and relative position up to a scale factor) are estimated using the singular value decomposition (SVD) of the homography. This homography-based VO was evaluated offline on monocular image sequences gathered by an unmanned helicopter. Researchers from Draper Laboratory and MIT have developed a vision-aided navigation system for UAS using monocular images (Madison, Andrews,

DeBitettoand, Rasmussen, & Bottkol, 2007). The proposed system architecture allows the estimation of the UAS absolute position and velocity when GPS is lost. This is achieved by initially estimating the 3D positions of landmarks when GPS is available, and then using these 3D points and visual line-of-sight measurements to these landmarks to continue the estimation of the vehicle pose long after losing both GPS and the features that were localized using GPS. This strategy was validated in simulations as well as in flight tests using the MIT quadrotors and VICON system to provide pseudo-GPS updates. This VO system was also used in Ahrens, Levine, Andrews, and How (2009) for vision-based flight control and obstacle avoidance in GPS-denied environments. When equipped with the developed VO system, the MIT quadrotor UAS achieved drift-free hover for 96 s and a collision-free forward flight of 5 m. A different approach for estimating the range and recovering the 3D position of a RUAS from monocular images in a visual odometer framework was proposed in Kendoul et al. (2009). The authors proposed a real-time adaptive VO-inertial system that recovers the 3D position of the rotorcraft using an onboard camera, IMU, and pressure sensor. The vision algorithm computes the optic flow, tracks visual features, and integrates the image displacement to compute the traveled flight distance in pixels. These visual measurements are then fused with IMU data to compensate for undesired rotation effects. This vision algorithm has been augmented by an adaptive observer that identifies the scale factor (range) from optic flow and INS measurements and recovers the 3D position and velocity of the vehicle. For more accuracy and robustness, a Kalman filter is used to fuse these estimates with inertial data, enhanced by pressure sensor measurements to overcome scale factor identification errors during hovering and slow flights. The performance of the adaptive VO was demonstrated in outdoor and indoor flight tests using a quadrotor MAV and real-time architecture. The VO estimates were used in a closed loop to control the MAV motion for automatic takeoff and landing, hovering, trajectory tracking, and tracking of a moving ground-based target.

A descent image motion estimation system (DIMES) was developed by NASA JPL and used for horizontal velocity estimation during the Mars exploration rover landings (Johnson, Willson, et al., 2007). Although this VO has been developed for a spacecraft, it has been flight-tested using a helicopter, which makes it suitable for this survey. The DIMES algorithm combined measurements from a descent camera, a radar altimeter, and an IMU. The horizontal velocity is estimated by tracking features in three images after rectifying them to counter for changes in scale and orientation. The system was evaluated on Monte Carlo simulations and flight-tested using a full-scale helicopter. Finally, the DIMES system was successfully used to compute and reduce the velocity during the Opportunity and Spirit rover landings on Mars. Two templates are tracked between three images taken at roughly 2000, 1700, and 1400 m above the surface.

Omnidirectional-vision-based odometry: There are also some examples of using omnidirectional vision for helicopter attitude estimation in unstructured environments. In Mondragon, Campoy, Martinez, and Olivares (2010), catadioptric images are used to estimate pitch and roll from skyline detection. A visual compass on appearance images is also proposed to estimate relative yaw by unwrapping the spherical image and finding the best match based on the images column shift using the Euclidean distance. Different tests have been performed on a helicopter in different seasons and conditions. When compared to IMU measurements, the VO estimates have an RMSE of (2.86°, 4.3°, 10.7°) for (roll, pitch, yaw) in the worst case and (0.18°, 1.9°, 1, 2°) in the best case.

• **Target Relative Navigation:** One of the important applications of vision for RUAS is target detection and tracking. The objective is generally to detect a specific target and to estimate its position relative to the rotorcraft. This relative position may be expressed in an inertial frame in meters or in the image frame in pixels. Potential applications of visual TGRN are precise landing on a specified target and tracking of moving air/ground targets such as cars, persons, or other UAS. In contrast to VO approaches, there is the notion of "target" in TGRN methods which must be kept in the camera field of view. This target may be static or moving, a priori known or selected during flight by the operator or an automatic perception system. TGRN systems, developed for RUAS, can be further divided into four categories: (1) vision-based landing on a known target, (2) vision-based landing on an unknown target, (3) vision-based static target tracking, and (4) vision-based mobile target tracking.

Observation 5. In this paper, we make a distinction between target-based visual state estimation and target-based perception (described in the next section). In the first case, vision-based estimates are directly used as measurements for the controller, replacing GPS data. In the second case, target-based visual perception provides inputs to the guidance system, which generates reference position/velocity commands to the controller, but the control feedback is based on GPS data.

Vision-based landing on a known target: TGRN approaches have been actively researched for automatic landing on some artificial landmarks with known attributes such as size, color, and pattern. The landing helipad or target is generally designed to simplify image processing and to allow accurate motion estimation from different viewpoints. Generally, the pattern and size of the target are known, which allows both the orientation and the translation of the helicopter relative to the target to be recovered without scale factor ambiguity. These vision-based landing systems

are generally practical when the landing point is fixed or known, such as the base station.

In our literature review, we found that research on vision-based landing of unmanned rotorcraft goes back to the nineties, but only results from simulation or offline processing of real images were presented (Bosse, Karl, Castanon, & Debitetto, 1997). There are also some theoretical works such as (Marconi, Isidori, & Serrani, 2002), where the problem of automatic landing of RUAS on an oscillating platform has been approached as a robust nonlinear regulator problem. In Shakernia, et al. (2002), a vision algorithm that detects a known target (white squares) and computes the helicopter's relative position and orientation is presented. The algorithm is based on multiple view geometry, which exploits the rank deficiency condition of the multiple view matrix. The proposed system has been implemented on the Yamaha R-50 helicopter and vision-based hovering (18 s hovering) above the target was achieved, but no automatic landing was attempted. The work presented in Saripalli, Montgomery, and Sukhatme (2003) combines vision and GPS for automatic landing on a known target. Therefore, it is not considered a pure vision-based landing system but rather a vision-based perception system, which will be described in the next section. The paper by Saripalli and Sukhatme (2003) about vision-based landing on a moving target also uses vision and GPS and only offline processing results were presented (no automatic landing).

The first experimental results on vision-based automatic landing (without GPS aid) of an unmanned helicopter were published in 2004 by Merz, Duranti, and Conte (2004). Their vision-based landing system is based on a single pan/tilt camera and a specially designed landing pad with black circles on a white board. The image-processing algorithm computes the helicopter's relative position and attitude by minimizing the reprojection error of the extracted center points and semiaxes of the three ellipses. Using a Kalman filter, these estimates are then fused with the IMU data to improve the system robustness, especially when the vision system is "blind" for short periods. Several autonomous landings of the Yamaha RMAX helicopter were conducted from different relative positions, on grass and snow fields, under different wind and illumination conditions. Successful accurate landings were achieved with an average touchdown precision of about 42 cm, a vertical velocity at touchdown that ranged between 18 and 35 cm/s, and a horizontal velocity of about 15 cm/s.

Vision-based landing on an unknown target: Landing on unknown and unprepared sites has also been an active research topic because of its importance for real applications. The first step in vision-based landing in unknown areas is safe landing area detection (SLAD); this will be described in the next section. In this section, we are interested in vision-based landing (without GPS) on the selected target once it is detected by the SLAD system or selected by the operator during flight. In contrast to previous approaches with known targets, there is no information about the size, color, or shape of the landing area, which makes position estimation challenging because of the scale factor ambiguity unless another sensor is used. The landing area is generally flat and homogeneous, with poor texture, which makes feature tracking difficult, too. The motion estimation problem here is very similar to the visual odometry problem.

One of the advanced works in this area was carried out in the framework of the precision autonomous landing adaptive control experiment (PALACE) project by the U.S. army/NASA Autonomous Rotorcraft Program (ARP), Brigham Young University, and the Mobility and Robotics Group of JPL. In Theodore, et al. (2006), a vision-based landing system at noncooperative sites without the aid of GPS is presented. The proposed approach includes three main parts: (1) stereo-based mapping for safe landing area detection, (2) monocular-vision-based position estimation, and (3) flight controller. The position estimation algorithm takes as inputs the feature-tracking data, the LIDAR range to the image center and the IMU attitude measurements, and outputs the 3D position of the helicopter relative to a fixed point in the selected landing area. Closed-loop vision-based control has been evaluated in flight for a number of different conditions and heights (up to 30 m AGL) using the Yamaha RMAX helicopter. The tracking drift observed in each case was between 15 and 20 cm/min. The complete SLAD and vision-based automatic landing system has also been demonstrated in flight, with a total of 17 successful landings on various surfaces and obstacle fields, with a landing accuracy of better than 1 m. In Yu, Nonami, Shin, and Celestino (2007), a vision-based landing system for unknown flat areas is developed for a robotic helicopter. Stereo vision is used to build a depth image of the area below the helicopter. Assuming that the ground is flat, a least-mean-square (LMS) technique is used for plane fitting and height estimation. The system has been implemented on the Hirobo SR-40 helicopter and demonstrated in vision-based closed-loop flights to perform automatic landings from a height of 5.5 m. The visual odometer, described previously in Kendoul et al. (2009) has also been successfully used for tracking and landing on unknown static and moving ground targets. The VO system was extended to include target tracking by just selecting in the image the desired target as the area where visual features will be tracked. The outputs of the vision system are then the position and velocity of the quadrotor relative to the target. The flight controller regulates these visual estimates to zero, resulting in vision-based target tracking and automatic landing on unknown targets without the aid of GPS.

Vision-based static target tracking: These systems share many characteristics with vision-based landing approaches. However, the objective of the approaches described here is not to land on the target but to fly toward that or hover near it. The target is not necessarily on the ground, because it can be on a vertical wall, for example.

An example of a such system is described in Hrabar and Sukhatme (2003), where the objective was to develop a sideways-looking system that will allow a helicopter to locate and move toward the centroid of a number of known targets, using omnidirectional vision. A vision-aided inertial navigation system is presented in Proctor and Johnson (2005), where the helicopter's position and attitude relative to a known target (dark square with known position, orientation, and size) are computed from images and inertial data using an extended Kalman filter. Experimental results from a flight test where the vision-based EKF estimates are used in the closed-loop control of an RMAX helicopter are presented. The helicopter started at a location 36 m from the target and was given a step input of 2 m up in altitude, a 3 m step parallel to the plane of the window, and finally a 3 m step toward the window, which is a mock square window with an area of 4 m^2. In Guenard et al. (2008), image-based visual servoing of a quadrotor UAS is described. Visual features are mapped into control inputs via the inverse of an image Jacobian matrix without estimation of the rotorcraft position. The target consists of four black marks on the vertices of a stationary planar square. An image-based error signal is regulated to zero using a nonlinear backstepping controller, thereby resulting in position control using vision and angular rates. This visual servoing algorithm was evaluated in real-time experiments using a small quadrotor with downward-looking camera. The vehicle was tasked to hover 1.4 m above the target (the yaw was controlled manually). The rotorcraft performed a 50-s–long hover flight with a maximum regulation error of 10 cm.

Vision-based mobile target tracking: Vision-based target relative navigation includes also moving ground or air target tracking. The challenge in visually tracking a moving target is that it is difficult to estimate both target and rotorcraft velocities, because only relative velocity can be estimated. The image processing algorithm also needs to be robust to track a dynamic target in image sequences. Several vision-based navigation systems have been developed at Georgia Tech. and implemented on RUAS. In Ludington, Johnson, and Vachtsevanos (2006), three vision systems that were developed and tested on the GTMax unmanned helicopter are presented: (1) vision-based state estimation, (2) automated search of stationary ground targets, and (3) mobile ground target tracking. The first system fuses inertial data and visual measurements from the detection of a known dark square on the ground to estimate the helicopter state using an EKF. This vision-aided inertial system has been used in the closed-loop control of the RMAX helicopter and has demonstrated autonomous vision-based hovering and forward flight with good accuracy when compared with DGPS measurements. The second vision-based system is dedicated to automatic identification of buildings of interest based on some a priori information. When the RMAX helicopter was equipped with that system, it was able to autonomously search an urban region of 15 buildings for a particular building and then identify the openings to the building. The third system was developed for mobile ground target tracking using a particle filter algorithm that tracks the target in the incoming video frames and estimates its relative position. A closed-loop tracking system was tested in simulation and onboard the GTMax helicopter to track a white van from altitude 30 m. However, the flight controller relied on GPS data for vehicle stabilization and trajectory tracking. Another vision system has been developed at Georgia Tech. for UAS navigation relative to an airborne target using only visual information from a single onboard camera (Johnson, Calise, Watanabe, Ha, & Neidhoefer, 2007). The image processing uses an active contour to calculate the location of the target. Two distinct vision strategies are proposed for range and hence position estimation from monocular images. The first method, called center only relative state estimation (CORSE), tracks only the target center position and uses an EKF to compute the relative position and velocity. To maximize the accuracy of range estimation, optimal control is used to generate a sinusoidal trajectory, making range estimation possible from visual measurements and rotorcraft accelerations or control inputs. The second approach, called subtended angle relative state estimation (SARSE), uses a more computational image processing algorithm to compute the target center as well as the locations of the wing tips in the image. The range to the aircraft target and wingspan are estimated from an EKF based on visual measurements and helicopter accelerations (the target acceleration is assumed to be zero). Both techniques have been tested and evaluated in simulations as well as in closed-loop flights. The SARSE algorithm was used to perform relative navigation and formation flying between the GTmax helicopter and the GTEdge fixed-wing UAS. Although successful autonomous flights were achieved, inaccuracies in depth estimation were observed. Later, in Ha, Johnson, and Tannenbaum (2008), the vision-based tracking system was improved by using fast geometric active contours and particle filters for multiple target tracking. However, the shape and size of targets were assumed to be known in order to simplify range and position estimation from monocular images. A leader–follower formation flight was demonstrated using vision.

In the framework of the French project ACTION (cooperation of heterogeneous autonomous vehicles), a visual air-to-ground target tracking system was developed by ONERA (Watanabe, Lesire, Piquereau, & Fabiani, 2010). The system includes three main components, which are target tracking and relative motion estimation, optic flow computation for rotorcraft self-motion estimation, and a guidance law for vision-based tracking. The target detection and tracking problem is made simpler by assuming a priori knowledge of the target color and size. Sparse optic flow is also computed around the target by assuming the surface is flat. By fusing target visual measurements,

optic flow, and inertial data, the position and velocity of both the helicopter and of the target are estimated using an EKF. This vision-based tracking system has been implemented and tested on the ONERA ReSSAC unmanned helicopter (Yamaha RMAX). Successful target (car) tracking was achieved from an altitude of 40 m and overflight course of about 50–100 m using vision for target tracking and a guidance law and GPS for flight control and trajectory tracking. Optic-flow-based self-motion estimation is implemented but not yet tested in flight (it is not yet pure vision-based target tracking).

• **Terrain/Landmark Relative Navigation:** TRN refers to a vehicle's position and/or velocity estimation by comparing terrain measurements from onboard sensors with an a priori terrain map. TRN can be achieved using passive imaging or active range sensing. The reference map can be a digital elevation map, satellite images, topographical map with landmarks, etc. The main purpose of TRN is to provide a drift-free navigation tool by estimating position (global or local) or bearing relative to known surface landmarks, yielding a powerful alternative to GNSS navigation. It is important to note that some animals, such as birds and bees, seem to use landmark-based cues as well as visual odometry to navigate to a goal location. Before describing vision-based TRN systems developed for RUAS, let us review some TRN algorithms (visual and nonvisual) that have been implemented successfully on other aerial vehicles such as military aircraft, cruise missiles, and spacecraft. This can help to gain more insights in TRN approaches and to have ideas for adapting existing TRN systems to RUAS navigation.

Non-RUAS terrain-relative navigation systems: TRN systems were first developed for cruise missile guidance before the advent of GNSS, and they are still used independently or in conjunction with GPS navigation. One of the earliest TRN systems was developed for cruise missiles in the late 1970's (Golden, 1980). The algorithm, called TERCOM (terrain contour matching), uses a radar altimeter to continuously sense the contour of the land directly beneath the missile, and compares the sensed elevation with an a priori contour map containing the planned route. Another similar system developed by Atlantic Inertial Systems (UK) is called TERPROM (terrain profile matching) and is used on a number of jet fighters and cruise missiles. It also combines a radar altimeter and navigation data with a digital terrain map to provide a ground proximity warning system (GPWS) and advanced terrain avoidance cues (ATAC). DSMAC (digital scene mapping and area correlation) (Carr & Sobek, 1980) is a passive TRN system that uses a single digital camera and a set of preregistered images. An image correlation technique is used to determine the optimal location that would correct the sensed image so that it would best match the planned image. These visual estimates occur periodically throughout the flight, and are used to correct the drift of the inertial navigation system.

TRN systems are also of great interest for pinpoint or accurate landing of spacecraft on Mars, the Moon, and other celestial objects, where GNSS systems are not available. In Adams, Criss, and Shankar (2008), a passive optical TRN system for spacecraft landing on the Moon is derived from DSMAC. The proposed system uses two orthogonal cameras and an images-to-map correlation method. The NASA JPL computer vision group developed different visual TRN systems for Mars landers. In Trawny, Mourikis, Roumeliotis, Johnson, and Montgomery (2007) described an EKF-based TRN system that fuses IMU data with visual observations of a priori mapped landmarks such as craters to estimate the spacecraft attitude, position, and velocity during descent and landing on a planet. This system, called a descent image motion estimation system (DIMES), was evaluated using data sets from NASA subsonic parachute drops (8,000 m AGL) and field testing on a helicopter (1,000 to 2,000 m). In all these tests, the final error bounds obtained (3σ) were on the order of ± 3 m for position and $\pm 0.5^\circ$ for orientation. Another vision-aided inertial TRN system for pinpoint landing on Mars is presented in Mourikis et al. (2009). The algorithm extracts 2D-to-3D correspondences between descent images and a surface map (mapped landmarks with known global coordinates) and 2D-to-2D feature tracks through a sequence of descent images. These visual estimates are then fused with inertial data using an EKF in order to estimate in real time the lander's terrain-relative position, attitude, and velocity. This system was experimentally validated on a sounding-rocket test flight with estimation errors of 0.16 m/s for velocity and 6.4 m for position at touchdown. For more work and comparison of different TRN systems for planetary landing, the reader can refer to the review paper by Johnson and Montgomery (2008) and the references therein.

Visual terrain-relative navigation systems for RUAS: Although TRN is crucial for UAS as a backup system for GPS navigation or as a main navigation system for GPS-denied environments, very few TRN systems have been developed and implemented on UAS. Visual TRN seems more appropriate than active TRN for RUAS flying at low speeds and low altitudes. This is because the altitude variation is generally quite poor in terms of allowing ground profile matching using active range sensing. The most advanced and tested visual TRN system for RUAS is probably the one developed by the UASTech Lab. at Linköping University (Conte & Doherty, 2009). The proposed navigation architecture combines inertial sensors, visual odometry, barometric altimeter, and image to georeferenced map correlation data to estimate the rotorcraft absolute position and attitude using a 12-state Kalman filter, Figure 9. First, the horizontal absolute position is estimated by fusing the VO estimates (homography-based VO) with the image correlation information using a Bayesian point-mass filter. These position estimates, together with height measurements, are then used as position measurements to update an error

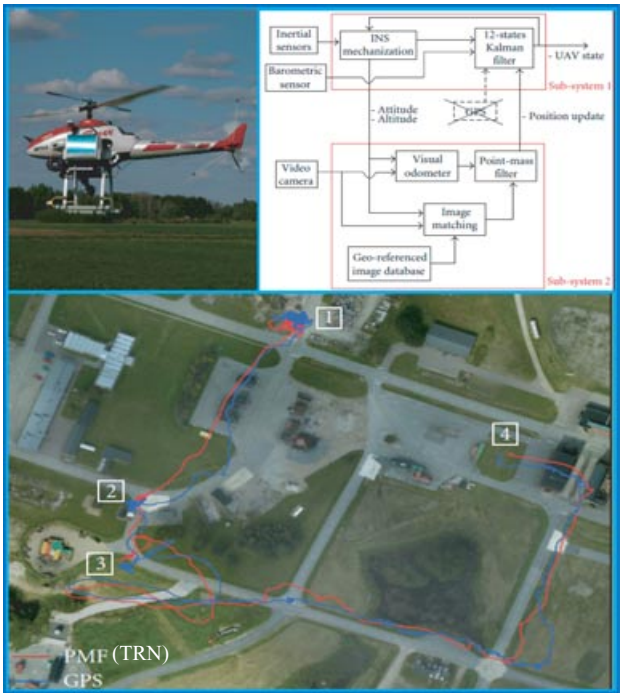


Figure 9. Architecture and real-time results of the visual TRN system developed at Linköping University (Conte & Doherty, 2009).

dynamics Kalman filter that estimates the INS errors and to correct its navigation solution. The developed TRN system has been tested offline on real flight data as well as online during vision-based flights of the Yamaha RMAX helicopter. Offline experimental results from a 1-km flight at 60-m altitude showed a maximum position error of 5 m and a maximum velocity error of 1.5 m/s when comparing visual TRN estimates with GPS/INS measurements. The vision-based TRN state estimator was implemented onboard the helicopter PC104 PentiumIII-based computer and used for vision-based waypoint navigation at an altitude of 55 m, Figure 9. The maximum error of position estimates obtained was less than 30 m. In Lindsten, et al. (2010), probabilistic template matching is used to compare flight images with a georeferenced map that is environmentally segmented and classified. The system has been validated offline using data collected during a 400-m test flight of the Yamaha RMAX helicopter. Error in horizontal position estimates was about 15 m in the direction parallel to the flight path and 5 m in the orthogonal direction. A different TRN approach was proposed in Courbon, Mezouar, Guenard, and Martinet (2010) and implemented on a small quadrotor UAS. First, a visual memory and route are built by sampling the environmental images and organizing them as a set of ordered target images (offline processing step). During flight, the captured images from an onboard camera are compared with the stored visual memory images to localize the vehicle and to regulate its relative position. This system was evaluated indoors using a small instrumented quadrotor that performed a vision-based flight of a few meters distance.

• **Concurrent Estimation of Motion and Structure:** Other emerging frameworks for rotorcraft localization and scene mapping are the SFM and SLAM methods. There are strong similarities between SFM and visual SLAM algorithms, because both estimate the vehicle's 3D pose, as well as the structure or map of the perceived environment. These approaches play roles of both state estimation and perception, but in some cases, they are used as pure VO systems (no mapping), where only the vehicle's motion is recovered, by assuming that feature depths are known or measured by another sensor. Issues common to SFM and visual SLAM include robust feature tracking in real time, translation–rotation ambiguities, unknown range (scale factor), estimate drift over time, and high computation requirements, especially when extended to 3D navigation.

Structure from motion (SFM): The problem of SFM concerns the estimation of the camera ego-motion and the reconstruction of the 3D structure of a scene from its projection onto a moving two-dimensional surface (image sequences). SFM has been a central problem in computer vision for many decades, and the literature comprises a variety of schemes. Although SFM approaches have been widely used in computer vision, there are very few works about their real-time implementation on UAS. This is because of their computational requirements and lack of robustness in natural environments. A vision-based state estimator for UAS using the SFM approach is presented in Kanade, Amidi, and Ke (2004). It is based on feature tracking and uses the well-known linear Eight Points algorithm to compute the essential matrix parameters and to recover motion parameters up to some scale factor. However, the system has not been implemented on a UAS and no experimental results have been presented. In Kendoul, Fantoni, and Nonami (2009), the SFM method has been applied successfully for state estimation and flight control of a small quadrotor UAS. The proposed implementation is based on three nested Kalman filters (3NKF) and involves real-time computation of optic flow, fusion of visual and inertial data, and recovery of translational motion parameters. The scale factor ambiguity is solved by using height measurements from a pressure sensor and assuming flat ground. The system has also been augmented by a visual odometer that estimates the horizontal position by integrating image displacements. The developed system was first validated on a ground vehicle and then on a quadrotor UAS through vision-based autonomous flights including automatic takeoff and landing, outdoor and indoor hovering, and velocity command tracking.

Simultaneous localization and mapping: SLAM techniques are closely related to SFM approaches, and aim to

build up a map within an unknown environment or to update a map within a partially known environment, while at the same time localizing the vehicle on the map. SLAM algorithms have advantages over SFM techniques when the vehicle revisits the same places (loop closure). Visual SLAM (vSLAM) implementations mainly rely on feature (or landmark) correspondences and use extended Kalman filters (EKF) or Rao–Backwellized particle filters (RBPF). Although vSLAM has been widely used on ground robots, only a few applications have been implemented on UAS. vSLAM algorithms developed for RUAS can be further classified into stereo vision SLAM (svSLAM) and bearing-only SLAM (boSLAM), which is based on monocular images. boSLAM is a partially observable problem where the vehicle's motion and feature positions (map) can be recovered up to some unknown scale factor, unless a priori information or other ranging sensors are used. Landmark initialization for boSLAM has also proven to be a difficult issue to resolve in a consistent manner. Lemaire, Berger, Jung, and Lacroix (2007) describe svSLAM and boSLAM algorithms that both rely on a robust interest-point matching algorithm and the EKF framework. The performance of the two approaches was compared offline using data acquired with a ground rover and an aerial blimp. Other svSLAM (Nemra & Aouf, 2009) and boSLAM (Kima & Sukkarieh, 2007) algorithms have been proposed and evaluated in simulations or using fixed-wing UAS.

There are also some works on applying SLAM to unmanned rotorcraft, but they are still premature, because only offline results have been reported. In Artieda, et al. (2009), the implementation of a vSLAM technique on images taken from a RUAS is presented. An EKF is used to reconstruct the UAS motion as well as feature positions (map) up to some unknown scale factor. This monocular vSLAM algorithm has been tested offline using images collected from unmanned and manned helicopters. The obtained MSE of the differences between the vSLAM and the GPS horizontal coordinates for a 35-s sequence was about 2 m. The VO system, previously described (Caballero et al., 2009), has been augmented by a vSLAM algorithm to reduce the drift in position estimates. The proposed technique has also been evaluated offline on images gathered by an unmanned helicopter during a 90-m flight at 15 m altitude. The obtained results are promising, with a maximum error of (7, 9, 5 m) for (x, y, z) without IMU data in the SLAM algorithm and about (6, 4, 1.5 m) when are integrated the IMU data in the EKF. Another vSLAM technique for RUAS navigation is presented in Tornqvist, Schon, Karlsson, and Gustafsson (2009). The proposed algorithm uses the Rao–Blackwellized particle filter to fuse data from inertial sensors, barometer, and vision. The FastSLAM algorithm has been adapted to handle high-dimensional state vectors efficiently. The derived algorithm is applied offline to experimental data from an unmanned RMAX helicopter flying at an altitude of 60 m and a forward speed of 3 m/s. The vSLAM horizontal position error was about 12 m after 280 m flight and the height estimate was approximately 2.5 m error. An interesting work about vSLAM for full-scale helicopters is presented in Kim, Lyou, and Kwak (2010). The conventional GPS/INS navigation system has been augmented by a vSLAM algorithm to compensate for position drift during GPS loss and blockage. The imaging system uses two cameras that are perpendicular to each other and observe the forward and downward feature points, respectively. The vehicle's position, velocity, and attitude are recovered by a KF-SLAM algorithm based on the fact that the 3D position of initial features is known.[18] The scale factor ambiguity is solved by assuming that the precise altitude of the rotorcraft relative to the ground surface is known from a radio altimeter, a pressure sensor, or GPS. The GPS–INS–vision algorithm has been demonstrated in three different scenarios (forward-looking only, downward-looking only, forward–downward-looking) using a manned helicopter flying at 110 knots. When the GPS signal was lost for 60 s, the GPS/INS solution error was about 320 m, where as this error was reduced to 20 m when the vSLAM algorithm was used with downward and forward-looking cameras.

Observation 6. Despite the popularity of SLAM techniques, we have not found a single paper describing autonomous flight of a RUAS using visual SLAM estimates. All reported results have been obtained from offline processing or simulations.

• **Bio-inspired Optic Flow Navigation:** In most vision approaches previously described, the rotorcraft position and/or velocity are explicitly recovered and used as measurements in the flight controller. Recent experimental research in biology has discovered a number of different ways in which flying insects use cues derived from optical flow for navigational purposes without explicit estimation of motion and environment structure. Indeed, flying insects such as bees and flies have developed alternative, simple, and ingenious stratagems for dealing with the 3D vision problem to perform navigational tasks such as attitude and course stabilization (Hengstenberg, 1993), centering response, visual odometry (Srinivasan, Zhang, & Bidwell, 1997), flight speed regulation (Barron & Srinivasan, 2006), landing (Wagner, 1982), altitude control and terrain following (Ruffier & Franceschini, 2005), and obstacle avoidance (Tammero & Dickinson, 2002). It is clear that systems using these bio-inspired strategies play at the same time the role of state estimation systems, perception systems for obstacle detection, and guidance systems for reactive obstacle avoidance. Bio-inspired systems thus have

[18]In the first acquired image, a georeferenced map is created (features positions) based on the vehicle's position and attitude information from GPS/INS (before GPS loss).

the potential to provide the RUAS with higher levels of autonomy such as level 2 and level 4 of the ALFURS chart in Figure 4. Recently, there has been an increasing interest in developing bio-inspired GNC systems for mini flying machines, especially for terrain following, landing, and obstacle avoidance.

Terrain following: Altitude control and terrain following can be achieved by regulating the optic flow obtained from a downward-looking camera to a constant value. Based on this insect-inspired behavior, Franceschini's team (Ruffier & Franceschini, 2005) has developed an autopilot, called OCTAVE (optical altitude control system for autonomous vehicles), for navigation and flight control of small rotorcraft. The optic flow is computed by a small fly-inspired sensor developed by the authors, and is used by PID control loops to achieve the desired maneuvers. The effectiveness of this strategy has been demonstrated in a number of scenarios using a small tethered rotorcraft. In Garratt and Chahl (2008), an optic-flow-based terrain-following system has been developed and demonstrated in real flights using a robotic helicopter. Optic flow was computed in real time using the image interpolation algorithm (I2A) and then fused with IMU data to compensate its rotational component. The height above the ground is explicitly estimated using optic flow and GPS velocities. The terrain-following system has been implemented onboard the RMAX helicopter and demonstrated in closed-loop flights. During 100 s of closed-loop flight at a speed of 5 m/s, the helicopter maintained 1.27 m clearance from the ground using the estimated height from optic flow. An optic-flow-based terrain-following algorithm for mini rotorcraft is proposed in Herisse et al. (2010). The developed system computes optic flows at multiple observation points, obtained from two onboard cameras, using the LK algorithm and combines this information with forward speed measurements to estimate the height above the ground. A backstepping controller is used to regulate the height to some desired value. Indoor closed-loop flights have been performed over a textured terrain using the CEA quadrotor vehicle flying at a forward speed of 0.3–0.4 m/s. The system was able to maintain a desired height of 1.5 m above a ramp and 2D corner textured terrain of about 4 m length.

Automatic Landing: One important consequence of holding the optic flow beneath the vehicle constant is that the horizontal speed is automatically reduced as the height decreases. Therefore, height, and forward and descent velocities will decrease exponentially with time and become zero at touchdown, by just regulating the optic flow to some constant value. Bees use this simple strategy (two rules) for smooth landing without explicit measurement or knowledge of the flight speed and height above the ground. This insect-inspired landing strategy was used in Chahl, Srinivasan, and Zhang (2004) to control the altitude of a fixed-wing UAS during the approach phase. The same landing strategy was also demonstrated on a small fixed-wing MAV equipped with a 1D optic flow microsensor (Oh, 2004). In Beyeler, Zufferey, and Floreano (2009a), an optiPilot for automatic takeoff and landing of a fixed-wing MAV using optic flow is described. Sparse optic flow measurements are obtained from optic mouse sensors pointed at 45° with respect to the longitudinal axis. Angular rates from gyros are used to cancel the rotational optic flow, and the remaining translational flow is directly mapped into pitch and roll commands. Automatic takeoff and landing from 10 m height were successfully achieved using a fixed-wing MAV equipped with the optiPilot. Automatic landings of a 1.5-m-wingspan MAV were also achieved using a barometric altimeter and optic flow sensors (Agilent ADNS-2610) (Barber, Griffiths, McLain, & Beard, 2007). Optic flow has also been used to land mini RUAS, as described in Herisse, Russotto, Hamel, and Mahony (2008), where divergent optic flow has been combined with inertial data and used to control the vertical motion of a quadrotor. The approach has been validated through autonomous hovering and landing from 1 m height.

Obstacle avoidance: The main strength of optic-flow-based strategies is their ability to avoid obstacles without mapping the environment or explicitly detecting obstacles. The idea is to interpret regions with high optic flow as imminent obstacles. Indeed, translational optic flow is proportional to the magnitude of the vehicle velocity and inversely proportional to the distance to obstacles in the environment. The most investigated strategy is the use of divergent optic flow for frontal obstacle avoidance and right and left optic flows for centering response and lateral obstacle avoidance. This bio-inspired behavior has been successfully implemented on some small fixed-wing UAS. In Griffiths et al. (2006), three Agilent ADNS-2610 optical mouse sensors were mounted on a small fixed-wing UAS (right-forward, left-forward, and down) and used to sense the distance to obstacles by combining GPS velocities and optic flow signals. The avoidance controller generates a path offset based on the difference between the computed left and right distances. This approach has been demonstrated in real flights while the UAS was flying a GPS-based path through the Goshen Canyon. Successful lateral obstacle avoidance using optic flow sensors is demonstrated on a fixed-wing MAV and described in William, Green, and Oh (2008). Two 1D optic flow sensors from Centeye were mounted on the sides of the vehicle. By assuming that the forward velocity is constant, a full deflection of the rudder is performed when the computed optic flow from one sensor is higher than some prefixed threshold. Researchers from EPFL (Switzerland) have also developed and demonstrated an embedded OF-based system allowing an ultralight aircraft (30-g prototype) to detect and avoid frontal obstacles (Zufferey & Floreano, 2006), using the divergent OF signals obtained from the left and right linear cameras. The obstacle avoidance strategy aims at triggering a saccade maneuver in the direction of low optic flow

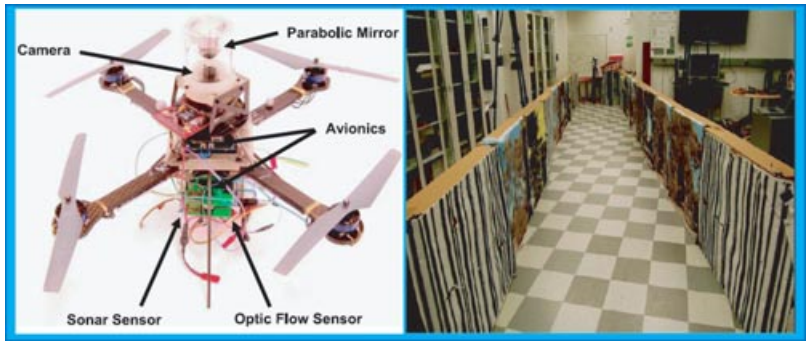


Figure 10. The Maryland University quadrotor during corridor navigation using optic flow obtained from omnidirectional vision (Conroy, Gremillion, Ranganathan, & Humbert, 2009).

when the divergent optic flow exceeds some threshold. In a recent paper (Beyeler, Zufferey, & Floreano, 2009b), the authors described another optic-flow-based autopilot (optiPilot) for an outdoor fixed-wing MAV . The optiPilot is based on seven optic mouse sensors (Avago ADNS5050), MEMS rate gyroscopes, and a pressure-based airspeed sensor. The estimated translational optic flow is directly mapped into control signals to steer the pitch and roll angles, whereas the airspeed measurements are used to control the forward speed. The optiPilot has been validated in simulations and demonstrated in real flights to avoid a group of tall trees (lateral avoidance) and small trees (flyover avoidance). Although optic-flow-based obstacle avoidance has been successfully demonstrated on fixed-wing UAS, a very limited number of papers have been published on the application of optic flow for obstacle avoidance onboard RUAS. In Hrabar and Sukhatme (2004), the use of optic flow for an unmanned helicopter centering in an urban canyon is investigated, but only offline results were presented. In Conroy, Gremillion, Ranganathan, and Humbert (2009), a bio-inspired optic flow navigation system has been implemented on a quadrotor UAS and demonstrated in an indoor textured corridor. The inner-loop pitch and roll stabilization is accomplished using rate gyros and accelerometers. Altitude control is achieved via fusion of measurements from sonar and an accelerometer. A ventrally located optic flow sensor from Centeye is utilized to augment longitudinal and lateral damping, thus improving vehicle stability. An omnidirectional visual sensing and processing based on a Surveyor camera board and a parabolic mirror is used for optic flow estimation and outer-loop control. The navigation strategy consists of decomposing patterns of translational optic flow (magnitude, phase, and asymmetry) with weighting functions in order to extract signals that encode relative proximity and speed with respect to obstacles in the environment, which can be used directly for outer-loop navigation feedback. The flight tests were performed in a textured corridor about 1.5 m wide and 9 m long, Figure 10. The quadrotor successfully avoided corridor walls and finished its course without collisions.

Observation 7. In all reviewed works, bio-inspired optic-flow-based navigation was demonstrated on fixed-wing platforms or on small quadrotors performing very short flights in structured indoor environments. It is thus clear that applying optic flow (without GPS aid) for outdoor navigation of RUAS is still an open research problem because no experimental results have been published yet.

5.2.3. State Estimation Using Ranging Sensors

In addition to conventional and vision-based state estimation systems, ranging sensor technologies have been also researched, with the objective of using them onboard RUAS for state estimation. Although active range sensors such as LIDARs, radars, and ultrasonic and IR sensors are generally used for perception, there are some works where these sensors have been used mainly for state estimation and indoor flight control.

• **State Estimation Using Ultrasonic and Infrared Sensors:** Ultrasonic sensors have been used for the stabilization of a quadrotor vehicle relative to office walls (Kendoul et al., 2007). Three perpendicular ultrasonic range finders have been mounted on a quadrotor to provide 3D position measurements in order to test and evaluate the performance of a nonlinear controller. Autonomous hovering and automatic takeoff and landing have been achieved. A similar system is described in Bouabdallah and Siegwart (2007), where ultrasonic and IR sensors are also used for altitude control and obstacle avoidance. In Shin et al. (2011), an ultrasonic positioning system (UPS) has been developed and used for indoor hovering control of a small ducted-fan UAS. The UPS consists of four transmitter nodes onboard the vehicle, eight receiver nodes on the ground, and a server node. The obtained precisions were 2 cm RMS (root mean square) for position estimation using UPS and about

10 cm RMS for hover control using PD for attitude control and LQI for position control. Recently, IR sensors have been used on a quadrotor UAS as the main position sensors for stabilization and precise landing on small rectangular or square objects such as tables or car roofs (Nonami et al., 2010, Chapter 13). The height is estimated by an ultrasonic range finder, whereas the horizontal 2D position is estimated from four actuated IR sensors that detect the edges of the landing target. This system has been demonstrated in real-time for autonomous hovering and automatic precise landing on a 52 × 52 cm table with an accuracy of 20 cm during hovering and 10 cm during landing. A combination of low-cost sonar range and infrared sensors was also used to perform mapping of indoor environments and localization of small RUAS (Sobers, Chowdhary, & Johnson, 2009). A SLAM-like algorithm was developed by alternating between mapping and localization. First, a map of an indoor environment is constructed and stored. The range sensors are then used to provide vehicle position relative to the stored map of the room being explored. A small coaxial rotorcraft equipped with this system was used to complete mission requirements for the International Aerial Robotics Competition. During the competition, the vehicle was successfully able to enter the arena, negotiate obstacles, and fly down a hallway for approximately 12 m via wall following.

• **LIDAR-Based SLAM Methods:** A LIDAR-based SLAM algorithm for autonomous indoor navigation of a quadrotor UAS is presented in Grzonka, Grisetti, and Burgard (2009). A Hokuyo LIDAR (URG-4) and an Xsens IMU were used as the main navigation sensors to localize the vehicle and to update the map. The height is estimated by deflecting some laser beams down using a mirror. The UAS motion is computed by incrementing the displacement between two subsequent scans by means of a scan matching algorithm (LIDAR odometry). A Monte Carlo localization algorithm has been used to estimate the 2D position in a given map and to reduce drift in position estimates. A mapping problem was also formulated as a graph-based SLAM. These navigation components have been validated separately on an instrumented quadrotor, but no closed-loop flights have been achieved, except for yaw and height control. A similar LIDAR-based localization and mapping system for quadrotor indoor navigation in unknown environments was described in Bachrach, He, and Roy (2009). The navigation system is based on a Hokuyo LIDAR (UTM-30LX), an IMU, a monocular camera (later upgraded to two monochrome USB cameras) and a Gumstix processor (later upgraded to Intel Atom processor). As in Grzonka et al. (2009), some of the LIDAR beams were deflected downward to estimate and control height above the ground plane. A LIDAR scan-matching algorithm is used to estimate the rotorcraft horizontal 2D position and heading. These estimates are then fused with IMU accelerations using an EKF, yielding accurate estimates of position and velocity. To reduce drift in position estimates, a 2D SLAM algorithm that is based on the GMapping algorithm[19] has been used. Throughout 1 min of flight in a closed room, the scan-matching odometer estimated the position and velocity of a robotic quadrotor with an average error of 1.5 cm for position and 0.02 m/s for velocity. The complete navigation system has also been demonstrated in three different unstructured environments. Indeed, when this navigation system was used with a planning and exploration algorithm, the quadrotor was able to navigate autonomously (motion estimation and mapping) for 8 min (208 m) in open lobbies, 6 min (44 m) in cluttered environments, and 7 min (75 m) in office hallway environments. Flight control was done onboard, whereas scan-matching and SLAM algorithms ran offboard.

Observation 8. Although LIDAR-based localization systems have been successfully applied for ground robot navigation in outdoor environments, we did not find any work of adapting this technology to UAS outdoor localization without GPS aid. This is mainly because UAS generally fly in relatively large open spaces without sufficient environmental structure for the relative position estimation. Therefore, algorithms such as scan-matching and SLAM will fail to provide reliable state estimates for flight control.

5.3. Perception Systems

To achieve real-world applications in natural environments, the RUAS may need to detect and avoid obstacles in real-time, recognize and track targets, map the environment, etc. These perception capabilities are essential for reaching autonomy level 4 of the ALFURS chart in Figure 4. Environment perception technologies used within RUAS include passive sensors such as cameras, and range sensors (active sensors) such as LIDARs. The selection of appropriate sensors depends heavily on the RUAS payload and intended applications. Perception systems for RUAS can thus be divided into two main categories: (1) vision-based perception (passive) and (2) LIDAR-based perception (active).

5.3.1. Vision-Based Perception

Some visual perception systems, developed for RUAS, have already been presented in Section 5.2.2, especially for SLAM, SFM, and optic-flow-based approaches, where both state estimation and perception are performed with the same algorithm. The algorithms presented in the previous section focus more on state estimation, whereas the algorithms presented here assume that state estimates are available (from the onboard GPS-IMU avionics) and focus on perception tasks as defined in Definition 8. The extracted information is not directly used as measurements in the

[19] A publicly available algorithm that uses a Rao–Blackwellized particle filter.

flight controller, but as inputs to higher-level guidance systems that generate trajectories, tasks, or commands for the controller. We classify visual perception systems based on the navigation task they achieve. By reviewing the RUAS literature, we found that most vision-based perception systems developed for RUAS have mainly been used for

- Target detection and tracking
- Obstacle detection without mapping (mapless approaches)
- Mapping-based methods for navigable space searching or safe landing area detection
- Mission-oriented perception.

• **Target Detection and Tracking:** From the image processing and navigation points of view, these perception algorithms are similar to those presented in the previous section, "Target-Relative Navigation". The main difference in the systems presented here is that the visual estimates are used to guide the vehicle whereas the controller relies on GPS or another state estimator to control the rotorcraft flight.

Target tracking for automatic landing: In Saripalli et al. (2003), such a system is used for landing a helicopter on an artificial helipad (letter *H*). A vision algorithm is used to detect and recognize the target by comparing the computed moments of inertia with stored ones (obtained offline). The horizontal 2D position and heading of the helicopter relative to the target are obtained by using the target coordinates in the image frame and the helicopter altitude above ground from a differential GPS. The landing guidance strategy consists of generating heading and velocity commands to the flight controller that tracks these trajectories based on GPS measurements. Sonar is also used to estimate and control height during the last 2 m of descent. Fourteen automatic landings were performed from an initial position of about (8, 10, 10 m) on a stationary target and on a momentarily hidden target. The helicopter achieved automatic landings with an average position error of 42 cm and an average orientation error of 7°. The same authors investigated vision/GPS automatic landing of a helicopter on a moving target, but only offline processing results have been presented (Saripalli & Sukhatme, 2003). A similar approach was used in Hermansson, Gising, Skoglund, and Schon (2010) to land a helicopter on a stationary target using vision and GPS. An EKF is used to fuse data from IMU, GPS, and a vision system in order to estimate the helicopter 3D pose as well as its relative position and heading to the target. These relative estimates are then used to generate reference positions for the GPS-based controller in order to keep the helicopter above the target while decreasing its altitude. Fifteen automatic landings from an altitude of 5 m were performed with a maximum landing error of 60 cm and an average error of about 34 cm. Another system that combines vision and LIDAR data to land a helicopter on a ship deck is described in Garratt, Pota, Lambert, Eckersley-Maslin, and Farabet (2009). The LIDAR determines both the distance and the orientation of the deck relative to the helicopter. This system has been augmented by a visual tracking system that detects a light source on the deck and tracks it over time. Automatic landing on the deck has not been attempted yet. Recently, Tom Richardson, from the University of Bristol, has won a national award for his part in developing a vision system for automatic landing of an unmanned helicopter on a moving target (roof of a car). To our knowledge, the obtained experimental results have not been published yet.

Horizontal target approach: Vision and GPS have also been used to approach a frontal target and to hover at some distance from it. In Mejias, Saripalli, Campoy, and Sukhatme (2006), an image-based velocity control approach to frontal target tracking is presented. The image processing algorithm detects and tracks features (building windows) using segmentation and square finding for detection, with template matching and a Kalman filter for tracking. The target coordinates in the image frame are used directly to generate lateral and vertical velocity commands to the flight controller. By using GPS measurements, the helicopter follows these commands in order to align itself with the window target. The developed system has been demonstrated on two different helicopters (USC AVATAR and COLIBRI). In both experiments, the helicopter successfully tracked the target window and performed automatic hover flight.

Moving ground target tracking: Vision is also the sensor of choice for mobile ground target detection and tracking. A vision-based pursuit–evasion game has been implemented on a helicopter UAS and unmanned ground vehicles (UGV) (Vidal, Shakernia, Kim, Shim, & Sastry, 2002) at the University of California, Berkeley (UCB). The Yamaha R-50 helicopter and two UGV pursuers were equipped with a vision system that can actively track many colored objects at 30 fps. These visual estimates are then used as inputs for a distributed hierarchical pursuit planner that generates reference trajectories for each agent. The navigation and pursuit policies have been validated in real-time experiments using a RUAS, two UGV pursuers, and one UGV evader. In the framework of the WITAS project at Linköping University, an onboard vision-based perception system was developed for ground vehicle tracking and surveillance by a robotic helicopter. We did not find any published experimental results, but the video clip available at Linkoping (2003) shows the Yamaha RMAX helicopter autonomously tracking a moving car in a natural environment for a period of 9 min. Cooperative vision-based tracking of ground vehicles by indoor quadrotors is presented in Bethke, Valenti, and How (2007). The vision tracking algorithm uses an optimization technique and a KF that combines the instantaneous observations of all UAS, allowing very accurate estimation of target location and velocity. The 3D position of the UAS is assumed to be known from its onboard avionics

(here, by the VICON system). The guidance system generates waypoints to keep the UGV in the RUAS FOV. The performance of this system has been evaluated using two quadrotors and a small RC car as a target. Another example of vision-based ground target tracking by a rotorcraft MAV is presented in He, Bachrach, Achtelik, Geramifard, Gurdan, et al. (2010). Images from a single onboard camera were used to identify obstacles and mines and to track a ground vehicle. These ground targets were then geolocated using the known position of the MAV and the camera. Ground targets were first detected by a human operator and then passed to the vision algorithm for automatic tracking in successive images. The tracker was based on an object appearance classifier that used a Bayesian particle filter. For robust tracking, optical flow was also used to refine the camera ego-motion. This tracker was evaluated offline on a 17-s video collected by a rotorcraft MAV during the MAV'08 competition in India. The algorithm successfully tracked a moving ground vehicle and a walking person. During mission execution, the system (assisted by a human operator) was also able to geolocate two mines and two ground obstacles.

• **Obstacle Detection without Mapping:** Computer vision can be used to detect and negotiate obstacles without mapping the environment by using: (1) image motion (optic flow) as described in Section 5.2.2, "Bio-Inspired Optic Flow Approaches," (2) a priori known characteristics of a specified object such as color and shape, or (3) depth information from stereo vision.

At DLR (the German Aerospace Centre), there is active research on using vision for autonomous rotorcraft perception. In Andert, Adolf, Goormann, and Dittrich (2010) considered the problem of flying through a course with gates that are slightly larger than the vehicle (narrow passage). The gate global position is estimated by tracking some colored flags on the gate (assuming that their size and appearance are previously known) and using the navigation solution from GPS and IMU. Once the gate position is estimated, the guidance system generates two waypoints that define a flight path (straight segment) that goes through the gate center. The developed system was demonstrated in real time using a 12-kg robotic helicopter that autonomously crossed gates of 6×6 m at a speed of 1.5 m/s without collisions. The use of optic flow and stereo vision onboard a helicopter for 2D navigation through urban canyons without mapping is investigated in Hrabar and Sukhatme (2009). An LK tracker is used to compute optic flow from a pair of sideways-looking fish-eye cameras. The translational optic flow is then used for the centering response by changing the vehicle's turn rate to balance optic flow on both sides. The perception system has been also augmented with a forward-facing stereo camera to detect frontal obstacles. Based on a 3D-point cloud representation, obstacles are detected in the upper half of the image using a distance threshold and region-growing algorithm. Once the obstacles have been detected, an appropriate evasive control command (turn away, stop) is generated. The combined stereo and optic-flow-based perception and control system have been evaluated in simulations as well as in flight experiments using an autonomous tractor and a robotic helicopter. When tested on a helicopter, the optic flow system was able to steer the helicopter away from obstacles with a success rate of five out of eight flights. For frontal obstacle avoidance, three flights were performed and the stereo vision allowed the helicopter to avoid the obstacle in one flight only. The combined stereo-optic flow was tested on a ground vehicle but not on a helicopter.

• **Mapping-Based Methods for Navigable Space Searching or Safe Landing Area Detection:** In the following paragraphs, we present vision-based perception systems for RUAS that are based on the mapping framework. In mapless approaches, obstacle avoidance strategies are generally reactive, whereas mapping-based methods allow the use of more sophisticated path planning algorithms. Mapping the environment consists of building some internal representation of the scene in the form of a depth image, digital elevation map, occupancy grid, feature database, set of landmarks, etc. The existing systems for mapping-based visual perception onboard RUAS can be classified into three categories: (1) simultaneous localization and mapping (SLAM), (2) simultaneous mapping and planning (SMAP), and (3) safe landing area detection (SLAD).

SLAM: As described previously (see Section 5.2.2, "Concurrent Estimation of Motion and Structure"), these algorithms estimate the vehicle's self-motion and build a map of the environment which is generally represented by a set of features or registered point clouds. Most work on SLAM-based perception has focussed on the vehicle's localization, and the generation of accurate maps of the environment. In these works, the generated maps have not been used to detect obstacles or to guide the vehicle through free spaces (Artieda et al., 2009; Bryson & Sukkarieh, 2009; Caballero et al., 2009; Kanade et al., 2004; Kendoul, Fantoni, & Nonami, 2009; Kima & Sukkarieh, 2007; Lemaire, Berger, Jung, & Lacroix, 2007; Nemra & Aouf, 2009; Tornqvist et al., 2009).

Simultaneous mapping and planning: Unlike SLAM, SMAP focuses more on building maps that can be used for obstacle avoidance and path planning. In these approaches, state estimates are generally available from the GPS–IMU, and efficiency and robustness are more important than accuracy. Probabilistic occupancy grids are the representation most widely used to address these issues. Most successful work on SMAP for RUAS has been done using LIDAR, as will be described in the next section. However, there is some interesting work on applying visual SMAP for rotorcraft, such as the system presented in Andert and Adolf (2009). The proposed world representation combines occupancy grids and polygonal features. First, a 3D occupancy grid around the vehicle is created

incrementally using stereo-based range measurements and GPS/INS data. Actual sensor data and previously stored features are then inserted into the grid. Obstacle features are constructed by clustering occupied grid cells, and a polygonal shape of these features is finally approximated by prisms. This mapping algorithm has been implemented on a 25-kg helicopter and has been demonstrated in real-time experiments. Another stereo-vision-based perception system for rotorcraft is described in Byrne, Cosgrove, and Mehra (2006). The VISTA (visual threat awareness) system was developed by Scientific Systems Company Inc. for collision detection on an unmanned helicopter. It combines block matching stereo (depth image) with image segmentation based on a graph representation appropriate for obstacle detection. Detected obstacles are then tracked using a KF, and the result is a state estimate for each obstacle, which is then passed to the guidance system for path planning. Nineteen flight experiments were performed using the Georgia Tech. RMAX helicopter. The vehicle had to fly a collision trajectory approaching either a "sign" obstacle or a "pole" obstacle at speeds up to 10 m/s. The obstacles were correctly detected at various collision distances.

Safe landing area detection: Unmanned rotorcraft may be commanded to land on unprepared and unknown terrains to accomplish some mission or to achieve an emergency landing. Therefore, safe landing area detection (SLAD) is necessary for mission achievement and vehicle survival. A safe landing area can be defined as a terrain without hazards, large enough to fit the RUAS, and suitable for landing a specified machine (slope, roughness, etc.). Vision and LIDAR are the sensors most investigated for solving the SLAD problem. We present here vision-based SLAD algorithms; LIDAR-based approaches will be described in the next section.

Early work on visual SLAD for an unmanned helicopter was presented in Garcia-Pardo, Sukhatme, and Montgomery (2002). A single downward-pointing camera was used to detect an obstacle-free circular area by assuming flat terrain and high contrast between obstacles and underlying terrain. Four different strategies were proposed and evaluated offline on images gathered from 10 flights of an instrumented helicopter. A stereo vision-based system for terrain mapping and SLAD for an unmanned helicopter is presented in Meingast, Geyer, and Sastry (2004). Given the motion of the vehicle, a digital elevation map is built using a multiframe planar parallax algorithm. A landing function is defined to analyze and classify each image pixel. Neither simulations nor experimental results were presented in this paper.

Later in 2007, the same authors presented another SLAD algorithm based on a single camera (Templeton, Shim, Geyer, & Sastry, 2007). The vision algorithm is similar to the previous one and uses a recursive multiframe planar parallax algorithm to build a DEM in world coordinates. The map is divided into blocks that are declared as landing candidates if they are below given thresholds on each element of the landing quality equation. The developed system was evaluated offline only. The SLAD problem has been also an active research topic for safe landing of spacecraft on unknown terrains such as Mars. JPL researchers have developed an autonomous helicopter to evaluate and validate some technologies that can be useful for spacecraft as well as for UAS (Montgomery, Roumeliotis, Johnson, & Matthies, 2006). A digital elevation map of the terrain is generated from monocular images using *motion stereo* and a narrow beam altimeter (Johnson, Montgomery, & Matthies, 2005). Once the DEM is built, local operators are applied to detect hazards by generating slope and roughness maps. Finally, safe landing areas are detected by generating binary images from the slope and roughness maps and applying a grassfire transform. The SLAD algorithm has been implemented on an unmanned helicopter and demonstrated in closed-loop flight experiments using vision for SLAD and GPS/IMU for flight control. Four successful autonomous landings were achieved in unknown and hazardous terrains with an average position error of 1 m. Other SLAD flights in more complex terrains have also been performed. SLAD and automatic landing problems have been also investigated in the US Army's Precision Autonomous Landing Adaptive Control Experiment (PALACE) project (Theodore et al., 2006). The landing site selection used for the PALACE project consists in creating a stereo range map of the terrain, and then running a SLAD algorithm. This algorithm applies a set of landing point constraints (slope, roughness, distance to obstacles) to the range map to find all safe landing regions, and then to choose the optimum landing point. The initial landing point is chosen from 30 m above second level (AGL) and then verified and refined at 24, 18, 12, and 6 m AGL. Position control is done using GPS until 12 m AGL, and then a visual odometer until 2 m AGL, and finally inertial navigation for the last 3 m. The complete system (SLAD, VO, automatic landing) has been implemented on a Yamaha RMAX helicopter and demonstrated in closed-loop flights. Stereo ranging and SLAD algorithms were first evaluated under different conditions (surface texture, altitude, pattern direction, obstacle height), and very satisfactory results were obtained. More than 17 successful landings were achieved on various surfaces and obstacle fields with a landing accuracy of better than 1 m and a success rate greater than 95%, Figure 11.

In a recent paper (Sanfourche, Besnerais, Fabiani, Piquereau, & Whalley, 2009), researchers from the French Aerospace Lab (ONERA), compared two different algorithms for terrain characterization and SLAD. One algorithm was developed by NASA and is based on hemispherical LIDAR (described in the next section), and the other was developed by ONERA and uses monocular vision. Feature correspondences are used to estimate the parameters of a homography and bundle adjustment is

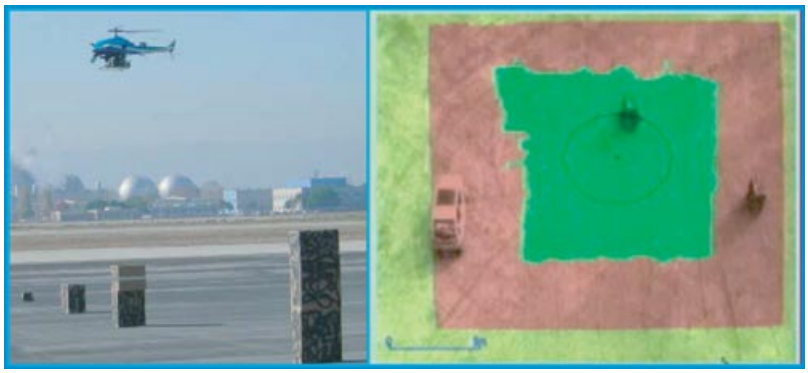


Figure 11. The NASA robotic helicopter during autonomous identification of safe landing area using stereo vision (Theodore et al., 2006).

employed to refine ego-motion and feature positions (structure) estimation. By assuming that the ground is flat with some obstacles, a DEM is built by classifying the 3D points as ground or obstacles. This vision-based mapping algorithm has been evaluated offline on a video sequence taken by the helicopter flying at a height of 10 m above flat ground with the same common obstacle field (COF) as in Takahashi, Schulein, and Whalley (2008). All obstacles (boxes) taller than or equal to 40 cm were correctly detected, with a standard deviation error of 5 cm. A comparison with the NASA LIDAR-based method revealed that the two approaches yielded comparable results. However, the LIDAR-based SLAD algorithm runs in real time onboard the helicopter, and it is effective even for night flights. In Cesetti, Frontoni, Mancini, Zingaretti, and Longhi (2010), optic flow has also been used to select a safe landing area without building a dense DEM of the environment. Sparse SIFT features are used to compute optic flow, which is then used to recover depth information in the landing area. By assuming pure translational flight at a constant height, a simple classifier based on a binary threshold is used to decide if the surface has variable depth or not and, consequently, if it can be used as a landing area. This approach has been evaluated in simulations and offline based on images collected by an unmanned helicopter.

Observation 9. From the reviewed papers, we found that vision-based perception technologies for UAS are not yet mature and very few experimental results have been reported from closed-loop flights. Indeed, most of the proposed approaches have been evaluated in simulations or in offline processing. In some other work, perception algorithms were running in real time but not used in a closed-loop to guide the vehicle. When closed-loop results are presented, they are basic, without proper evaluation of their performance in different environments and conditions. Furthermore, it turned out that there are no convincing results on using vision for SMAP and obstacles detection and avoidance. The most successful results on vision-based perception in unknown environments are probably the SLAD-related results obtained in the NASA *PALACE* project (Theodore et al., 2006), and the SMAP-related research by DLR Andert, Adolf, Goormann, and Dittrich (2011).

• **Mission-Oriented Perception:** In the work described previously, vision has mainly been used for the vehicle's navigation (localization and obstacle avoidance). However, perception systems are more general than that and may be used for environmental monitoring and mission achievement (camera as a mission sensor). In current systems, images are generally transmitted in real time to the ground station, where the operator analyzes them and makes appropriate decisions, or they are post processed. However, in some missions, it is more effective to automatically retrieve the required mission information from images in real time and guide the helicopter appropriately. There has been some work on using computer vision onboard rotorcraft for mission achievement purposes. A good example of that is the COMETS (EU project) project, where a cooperative vision-based perception system has been developed for multi-UAS and applied for automatic detection of forest fires (Merino, Fernando Caballero, Ferruz, & Ollero, 2006). In the WITAS project (Farnebäck & Nordberg, 2002), dense motion of the images has been applied to traffic monitoring with an autonomous RMAX helicopter. In more recent work, Rudol and Doherty (2008), demonstrated automatic detection of human bodies for search and rescue missions using an autonomous RMAX helicopter equipped with thermal and color imagery and an onboard image processing algorithm. The Australian Centre for Field Robotics (ACFR) has developed two vision-based perception systems for automatic detection and surveillance of aquatic weeds (Goktogan, Sukkarieh, Bryson, Randle, Lupton, et al., 2010), and classification of large farmland environments (Bryson, Reid, Ramos, & Sukkarieh, 2010). The

first system was demonstrated using an autonomous helicopter, whereas the second one was evaluated using images collected by a fixed-wing aircraft. There are also other interesting applications such as power line inspection, as described in Wang, Han, Zhang, Wang, & Li, (2009).

5.3.2. LIDAR-Based Perception

Although images contain rich information about the environment, their real-time processing to retrieve useful cues suffers from many issues, which are mainly related to high computational requirements and high sensitivity to the environment, such as light and, textures. On the other hand, LIDAR can provide accurate information about the environment structure with lower computational requirements. However, its weight and power consumption limit its use to RUAS with relatively significant payload, such as the Yamaha RMAX. Recently, progress in LIDAR miniaturization has motivated its use on mini quadrotor UAS as the main perception sensor. Most LIDAR-based perception systems for RUAS are based on building metric maps of the environment. Different types of maps have been proposed, but DEM and occupancy grids are the most widely used for RUAS perception. As for vision-based perception approaches, existing LIDAR-based perception systems can be classified into four main categories:

- Mapless-based approaches for obstacle detection
- Simultaneous localization and mapping (SLAM)
- Simultaneous mapping and planning (SMAP)
- Safe landing area detection (SLAD).

• **Mapless-Based Approaches for Obstacle Detection:** Obstacle detection and avoidance without mapping is an attractive solution because it can deal with uncertainty and run very quickly, providing a fast reactive system that can prevent last-minute collisions. CSIRO researchers have developed a LIDAR-based perception system for obstacle avoidance and infrastructure inspection beyond visual range (Merz & Kendoul, 2011). The proposed system uses only immediate LIDAR measurements to detect the ground and frontal obstacles/target, and computes just one reactive action based on the current context. The developed system has been implemented onboard the CSIRO unmanned helicopter and flight-tested in a number of mission scenarios and environments. More than 40 flights, totaling about 15 h of autonomous flight, have been performed to date. The system performed well with a high success rate and dependability. The developed and demonstrated capabilities proved to be sufficient to accomplish several real-world applications.

• **SLAM:** Some LIDAR-based SLAM algorithms for RUAS perception have already been described in Section 5.2.3. The systems described in that section use LIDAR for both localization and mapping, and are generally developed for indoor navigation of small quadrotors. In addition to these indoor systems, LIDAR has also been used on bigger helicopters for outdoor mapping. However, the helicopter state is generally available from GPS/INS because airborne LIDAR-only SLAM in outdoor and natural environments is very challenging. 3D terrain mapping from LIDAR onboard the Yamaha R-50 helicopter was demonstrated by CMU researchers (Kanade et al., 2004; Miller & Amidi, 1998). The helicopter was flown about 10 m above a 300 × 300-m site as the system scanned the surrounding environment. A digital elevation map with a 0.5 m^2 grid spacing was generated in real time from the scan data using known pose and position estimates. However, the generated maps were not used for guidance or navigation. Other researchers from CMU (Thrun, Diel, & Hahnel, 2003) proposed an airborne SLAM algorithm that builds accurate 3D maps of urban and natural terrains from 2D range data, GPS, and compass measurements. Real-time scan matching is used to improve the accuracy of pose estimates and the environment model. This system has been demonstrated on different environments using a remotely controlled Bergen Industrial Twin helicopter equipped with a SICK LIDAR. The generated maps appear to be visually and accurate, and locally consistent with spatial resolution in the centimeter range.

• **Simultaneous Mapping and Planning:** SMAP involves perception systems where mapping and planning are jointly performed to build a map of the environment that is used for obstacle detection and avoidance. Shim et al. (2006), proposed an obstacle detection and avoidance scheme based on building local obstacle maps and solving a conflict-free trajectory using model predictive control. Local obstacle maps are built by registering range measurements in a FIFO database and computing the nearest point in the surrounding. This nearest point (minimal distance) is considered as an obstacle, and is then used in the MPC algorithm for reference trajectory replanning with a safe distance from obstacles. The Yamaha R-50 helicopter, equipped with SICK LIDAR and avionics, was used to demonstrate the proposed SMAP algorithm. The vehicle flew from point A to point B at 2 m/s while avoiding obstacles, Figure 12. Local map building and trajectory generation were done in real time at the ground control station.

Researchers from NASA Ames Research Center and the U.S. Army have developed different LIDAR-based perception systems for robotic helicopters. In the framework of the Autonomous Rotorcraft Project (ARP), an active obstacle-sensing and mapping system was developed and demonstrated using 2D SICK LIDAR onboard a Yamaha RMAX helicopter (Freed, Fitzgerald, & Harris, 2005; NASA, n.d.). For effective 3D navigation in urban environments, 3D hemispherical LIDAR was developed by mounting 2D SICK LIDAR on a spinning mechanism (Takahashi et al., 2008), Figure 13. When bining range data and GPS and IMU measurements are combined, accurate 3D maps (registered point clouds) of the environment are generated. This 3D



Figure 12. BEAR helicopter during obstacle detection and avoidance using 2D LIDAR (Shim et al., 2006).

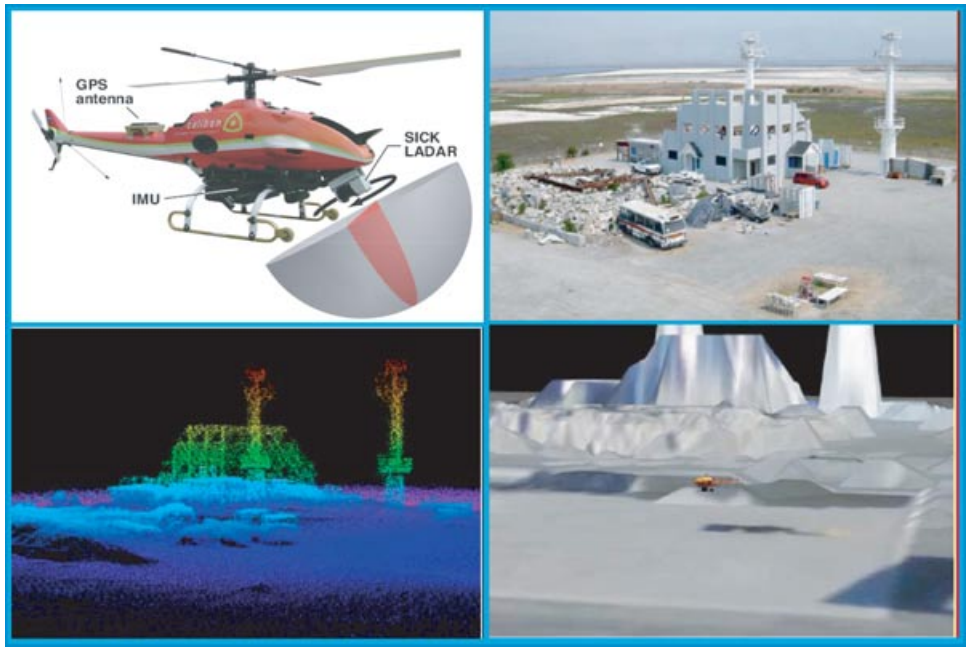


Figure 13. The U.S. Army/NASA obstacle field navigation system based on a 3D hemispherical LIDAR (Takahashi et al., 2008).

LIDAR system was then used to develop a complete 3D navigation system for an autonomous helicopter flying in urban environments (Tsenkov, et al., 2008; Whalley, Takahashi, Tsenkov, & Schulein, 2009). The chosen map format or terrain representation is a cell-based hybrid data structure with dual data access: directly through the data table, or hierarchically through a quadtree. Two 3D path planning algorithms have been proposed and used to guide the helicopter through obstacles. The first is based on an obstacle proximity map, whereas the second is based on a height map. This SMAP system has been implemented on the Yamaha RMAX helicopter and demonstrated in a number of real obstacle avoidance scenarios and environments; see Figure 13. Other interesting work about LIDAR-based SMAP is presented in (Scherer, Singh, Chamberlain, & Elgersma, 2008), where researchers from CMU have demonstrated safe autonomous flight with 3D obstacle avoidance capability. The developed method combines online environment sensing, global planning, and local planning for reactive obstacle avoidance. The perception system is based on a customized 3D LIDAR system from Fibertek Inc ($30^\circ \times 40^\circ$ field of view). The raw range sensor data are transformed to an occupancy probability and then mapped into a 3D evidence grid using position and orientation information from GPS and IMU. The developed system was tested extensively in different environments and scenarios, with more than 700 successful obstacle avoidance runs at commanded speeds up to 10 m/s. Miniaturization of LIDAR units led to their integration into mini platforms, such as quadrotors that weigh less than 1 kg. Indeed, researchers

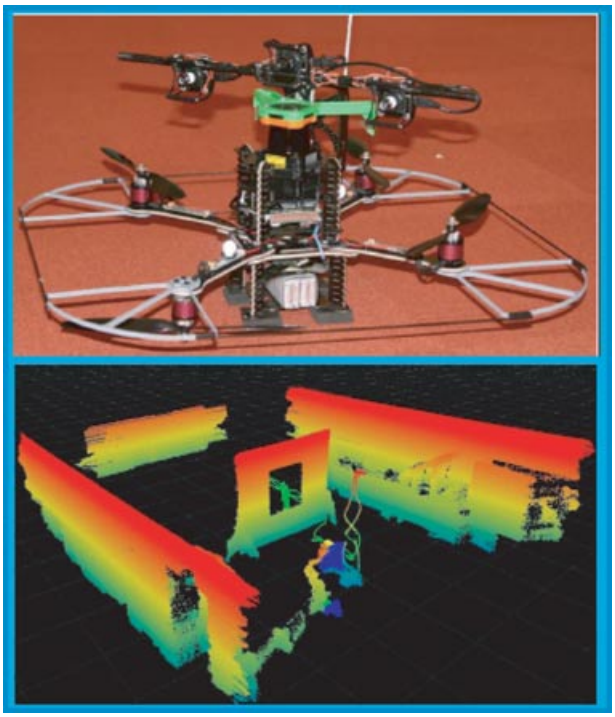


Figure 14. The MIT robotic quadrotor demonstrating autonomous indoor navigation and mapping using *Hokuyo* LIDAR (He, Bachrach, & Roy, 2010b).

from MIT (Bachrach et al., 2009; He, Bachrach, & Roy, 2010) have developed a robust autonomous navigation system that enables a quadrotor MAV to explore and map unstructured and unknown GPS-denied environments. Real-time SLAM and SMAP algorithms allowed the vehicle to estimate its relative position, to map the environment and to plan its obstacle-free path for exploration and goal-oriented navigation, Figure 14. Data from a Hokuyo UTM-30LX LIDAR, stereo vision, and IMU are fused through an EKF for state estimation, and a GMapping SLAM algorithm for position correction and occupancy grid updating. The system has been tested in a number of GPS-denied environments both outdoors and indoors, and it has allowed the team to win the AUVSI International Aerial Robotics Competition in 2009; see Figure 14.

• **Safe Landing Area Detection:** LIDAR-based SLAD systems outperform visual SLAD algorithms in many situations, such as textureless environments, complex terrains, or poor lighting conditions. Furthermore, LIDAR can provide very accurate information about the environment structure, allowing safe landing area selection even in challenging terrains. In the framework of the NASA/Army PALACE program, a SLAD algorithm that takes a range map from a 3D LIDAR and combines it with a set of landing point constraints was developed. The algorithm first calculates a safe landing map, and then chooses the optimum landing point. This algorithm was successfully tested on a helicopter using range maps from stereo vision (Theodore et al., 2006). In Takahashi et al. (2008) and Whalley et al. (2009), the same SLAD algorithm was applied to the 3D point cloud obtained from a hemispherical 3D LIDAR to determine the regions where the helicopter could land safely. The landing site selection constraints are surface slope, obstacle height, and hazard proximity. Approximately 110 runs of the LIDAR-based SLAD system were performed at Moffet Federal Airfield, the Fort Ord MOUT site, and Fort Hunter Liggett, using the NASA autonomous Yamaha RMAX helicopter with real-time onboard processing. This included surface types of concrete, asphalt, dirt roads, and natural terrain. The SLAD system generally performed well in natural and manmade environments, Figure 15. The same system was also applied to select a safe spot for dropping a slung load. In a recent paper (Takahashi, Abershitz, Rubinets, & Whalley, 2011), the authors presented the results of quantitatively evaluating two SLAD algorithms using a new test method that incorporates a detailed survey of the test sites. The first SLAD algorithm was developed under the PALACE program and is described in the previous paragraph. The second algorithm is a new approach that applies the inflating

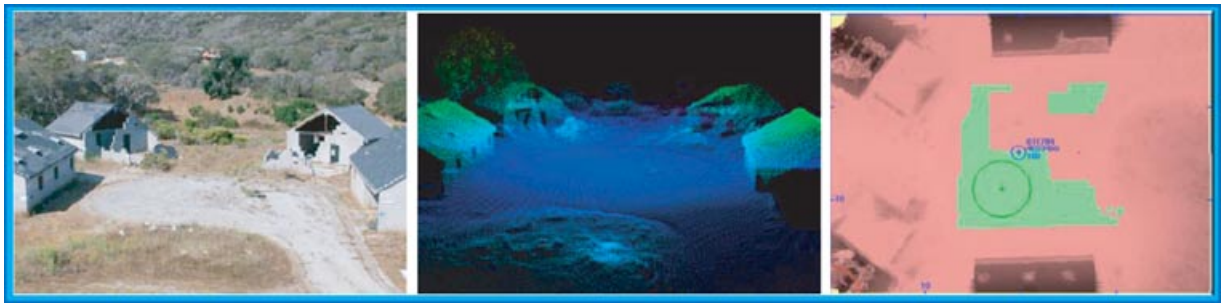


Figure 15. SLAD results of the U.S. Army/NASA autonomous helicopter equipped with a hemispherical SICK LIDAR (Whalley et al., 2009).

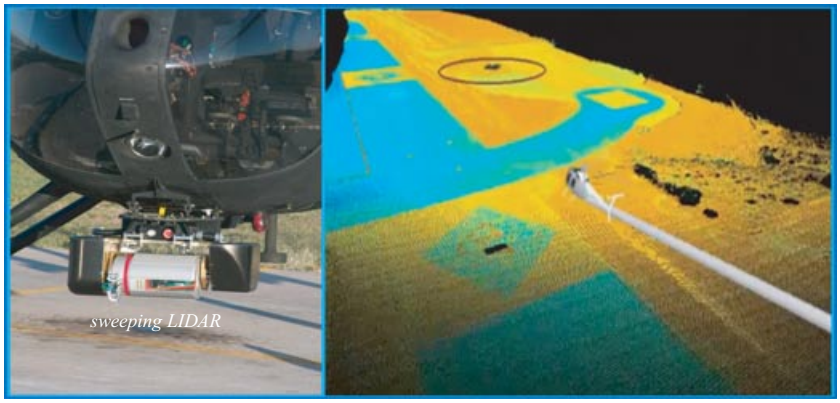


Figure 16. Results from autonomous landing of an unmanned full-scale helicopter in unknown zones, using the CMU LIDAR-based SLAD and obstacle avoidance system (Scherer, 2011).

circles technique to a smoothed height map to detect a set of potential landing points and uses fuzzy logic to rank them based on geometric (size, surface roughness, and slope) and mission (distance, azimuth, and elevation to a target point) criteria. The main contribution of the paper is the development of a test method for quantitatively comparing and evaluating the performance of SLAD algorithms in realistic environments. This test approach depends on generating benchmark sites or "truth" in conjunction with the receiver operator characteristic plots. Both SLAD algorithms were evaluated based on data sets collected at seven different sites using the NASA RMAX helicopter, equipped with a SICK spinning LIDAR (Takahashi et al., 2008). Experimental results indicated that both algorithms performed well with the simple sites but had more difficulty dealing with complex sites.

LIDAR-based SLAD has also been successfully applied to safe landing of autonomous full-scale helicopters. Carnegie Mellon University and Piasecki Aircraft Corp. have developed and flight-demonstrated a navigation system that enables full-size, autonomous helicopters to fly at low altitude while avoiding obstacles; evaluate and select suitable landing sites in unmapped terrains; and land safely using a self-generated approach path (CMU & Piasecki, 2010). Their algorithm, described in (Scherer, 2011; Scherer, Chamberlain, & Singh 2010), takes as input a set of 3D range points registered in a global coordinates system to select a safe landing area using a combination of coarse and fine evaluation. The coarse evaluation finds candidate areas by applying a fast plane fit to measure the slope, roughness, and other statistics in a terrain grid map. The fine evaluation then evaluates promising areas by fitting a 3D model of the helicopter and landing gear to a triangulated surface of the environment. In (Scherer, et al., 2010), primary experimental results from 3D point cloud datasets, collected by a manned helicopter on nine sites with a flight time of 8 h, were presented. The helicopter was equipped with a fixed Riegl LMS-Q140i-60 LIDAR, a Novatel SPAN INS/GPS, and an IMU. Experimental results showed that the SLAD system was able to detect safe landing areas from 100–200 m AGL in different vegetated and urban sites. In another experimental setup (CMU & Piasecki, 2010; Scherer, 2011), the Riegl LIDAR was mounted on a sweeping mechanism and integrated, together with GPS and IMU, into a full-scale Unmanned Little Bird helicopter; see Figure 16. This technology has been flight-tested at the Boeing Company's Rotorcraft Systems facility, and autonomous landings in cluttered environments have been repeatedly demonstrated. In each case, the system was able to map an unknown area, detect obstacles, select a safe landing area, plan its approach path, avoid obstacles and perform automatic landing. An interesting video of an autonomous flight test is available at CMU and Piasecki (2010).

Sevcik, Kuntz, & Oh (2010) investigated the effects of obscurants such as smoke and dust on SLAD when an airborne LIDAR was used. First, a probabilistic point cloud terrain map is created by fusing range points with the rotorcraft pose estimates and applying scan alignment for improving registration accuracy. Point clouds are converted to a grid map and then a cost map based on the slope and roughness of the surface. The lowest-cost area that fits the helicopter rotor is finally selected as a safe landing area. The system was evaluated offline using data collected by the SR100 Rotomotion helicopter, equipped with a SICK LIDAR. The algorithm successfully detected safe landing areas, and experimental results showed performance degradation in the presence of smoke.

5.4. Situational Awareness

There is confusion in using the term situation or situational awareness (SA) in the UAS literature. This term is

commonly used in aviation, and some technologies (obstacles detection and visualization, for example) have been developed to increase the situational awareness of the pilot or the UAS operator. In current manned and unmanned vehicles, the SA process is performed by the human crew or operator. We are interested in autonomous SA onboard the rotorcraft, which allows it to reach autonomy level "8" of the Rotorcraft Autonomy Levels chart, presented in Figure 4. In our literature review, we did not find any published papers about autonomous SA for UAS, which means that issues related to autonomy level 8 and above have not been addressed yet.

6. GUIDANCE SYSTEMS

As defined in Section 3.1, a UAS guidance system tends to replace the pilot's deliberative process and decision functions, resulting in commands for the flight controller to achieve a given mission safely. A guidance system is a key component in increasing the autonomy level of an unmanned rotorcraft; see Figure 4. As shown in Figure 3, a typical guidance system for UAS includes the following main components: (1) trajectory generation, (2) path planning and obstacle avoidance, (3) mission planning, and (4) reasoning and high-level decision making. Depending on the autonomy level of the rotorcraft, each of these components can be performed manually by human operator or autonomously onboard the vehicle. Currently, UAS are operated remotely by human operators from ground control stations, or use rudimentary guidance systems, such as following preplanned or manually provided waypoints. This is sufficient for UAS operating at high altitude for reconnaissance and exploration. However, UAS will require more advanced guidance capabilities to achieve more complex tasks and missions. Furthermore, RUAS are generally designed to operate at low altitude in cluttered and dynamic environments. Indeed, operation at low altitudes is where RUAS are most valuable, but also where they are most vulnerable. Therefore, advanced guidance and perception capabilities are necessary to negotiate obstacles and to plan the flight path. Currently, there is growing interest in increasing the vehicle's autonomy by developing guidance systems that are able to tackle a number of operational events without operator intervention: for example, trajectory replanning following the detection of an obstacle or changes in the mission or environment configuration. Guidance capabilities can also be increased to allow integration with other UAS and manned vehicles to accomplish coordinated missions. In this section, we review some guidance systems that have been successfully applied to unmanned rotorcraft systems and describe the most important milestones and achievements in both path and mission planning, focusing on practical works with experimental results. Figure 17 outlines methods and algorithms that have been developed for RUAS and presented in this section.

6.1. Path Planning Methods and Algorithms

Path planning is a fundamental aspect of autonomous UAS guidance and deployment and an essential component for reaching autonomy level 4 and higher levels; see Figure 4. The path planning problem has been extensively studied for manipulators and ground robots, and different methods and algorithms have been proposed. Classifying path planning algorithms is not a trivial task, but one can, for example, make distinctions between optimized and heuristic algorithms; global and local path planners; path planners with and without differential constraints; and 2D and 3D path planners. We attempt to briefly describe some methods that are commonly used for path planning, and present some successful implementations on unmanned rotorcraft. For a general understanding of the motion planning problem and existing approaches, the reader can refer to published surveys and textbooks in this field, such as (Latombe, 1991; LaValle, 2006). In addition to these traditional motion planning methods, other planning approaches have been proposed recently, such as a multilayered synergistic framework that can specify complex temporal goals (Bhatia, Kavraki, & Vardi, 2010). More recently, Goerzen et al. (2010) provided an overview of existing motion planning algorithms with a general perspective on the problem of UAS guidance. Many papers have been published on path planning, and many techniques have been applied to the UAS path planning problem. However, most proposed algorithms have been validated in simulations only. Therefore, this survey will not cover all these papers and techniques, but rather will present the most used and practical methods, with a particular focus on works with experimental results. From the literature of UAS path planning, we have identified six main classes of path planning techniques that have been implemented on RUAS:

1. Road maps (RM), including *visibility graph, Voronoi road map, probabilistic road map (PRM), and rapidly exploring random trees (RRT)*
2. Potential fields (PF)
3. Optimization methods (OM), including *mixed integer linear programming (MILP), receding horizon control (RHC), and motion primitive automaton (MPA)*
4. Heuristic search algorithms (HSA), including $A^\star$ and $D^\star$ *algorithms*
5. Planning under uncertainties (PUU)
6. Reactive and bio-inspired obstacle avoidance methods (RBIOAM).

Path planners based on methods 1 to 5 can be global, local, or a combination of both, whereas techniques in 6 are reactive. Global path planners generally assume complete knowledge of the world and in return provide a complete path to the destination with useful properties of correctness, optimality, and completeness. These algorithms are generally expensive to compute, especially if the trajectory has to be revised or replanned when new obstacles are

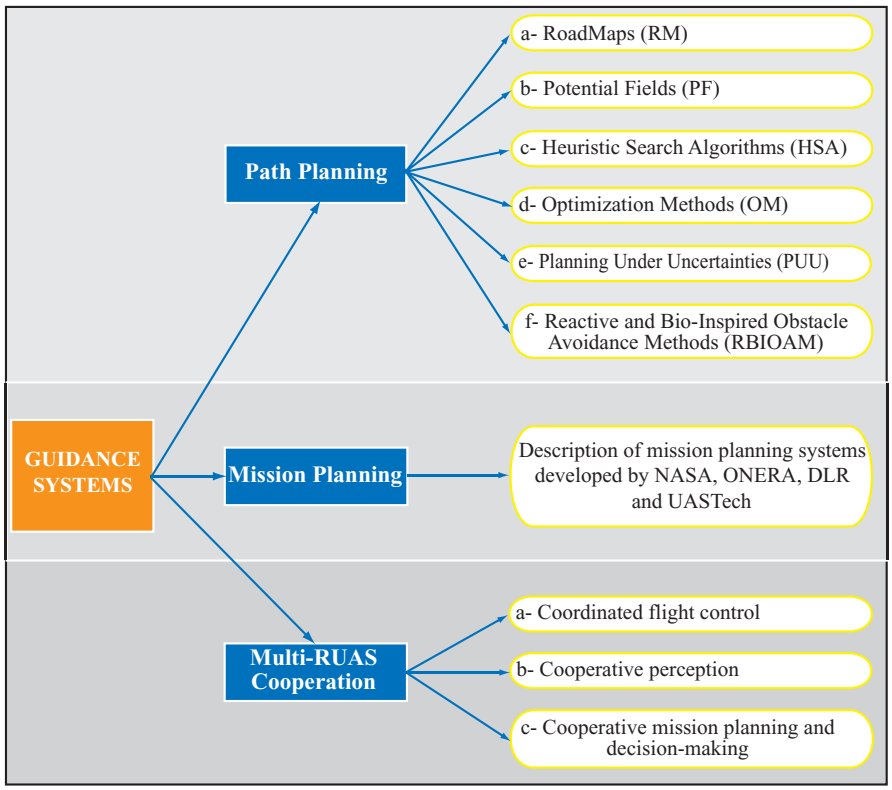


Figure 17. Classification of guidance systems developed for RUAS.

detected. Furthermore, most outdoor UAS missions do not have complete knowledge of the world. Local path planners can be defined as sensor-based incremental planning algorithms where sensor information is acquired during mission execution, a local model of the environment is built in the neighborhood of the vehicle, and the "best" local path is determined, based upon these local measurements or maps. Local path planners are better able to accommodate new sensor data than classical global planners. However, such path planning algorithms do not provide any guarantee of completeness and optimality. In contrast, reactive obstacle avoidance algorithms operate in a timely fashion and compute just one next action in every instant, based on the current context. They use only immediate measurements of the obstacle field to take a reactive response in order to prevent last-minute collisions by stopping or swerving the vehicle when an obstacle is known to be in the trajectory, which has been planned by a different algorithm. These reactive algorithms are important in dealing with uncertainty, and run very quickly. However, reactive methods cannot guarantee an appropriate solution to a route for a goal. A practical implementation of path planning may include the combination of these different path planning algorithms and reactive methods.

6.1.1. Road Maps

In this method, the motion planning problem is reduced to fitting a road map or a graph to the work space and searching for the best path. A road map is generally represented as a graph in which the nodes correspond to placements of the robot and the edges represent collision-free paths between these placements; see Figure 18. There are different road map methods, such as visibility graphs, Voronoi road maps, probabilistic road maps (PRM), and rapidly exploring random trees (RRT). Once the road map is produced, a graph search algorithm such as Dijkstra's method or A^* can be used to find a sequence of states (e.g., waypoints) that define the best path.

- **Visibility Graph:** The visibility graph was one of the earliest road map methods, which consists of straight line segments that connect nodes of polygonal obstacles. It is a complete and optimal path planner that is computable in only two dimensions. To our knowledge, there are no works on applying visibility graphs for rotorcraft path planning except the work presented in Hoffmann, Waslander, & Tomlin (2008). However, their work focused on trajectory generation for a quadrotor where a path is generated by a visibility graph-based algorithm and then smoothed to consider the vehicle's dynamics.

• **Voronoi Diagrams:** Voronoi diagrams are another popular mechanism for generating a road map from a configuration space (c-space). In this case, the road map consists of Voronoi edges, which are equidistant from all the points in the obstacle region. In contrast to visibility graphs, Voronoi paths are by definition as far as possible from the obstacles. Researchers from NASA and U.S. Army have developed an obstacle field route planner (OFRP) that is based on Voronoi roadmaps (Howlett, Schulein, & Mansour, 2004; Howlett, Tsenkov, Whalley, Schulein, & Takahashi, 2007; Whalley et al., 2005). The Voronoi-based OFRP algorithm provides a 2D solution using a four-phase approach: (1) Voronoi graph generation from obstacle edges, (2) graph culling using binary space partitioning, (3) shortest path search using Eppstein's search algorithm, and (4) path smoothing using binary space partitioning again. This path planner has been implemented on the NASA RMAX helicopter and flight tested at the NASA Ames Disaster Assistance and Rescue Team (DART) site. Flights were conducted at a constant altitude of 10 m AGL and at a maximum speed of 5 m/s. However, for these tests, obstacle locations were known beforehand and predefined on the 2D map of the test site. This Voronoi-based planner performed well, but experimental results showed that it has some limitations in the case of a changing obstacle field.

• **Probabilistic Road Maps:** A much more recent advance in road map methods is the probabilistic road map (PRM) approach that attempts to make planning in large or high-dimensional spaces tractable. A PRM is a heuristic sample-based approach that discretizes a continuous c-space, resulting in fewer states than the original c-space. In the first stage, the road map is generally generated offline by randomly sampling the larger c-space and then using a local planner to connect collision-free configurations. In the second stage (query phase), the start and goal points are added and a graph search algorithm uses the road map created in the first stage to search through the waypoint nodes in order to find the least-cost path between the start and goal configurations. Some problems with a standard PRM method are inefficiency for narrow confined spaces, slow convergence rate, and dynamic obstacles.

In the framework of the WITAS project, researchers from Linköping University have developed and evaluated a PRM path planner for a robotic helicopter (Pettersson and Doherty, 2004). Their planner uses an OBBTree-algorithm for collision checking and $A^\star$ for graph search. Constraints such as maximum and minimum altitude, no-fly zones, and ascent/descent rate have been introduced during the query phase (the $A^\star$ search phase) in order to adapt the approach to the RMAX helicopter. Different replanning strategies were presented which modify the path when a change in the sensed environment or mission is detected. The path planner has also been augmented by a trajectory generator that creates time-parameterized reference positions, velocities, and angles. The complete path planner and path-following system have been implemented onboard the helicopter and flight tested. The UAS contains an onboard GIS map that includes a terrain 3D model of the test area, including buildings, trees, and other obstacles, with an accuracy of decimeters. Successful experimental results from different flight scenarios are presented in (Pettersson and Doherty, 2004; Wzorek et al., 2006), where the helicopter was tasked to fly to a designated building and to photograph each of its façades. In a recent paper, Wzorek, Kvarnstrom, & Doherty (2010) investigated the use of machine learning techniques to automatically choose the best possible replanning strategy, but only HIL simulations were presented. Another PRM-based path planner for a robotic helicopter was presented in Hrabar (2008). The road map is constructed based on an occupancy grid representation, and a $D^\star$ Lite algorithm is used to search for the shortest collision-free path. The proposed approach has been validated in simulations as well as on the CSIRO air vehicle simulator cable-array robot. A variant of the PRM technique has been used in path planning for a small quadrotor UAS navigating in indoor GPS-denied environments. He, Prentice, & Roy (2008) present a belief road map (BRM) path planning algorithm, which is an information or belief-space extension of the probabilistic road map method. The originality of this algorithm is that the graph is constructed in the belief or information space rather than the state space. In doing so, the algorithm allows the vehicle to reach its goal while maintaining reliable localization during path following using an unscented kalman filter (UKF). This algorithm has been tested using a quadrotor equipped with an IMU and a 4-m-range Hokuyo LIDAR as the main localization sensor. The quadrotor achieved its task by detouring from the shortest path toward areas of high sensor information, successfully reaching its desired goal using LIDAR-based odometry.

One of the rare works on RM-based path planning for an unmanned helicopter with online stereo-based mapping is presented in Andert et al. (2011). The stereo-based mapping system, previously described in Section 5.3.1 (Andert and Adolf, 2009) has been augmented by a real-time path planning algorithm that is based on a quasi-random road map (QRM) approach, an $A^\star$ algorithm for initial path search, and an $AD^\star$ algorithm for replanning. Both mapping and path planning algorithms have been implemented onboard the DLR ARTIS helicopter and flight-tested in two scenarios, one with a single obstacle and the other one with buildings in an urban area, Figure 18.

• **Rapidly Exploring Random Trees:** The rapidly exploring random tree (RRT) is also a heuristic randomization-based approach, which is a variant of PRM methods. Rather than randomly sampling the configuration space as a PRM does, the planner begins at the start location and randomly expands a graph or tree (a tree here is a directed graph with no cycles) by pushing the search tree away from previously

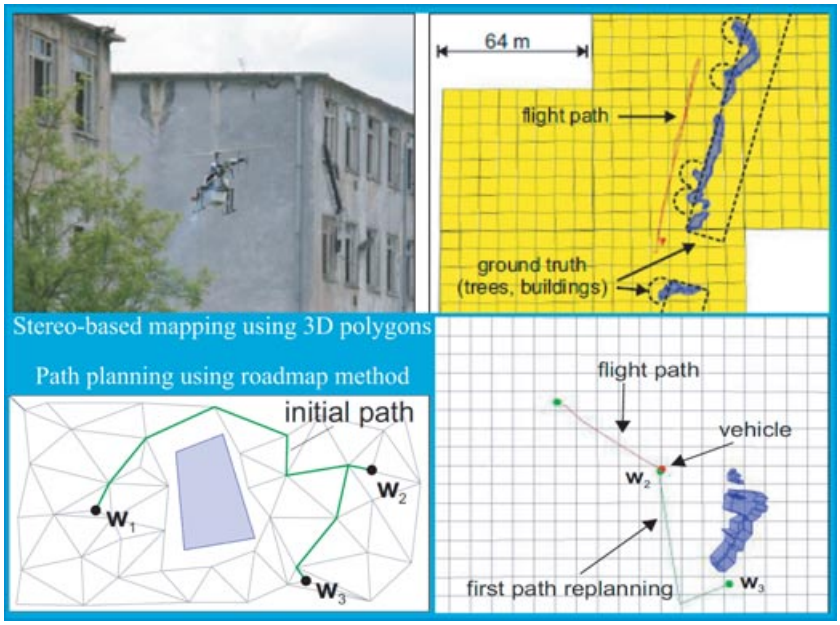


Figure 18. Example of stereo-based mapping and path planning by the DLR ARTIS helicopter (Andert et al., 2011).

constructed vertices. This allows it to rapidly search large and high-dimensional spaces, thereby outperforming PRM from the efficiency point of view. RRT are also well suited to the capture of dynamic or nonholonomic constraints and can be immediately extended to allow for moving obstacles. One method for planning in the RRT approach is to grow two RRTs, one from the goal and one from the start, and then search for states that are common to both, creating a linked path between the two. This sample-based path planner has gained much interest, and its application to UAS guidance in unknown environments has been investigated. Dahleh, & Feron (2002) extended the original RRT algorithm to deal effectively with the system's dynamics, in an environment characterized by fixed and moving obstacles. Different versions of the proposed path planner have been evaluated in simulations using a ground robot and a flying helicopter model in different scenarios. Wzorek and Doherty (2006) compared the performance of the PRM and RTT planners in terms of completeness, planning time, and path length. The helicopter was equipped with a digital map of the environment, and no-fly zones were randomly added during flight by the ground operator. Experimental results showed that the PRM algorithm performs better than the RRT algorithm in most situations if equipped with a suitable precompiled road map. However, the RRT planner is more appropriate for unknown environments, because it can build its road map online. There are also some other works on applying RRT methods to rotorcraft path planning, but only simulation results have been presented (Redding et al., 2007; Yang, Gan, & Sukkarieh, 2010).

6.1.2. Potential Fields

The potential fields (PF) approach is another major type of representation used in path planning, which was first introduced by Khatib and Mampey (1978) for manipulator control. This method treats the robot as a point under the influence of force fields generated by the goals and obstacles in the world. Obstacles generate repulsive forces that repel the vehicle and goals generate attractive forces; see Figure 19. These methods are characterized by their low computational complexity and easy implementation in real time. However, they are incomplete, because such approaches are prone to local minima. Much of the effort of adapting potential fields has been spent in overcoming this local-minima problem by generating so-called navigation functions such as depth-first and best-first techniques, wavefront-based methods, and harmonic potential functions. An interesting work on implementing and flight-testing a variant of PF-based path planning algorithm on a robotic helicopter is presented in Scherer et al. (2008). The authors addressed the problem of flying relatively fast at low altitudes in cluttered environments relying only on online LIDAR-based sensing and perception. Their approach combines a slower global path planner that continuously replans the path to the goal based on the perceived environment with a faster local collision avoidance algorithm that ensures that the vehicle stays safe. The global planner is based on an implementation of Laplace's equation that generates a potential function with a unique minimum at the goal. Binary occupancy grid and goal points are used to assign boundary conditions for Laplace's equation. The local

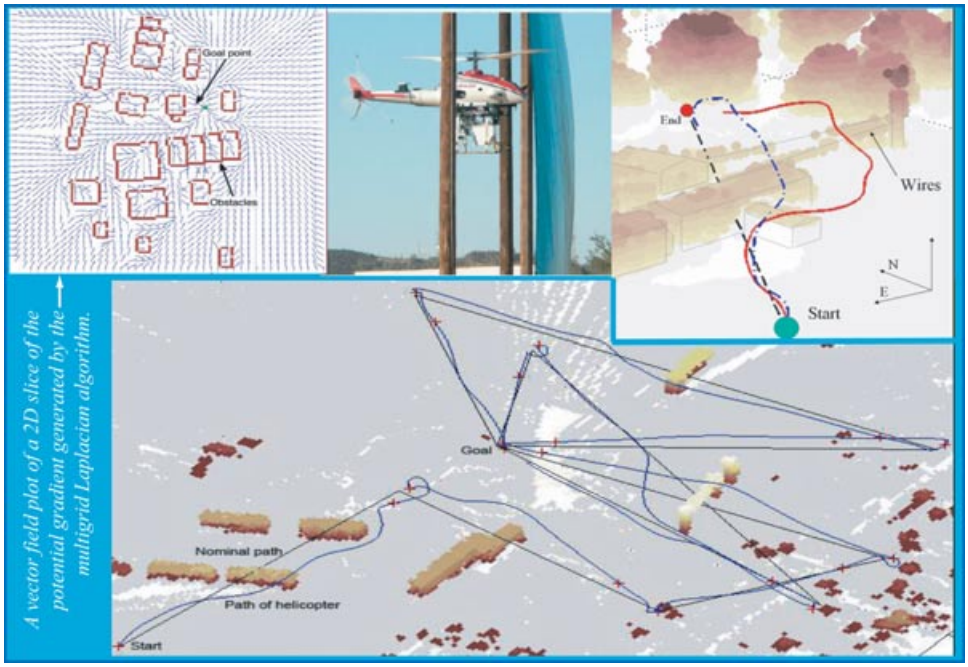


Figure 19. Path planning and obstacle avoidance by the CMU robotic helicopter using 3D LIDAR (Scherer et al., 2008).

path planner uses Fajen and Warren's model, where repulsive and attractive forces depend not only on distances (as in the original PF approaches) but also on angles (headings) relative to obstacles. Based on spherical representation of 3D obstacles, the reactive guidance law generates two steering rates in the azimuth and elevation directions. By constraining the lateral velocity to zero and the magnitude of the velocity vector to some defined value, the steering rates are mapped to a heading rate (horizontal avoidance) and a vertical velocity. The complete control–navigation–guidance system has been implemented on a Pentium M processor (1.6 GHz) onboard the CMU Yamaha RMAX helicopter. A custom 3D LIDAR unit from Fibertek Inc. was used as the main perception sensor. The system has been extensively flight-tested in different sites with different obstacles and flight speeds. More than 700 successful obstacle avoidance runs were performed where the helicopter autonomously avoided buildings, trees, and thin wires; see Figure 19.

6.1.3. Heuristic Search Algorithms

Most approaches to rotorcraft path planning problems are based on heuristic concepts, because of their ability to provide sufficiently good performance at a low computational cost. Heuristic search algorithms (HSA) use a heuristic, which is a rule for making a guess as to which path moves the vehicle closer to the goal, yielding a efficient but suboptimal planning. HSA tend to minimize route cost on a graph, as in RM approaches, or directly on a regular grid or any other world representation. These approaches can then be considered as a middle ground between heuristic and optimization methods. $A^\star$ and $D^\star$ algorithms, with their variants, are probably the most used HSA in robotics. The $A^\star$ algorithm evaluates the goodness of each node or cell by combining two metrics (distance from the start and estimated distance to the goal) to estimate the distance to the goal. $D^\star$ is an incremental HSA that can replan the path in real time when changes occur in the environment. It uses local sensor information to continuously update the map and dynamically repair the $A^\star$ paths affected by the change in the map.

A significant milestone in path planning for RUAS was achieved by NASA/Army researchers (Tsenkov et al., 2008; Whalley et al., 2009), who developed two 3D path planners based on heuristic planning concepts and the $A^\star$ search algorithm. The first one, called the plane-slicing 3D route planner, uses multiple invocations of the standard 2D $A^\star$ grid search algorithm on a set of obstacle edges obtained by slicing the terrain representation[20] with a horizontal, a vertical, or an off-axis plane. It first obtains a horizontal route between the origin and the goal points and then optimizes the path by replacing portions of it with up-and-over

[20]Hierarchical quadtree is used along with the obstacle proximity map that is constructed by Bresenham rasterization of line segments and circles on a regular grid.



Figure 20. Architecture of the U.S. Army/NASA Obstacle Field Navigation (OFN) system and example of flight paths at two different sites (Whalley et al., 2009).

shortcut hops. The final obtained 3D route is thus composed of a series of 2D pieces or paths. The other 3D route planner executes the $A^\star$ search algorithm on a height map. It plans directly in 3D and requires a single invocation of the $A^\star$ search algorithm. The proposed route planner is a 3D extension of a conventional 2D $A^\star$ algorithm on a height map extracted from the terrain representation. The $A^\star$-generated route is contained on the surface of that height map, which has been inflated with vertical and horizontal safety factors. Originally minimizing route length, the algorithm has been reworked to minimize flight time by accounting for speed variation caused by horizontal and vertical maneuvering. For both path planners, the route is refined and a trajectory generator is used to fit through the route waypoints a series of Kochanek–Bartels splines, resulting in a smoothed trajectory that remains within the operational corridor and dynamic limits of the vehicle. The developed obstacle field navigation (OFN) system was evaluated in Monte Carlo simulations as well as in flight experiments. After the OFN system was tuned and validated in simulation, it was implemented on the Yamaha RMAX helicopter and demonstrated in a number of real obstacle avoidance scenarios. As described in Takahashi et al. (2008), a spinning SICK LIDAR was used as the main perception sensor. Initially, 91 runs with a total duration of 44 h were flown in a virtual obstacle setup. In the second phase, more than 125 flight tests were performed at the DART site and the Fort Ord MOUT site with real natural and manmade obstacles at speeds that ranged from 1 to 4 m/s. The rotorcraft was able to fly a sequence of waypoints, sense the terrain, build a map online, replan the route, and follow the path simultaneously, with all processing performed onboard the helicopter. The OFN system was also successfully flown with a 5-m slung load on several occasions. Experimental results have also shown that the $A^\star$ 3D planner provides similar or better overall performance, compared to the plane-slicing 3D route planner, at a small fraction of the computational cost. Figure 20 shows two typical test runs at different sites.

6.1.4. Optimization Methods

Optimization methods (OM) consider the path planning problem as a numerical optimization problem where constraints on the problem (obstacles, vehicle dynamic limits, mission constraints, etc.) are translated into mathematical relationships. These methods are attractive because they theoretically produce optimal solutions and can also directly generate paths that consider the vehicle's dynamics. However, they are computationally expensive, especially when many constraints need to be satisfied. The computational cost of optimization-based methods is frequently

reduced by employing receding-horizon control techniques, or by using heuristics to simplify the optimization-based solution. Three major optimization methods have been investigated for designing path planning algorithms for RUAS. They are mixed integer linear programming (MILP), receding horizon control (RHC) or model predictive control (MPC), and motion primitive automaton (MPA). Because these methods are well described in the survey (Goerzen et al., 2010), we will not present them here. Nevertheless, it is important to mention that some optimization-based path planners have been demonstrated in real time using actual RUAS. Indeed, MILP and RH have been successfully applied for path planning for an indoor quadrotor (Culligan, Valenti, Kuwata, & How, 2007) and (Mettler, Dadkhah, & Kong, 2010), and a small indoor helicopter. (Mettler, 2010) using VICON system for navigation. The BEAR team from the University of California, Berkeley has successfully flight-tested an obstacle avoidance system that uses a LIDAR-based perception system and a hierarchical MPC-based path planner and controller (Shim et al., 2006); see Figure 12. (Mettler, Kong, Goerzen, & Whalley, 2010) presented a benchmarking framework for evaluating and comparing path planning algorithms for UAS. It is based on generating near time- or energy-optimal trajectories (using nonlinear and dynamic programming) that are then used as performance baselines to evaluate different algorithms. In a more recent paper (Kong & Mettler, 2011), an extension to that method is proposed based on spatial cost-to-go (SCTG) maps. The concept of SCTG maps allows studying the effects imposed by different airframe configurations and performance criteria. The paper also presented the simulation results obtained from applying this framework to a basic goal-directed guidance task in an urban environment for three small UAS types (fixed-wing aircraft, helicopter, and quadrotor).

6.1.5. Planning under Uncertainties

Uncertainties in localization, sensing, and control are unavoidable for the UAS planning problem. The common approach generally used to deal with uncertainties consists of treating them as a deterministic worst case by introducing a conservative safety corridor. However, there are some works that consider uncertainties directly in the planning algorithm. For example (Davis & Chakravorty 2007) formulated the problem of PUU as the adaptive control of an uncertain Markov decision process, consisting of a known control-dependent system state and an unknown control-independent environment. The feasibility of the approach was illustrated by testing it on an unmanned helicopter navigating through an urban environment in a 6-DOF flight simulation. PUU was also addressed in He et al. (2008) for autonomous indoor exploration using a quadrotor vehicle. A belief road map (BRM) path planning algorithm, which is a belief-space extension of the probabilistic roadmap method, was developed and flight-tested in different indoor environments. In He, Bachrach, and Roy (2010), the same group proposed a probabilistic planner for tracking targets in the presence of uncertainties in target pose. The PUU problem was treated using a multimodal Gaussian representation of the agent's beliefs and a forward-search algorithm. Experimental results for a quadrotor tracking small vehicles indoor were presented.

To allow RUAS to operate with a high degree of reliability and safety in the presence of many sources of uncertainties, NASA researchers have developed a new 3D path planning algorithm based on risk minimization (Goerzen & Whalley, 2011). The proposed planner (called RiskMinOFN) converts a 3D occupancy grid into a dynamic risk map by combining the various risk factors, such as sensing uncertainty, tracking error, uncertainty in helicopter position, quantization errors, altitude penalty, and other known risks such as mechanical failures. The RiskMinOFN relies on dynamic programming and the $A^\star$ algorithm to compute the minimum cost-to-go function over the field of view of the sensor, which is used as a navigation function. The velocity command module uses this 3D navigation function to generate reference velocities directly for the flight controller, causing the vehicle to move in the direction of the navigation function gradient. This path planner was first validated in simulations against the benchmarks described in Mettler et al. (2010), and then implemented on the NASA RMAX helicopter and demonstrated in urban and natural environments. The experimental results have shown that RiskMinOFN is useful for obstacle avoidance in the presence of uncertainties.

6.1.6. Reactive and Bio-inspired Obstacle Avoidance Methods

Most of the motion planning methods previously presented can be considered as global or local path planners where a global or local representation (map) of the environment is built. These path planners are generally computationally expensive and their implementation onboard mini and micro RUAS is challenging. On the other hand, reactive obstacle avoidance algorithms run very quickly and are crucial for preventing last-minute collisions. These characteristics motivated many researchers to investigate their implementation of reactive and bio-inspired obstacle avoidance methods (RBIOAM) onboard RUAS as main guidance systems or in combination with global and local path planning algorithms.

• **LIDAR-Based Reactive Obstacle Avoidance:** In a recent work, Merz and Kendoul (2011) have demonstrated the effectiveness of a LIDAR-based reactive obstacle avoidance system for infrastructure inspection by an autonomous helicopter. The perception system uses a COTS 2D LIDAR (Hokuyo) in combination with two special helicopter flight modes (Pirouette descent and Waggle cruise) to address the problem of 3D perception. Simple but effective

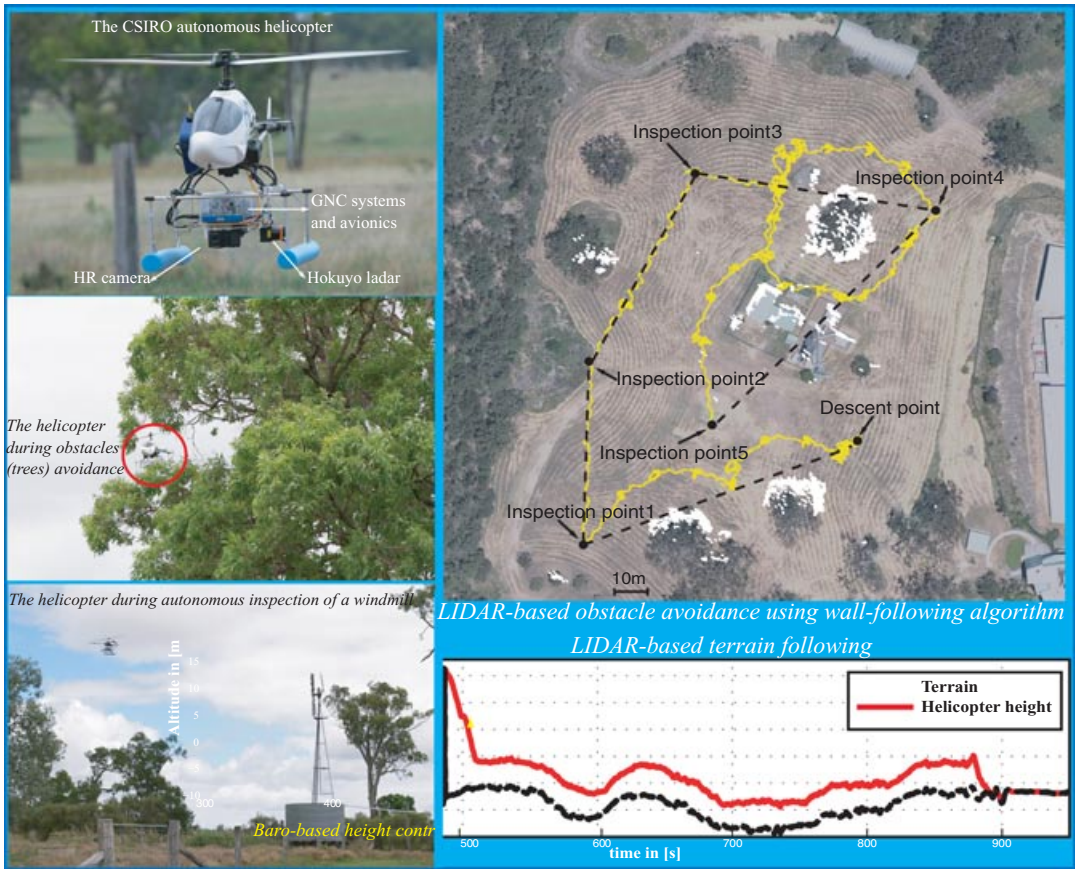


Figure 21. Obstacle avoidance and infrastructure inspection beyond visual range by the CSIRO autonomous helicopter using a vertically mounted Hokuyo LIDAR (Merz & Kendoul, 2011).

reactive strategies were used to detect the ground and regulate the height, perform terrain-following, detect and avoid frontal obstacles, detect and stop in front of the target, and take high-resolution pictures from a specified viewing angle. This system has been implemented onboard the CSIRO robotic helicopter using the real-time extended state machine (ESM) framework (Merz, Rudol, & Wzorek, 2006). Several flight tests of the perception and guidance systems have been performed in different environments, with a total of 14 h of autonomous flight to date. This system has also been successfully used to perform an autonomous inspection of a windmill beyond the visual range and without a safety pilot; see Figure 21. In another work from the same group, Hrabar (2011) developed and flight-tested a 3D reactive obstacle avoidance algorithm using Hokuyo LIDAR and a stereo camera. This algorithm finds an escape point (intermediary waypoint) by checking collisions within a cylindrical safety volume in a 3D occupancy map representation of the environment. The conducted research also included the evaluation and comparison of LIDAR and stereo vision systems for RUAS real-time perception. Successful collision avoidance results are presented from flight experiments with a CSIRO autonomous helicopter equipped with stereo and LIDAR sensors.

• **Vision-Based Bio-inspired Reactive Obstacle Avoidance:** Visual bio-inspired reactive methods have been popular because of their simplicity and low weight requirement. Furthermore, their effectiveness in nature has been proven in many flying insects. There are numerous ongoing works on applying insect-inspired approaches to obstacle avoidance in small UAS. In these methods, optic flow is directly used to compute an avoidance maneuver, yielding to obstacle avoidance without map building and path planning. As described in Section 5.2.2, some interesting results have been obtained for terrain following (Ruffier & Franceschini, 2005; Garratt & Chahl, 2008; Herisse et al., 2010; Ruffier, & Franceschini, 2005) and obstacle avoidance (Conroy, Gremillion, Ranganathan, & Humbert, 2009; Hrabar & Sukhatme, 2009; William et al., 2008; Zufferey & Floreano, 2006), when implemented on small UAS. These methods are very powerful and provide an interesting alternative for both perception and guidance onboard mini

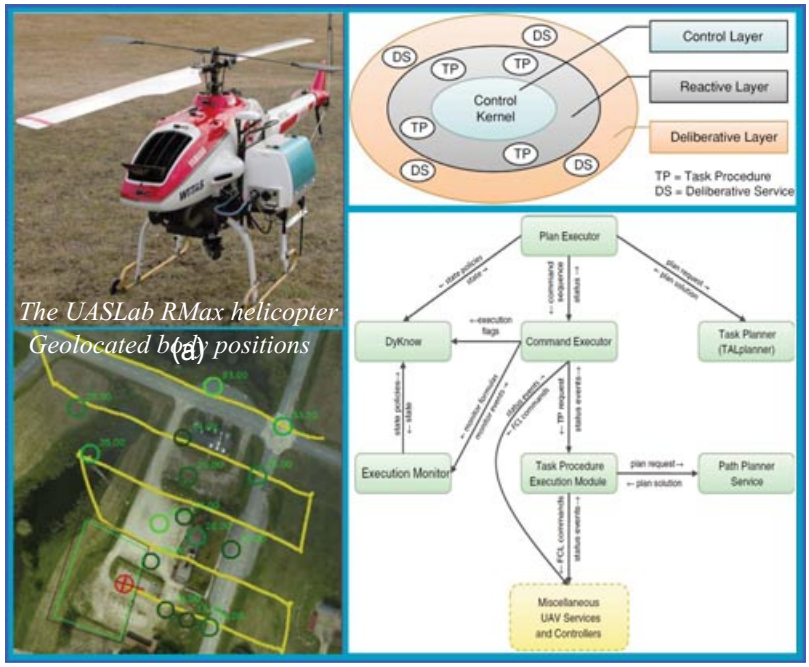


Figure 22. Architecture of the mission planning and execution monitoring system developed at Linköping University (Doherty, Kvarnstrom, & Heintz, 2009).

UAS with limited payload. However, the problem of robust optic flow computation in real time and obstacle detection is still a challenge. Moreover, these techniques need to be combined with some goal-oriented guidance to reach the assigned destination or to achieve the mission.

6.2. Mission Planning and High-Level Autonomy

As mentioned in the beginning of this section, guidance systems for UAS are not limited to path planning only. Research with UAS is reaching a new degree of sophistication and high-level autonomy where guidance systems include task-oriented planners and deliberative capability. This is an active research topic for ground robots that includes disciplines such as cognitive robotics, reasoning, and artificial intelligence. Mission planning for RUAS is defined in Section 3.2; it is mainly concerned with the selection of locations to visit (goal waypoints or trajectories) and other vehicle actions (loading/dropping a load, acquiring information), typically over a long time horizon and beyond sensing range. Functionally, mission planning resides above the process of path planning, where, for example, the mission planner generates a desired flight plan[21], and the path planner generates obstacle-free trajectories between the waypoints. A RUAS with mission planning capabilities is considered to have an autonomy level 6, according to the ALFURS chart presented in Figure 4. This section provides only a very short overview of mission planning and high-level autonomy systems that have been developed and implemented on single RUAS. Section 6.3 presents more work about multi-RUAS mission planning.

Although there were some attempts to implement sophisticated RUAS guidance systems that achieve mission planning and high-level autonomy, very few experimental results have been reported in the literature. Doherty et al. (2009) presented an architectural framework for mission planning and execution monitoring and its integration into a fully deployed unmanned helicopter, Figure 22. Planning and monitoring use the same logic formalism, a temporal action logic (TAL) for reasoning about actions and changes. The system has been implemented on the Yamaha RMAX helicopter and "partially" demonstrated in the context of an emergency services scenario. The first phase (body identification and flight-related monitor formulas) was achieved through real autonomous flights, whereas the second phase (package delivery and execution monitoring) has been evaluated through HIL simulations. In the framework of the NASA/Army autonomous rotorcraft project, a sophisticated guidance system has been developed for the autonomous surveillance planning problem for

[21] A series of waypoints or trajectories through which the vehicle must pass in order to collect specific information for the mission.



Figure 23. The guidance system developed by U.S. Army/NASA for autonomous surveillance planning (Whalley et al., 2005).

multiple and varying targets of interest (Whalley et al., 2005). High-level autonomy is provided by Apex[22], which is the intelligent behavior component of the guidance system, which generates mission plans using a decision-theoretic approach. It is based on three main layers: the deliberative layer (periodic surveillance planning), the goal executive layer (plan execution, monitoring, and human interaction management), and skills (autopilot and payload controllers); see Figure 23. This guidance system has been integrated into an RMAX robotic helicopter and flight-tested in more than 240 scenarios with comparative analysis of 2-Opt versus human in directing a surveillance mission. A similar project, called ReSSAC,[23] was carried out by the French Aerospace Lab (ONERA) with the objective of developing and demonstrating architectures and algorithms for high-level autonomy (decision-making and mission management) and information processing onboard RUAS in a search and rescue scenario (Fabiani et al., 2007). A decisional autonomy architecture for an exploration mission has been developed based on the idea of decomposing the mission into a sequence of tasks or macro-actions associated with rewards. The problem has been modeled using the Markov decision process (MDP) framework, and a symbolic focused dynamic programming (SFDP) algorithmic scheme for mission planning has been developed. Teichteil-Konigsbuch and Fabiani (2007) extended the guidance system and described its implementation onboard the RMAX robotic helicopter. The system has been tested in an exploration scenario where the mission consists in exploring an area, detecting an injured person, and selecting a suitable safe landing area. A mission management system for RUAS was also developed at DLR (Germany) using a combination of the 3T layered architecture (deliberative layer, executive/sequencer layer, and reactive layer) with ideas from the behavior-based paradigm (Adolf & Andert, 2010; Adolf & Thielecke, 2007). The developed mission management system has been integrated onboard the ARTIS helicopter and validated in simulations (Adolf & Thielecke, 2007) as well as in real flight tests (Adolf & Andert, 2010) in a number of scenarios, including waypoint following and a search and track mission.

[22]Apex is a NASA reusable autonomy software architecture that integrates artificial intelligence capabilities for monitoring events and for reactively selecting, scheduling and controlling actions, see http://ti.arc.nasa.gov/projects/apex/projectARP.php for more details.
[23]Search and Rescue by Cooperative Autonomous System.

6.3. Multi-Rotorcraft Unmanned Aircraft Systems Coordination and Cooperation

Cooperation between vehicles extends their capabilities beyond the capabilities of a single vehicle, and offers exciting new capabilities to perform a wide range of applications

efficiently and with greater fault tolerance and flexibility. The RUAS capability to coordinate and cooperate, as defined in Definition 10, with other RUAS or unmanned systems in general will increase its autonomy level according to the ALFURS chart in Figure 4. For the other autonomy-enabling functions, RUAS cooperation has also different levels. It may involve control, navigation, guidance, or all of them, and may be centralized or distributed (decentralized).

Definition 1. *In this survey, we make a distinction between coordination and cooperation. RUAS coordination involves low-level interactions for task or maneuver execution, mainly for sharing space, including collision avoidance and formation flight. On the other hand, cooperation or collaboration occurs at a higher level and may require, in addition, that the RUAS work toward a common goal or mission by sharing data and controlling actions together.*

Although many researchers have examined algorithms for UAS coordination and cooperation, few have included experimental results demonstrating the implementation of these algorithms on RUAS. The multi-RUAS coordination and cooperation problem has been approached in a variety of ways in the past. However, proposed architectures and algorithms for RUAS can be regrouped into three main categories: (1) flight control coordination (control level), (2) cooperative perception (perception level), and (3) cooperative mission planning and decision making (mission level).

6.3.1. Coordinated Flight Control

In the literature, much of the research on multi-RUAS flight control coordination has been primarily concerned with formation flight, collision avoidance, and load transportation. The coordination problem was formulated as an optimal control problem (Gillula, Hoffmann, Huang, Vitus, & Tomlin et al., 2011; Hoffmann et al., 2009; Nonami et al., 2010; Schouwenaars, Feron, & How, 2006; Shaw et al., 2007; Shim, Kim, & Sastry, 2003a) or as a distributed control problem (Bernard and Kon, 2009; Michael, Fink, & Kumar, 2011; Yun, Chen, Lum, & Lee, 2010). Formation flight of two unmanned helicopters was demonstrated in real flights (Nonami et al., 2010, Chapter 9) using a leader–follower configuration and linear MPC for position control of the follower helicopter. Other constraints such as communication range and collision avoidance were also considered in MPC calculations. The follower MPC algorithm was implemented off board and used state estimates of the leader helicopter. Decentralized nonlinear MPC, enforced with potential functions, was formulated for rotorcraft formation flight and collision avoidance and presented in Shim et al. (2003a). Later, this nonlinear MPC-based algorithm was extended or modified to compute and execute a plausible emergency evasive maneuver or trajectory based on the RUAS constraints and the predicted future path of other vehicles to avoid. For validation, flight tests were performed using two unmanned helicopters (Yamaha R-50) in a head-on collision course, which broadcast their positions. Experimental results are presented in Shim and Sastry (2006) for a decentralized architecture and in Shim and Sastry (2007) for a centralized architecture. Successful results about coordinated flight of two unmanned helicopters using optimization approach are also reported in Schouwenaars et al. (2006). Coordinated multivehicle connectivity-constrained trajectory optimization was tackled using both centralized and distributed receding-horizon planning and a MILP framework. Flight test results are presented for a centralized two-helicopter mission where the lead and relay helicopters are controlled in such a way that indirect line-of-sight connectivity between the leader and the ground station is always maintained. Other concepts and algorithms using optimal control for multi-RUAS control were demonstrated on the Stanford Testbed of Autonomous Rotorcraft for Multi-Agent Control (STARMAC) platform. STARMAC is composed of six quadrotors, each with its own onboard sensing and control. Hoffmann et al. (2009) briefly described the developed algorithms for multi-RUAS coordination and cooperation and provided references for more details and information. The demonstrated algorithms include two different decentralized cooperative collision-avoidance algorithms, one using rules derived from optimal control and reachability analysis, and the other based on a nonlinear, nonconvex optimization program and a Nash bargaining cost metric. In a more recent work (Gillula et al., 2011), a combination of hybrid decomposition and reachable set analysis was used to design collision avoidance algorithms for multiple unmanned vehicles. The algorithms use a distributed optimal switching control strategy, derived from a dynamic game formulation, to command avoidance action (transition from nominal control to collision avoidance) when any vehicle was on the boundary of the computed avoid sets. Proposed controllers were implemented on STARMAC and demonstrated in real time. Experiments were flown using both two- and four-vehicle fleets, where each quadrotor broadcasts its states to all other quadrotors.

The UAV research group at the National University of Singapore has successfully demonstrated formation flight of two small unmanned helicopters using a nonoptimization approach (Yun et al., 2010). Simple equations of motion were derived to model the leader–follower pattern in a fixed geometrical formation. A dynamic inversion technique was used to design a controller for the follower helicopter to maintain a fixed distance from the leader while tracking its prescribed reference trajectory. This control scheme has been validated using a manually piloted leader helicopter and a follower helicopter. An interesting application of coordinated flight is load transportation by multiple RUAS. Unlike the case in the abovementioned



Figure 24. The AWARE multi-RUAS platform during slung load transportation and sensor node deployment missions (Bernard & Kon, 2009).

works, where the interactions between RUAS were essentially information exchange, joint transportation of a single load requires physical coupling between the vehicles. In the framework of the AWARE[24] project, researchers from TUB (Germany) developed a generic system for slung load transportation using multiple small unmanned helicopters (Bernard & Kon, 2009). The focus was on the flight control system, which is based on the dynamics model of the whole system (all RUAS and load). The proposed distributed control scheme includes two control loops: (1) an inner loop (attitude controller) for each helicopter that accounts for the complete dynamics of the whole system, and (2) an outer loop (translation controller) that only accounts for the translational dynamics of the whole system and depends on the number of helicopters. This algorithm was validated in real flight experiments, where a load of 4 kg was transported by three identical unmanned helicopters flying in a triangular formation, Figure 24. A similar system for load manipulation and transportation by indoor quadrotors was developed and demonstrated at the University of Pennsylvania (Michael et al., 2011). The problem was addressed by solving the inverse or direct kinematics problem based on a mathematical model that models the kinematic constraints and the mechanics underlying stable equilibria of the whole underactuated system. The designed flight controller was augmented by a simple potential field algorithm to avoid interrobot collisions. By using a VICON motion capture system for state estimation, a 6-DoF load was transported and manipulated using three quadrotors. In recent years, several other projects have been launched with the objective of developing swarms of cooperative mini RUAS such as the AirShield (airborne remote sensing for hazard inspection by network enabled lightweight drones) project, the SUAAVE (sensing, unmanned, autonomous aerial vehicles) project, and the sFly (swarm of micro flying robots) project.

Observation 10. Not all coordinated flight controllers presented in this section, are purely based on a control-theoretic approach. Some of them are augmented with simple coordinated trajectory-generation subsystems that generate an appropriate reference trajectory for each vehicle.

6.3.2. Cooperative Perception

Cooperative perception is beneficial for multi-UAS systems, but it presents many challenging issues. One of the main projects that addresses multi-UAS cooperative perception is probably the COMETS[25] project (Ollero, Lacroix, Merino, et al., 2005; Ollero and Maza, 2007). The objective of COMETS was to design and implement a system for cooperative activities using heterogeneous UAS such as unmanned helicopters and blimps. Merino et al. (2006) presents results of field experiments on fire detection, localization, and monitoring with two unmanned helicopters and an automated blimp cooperating. The architecture of the cooperative system consists of (1) partially distributed low-level image-processing functions (image stabilization, segmentation, georeferencing), which are dedicated to the

[24]http://grvc.us.es/aware/.

[25]The COMETS (real-time coordination and control of multiple heterogeneous unmanned aerial vehicles) project was funded by the European Commission.

various sensors (visual and infrared cameras, fire sensor), and (2) two centralized subsystems that fuse processed data from the different vehicles for fire detection/alarm confirmation, localization, and monitoring. When a fire alarm is detected and localized, the mission is replanned, and more UAS are sent to confirm the alarm. Field experiments were performed and tasks related to fire search, fire confirmation, and fire observation were successfully achieved. Multi-RUAS cooperative perception was also addressed in the AWARE project (Maza, Kondak, Bernard, & Ollero, 2010). The Aerospace Control Lab. (ACL) at MIT has been very active in studying and testing multi-RUAS cooperation using the RAVEN[26] platform (How et al., 2008). In the cooperative perception area, they have addressed the problem of persistent vision-based search and track using multiple UAS (Bethke, 2007). The cooperative perception problem was addressed in the bearings-only estimation framework, where multiple bearing measurements from different vehicles are combined to estimate the target position and velocity. The method uses an optimization technique to combine the instantaneous observations of all UAS, allowing rapid and accurate estimation. A RAVEN platform was used to conduct different experiments including cooperative tracking of a small ground vehicle by two quadrotors, cooperative tracking of a flying vehicle (quadrotor) by two quadrotors, and a persistent search and track mission for three quadrotors.

6.3.3. Cooperative Mission Planning and Decision Making

Very few contributions have dealt with multi-RUAS problems according to a deliberative paradigm, where the RUAS exhibit cooperation at the mission level, such as cooperative task assignment and mission planning. Cooperative task assignment or allocation is concerned with the selection of conflict-free matching of tasks to vehicles. In other words, a higher-level component of a mission planner provides a list of tasks to the task assignment component, which decides which of the available vehicles should perform each task, based on information about the tasks and the capabilities of the vehicles. In Bethke et al., (2008), an algorithm for cooperative task assignment was proposed with a focus on the health management problem at the task level. The receding-horizon task assignment (RHTA) algorithm, developed at MIT, has been extended to include the fuel state in the vehicle model. The modified RHTS algorithm solves an optimization problem to select the optimal sequence of tasks for each UAS. A set of experiments were conducted to accomplish a multi-RUAS mission that consists of three RAVEN quadrotors searching for, detecting, estimating, and tracking an unknown number of ground vehicles. The mission was successfully accomplished by combining the modified RHTA algorithm with the cooperative vision-based perception algorithm described earlier (Bethke, 2007). The same group has developed other architectures and algorithms for decentralized cooperative task allocation, Figure 25. In Choi, Brunet, and How, (2009), the consensus-based auction algorithm (CBAA) and consensus-based bundle algorithm (CBBA) are described. Both algorithms produce conflict-free task assignment solutions by matching a list of tasks to multiple vehicles to maximize some global reward. Simulation results were presented in Choi et al. (2009) and some experimental results were reported in How et al., (2009).

In many realistic missions, such as search and rescue, a certain degree of centralization and flexibility in cooperative mission planning is necessary for those in charge

[26] RAVEN is an indoor multivehicle test bed composed of fixed- and rotary-wing vehicles and a VICON motion capture system.



Figure 25. Eight-vehicle cooperative flight test on MIT's RAVEN (Bethke, Valenti, & How, 2008).

(human operators) to understand and eventually sign off on potential plans. In the framework of the COMETS project, decisional architectures and algorithms for the cooperation of heterogeneous UAS with different autonomy levels were proposed (Lacroix et al., 2007). The proposed architecture is based on a human-centered central decisional node (CDN) in the ground station and a distributed decisional node (DDN) onboard each vehicle. The DDN encompasses a multilevel executive (MLE) module that handles the coordination and task execution issues and a deliberative layer (DL) that deals with mission and task refinement and distributed task allocation. The MLE was tested in simulations as well as in actual flights for the fire detection and monitoring scenario, whereas the DL components have only been tested in simulations. A similar architecture was developed for the AWARE platform, which considers the self-deploying of the network (communication equipment and nodes) by means of multiple autonomous helicopters (Maza et al., 2010). Each vehicle contains an onboard deliberative layer (ODL) and a proprietary executive layer (EL). The former deals with high-level distributed decision making, whereas the latter is in charge of the execution of the tasks. The ODL has interactions with its EL and with the ODL of other UAS, as well as with the human–machine interface. This architecture has been implemented on three unmanned helicopters and used in a mission consisting of deployment of a sensor node from one RUAS in a given location to repair the wireless sensor network connectivity, whereas other RUAS supervised the operation with the onboard cameras and also monitored the area. More field experiments are presented in Maza, Caballero, Capitan, de Dios, and Ollero (2011) for the following missions: (1) multi-RUAS fireman tracking, (2) multi-RUAS sensor deployment, (3) fire confirmation and extinguishing using two helicopters, and (4) multi-RUAS surveillance using two helicopters. The UASTech at Linköping University has investigated the problem of cooperative mission planning for UAS according to an AI approach. Different architectures and algorithms have been proposed, but to our knowledge none of them has been flight-tested with a cooperative fleet of RUAS. In a recent work, Kvarnstrom and Doherty (2010) proposed a new mission-planning algorithm for collaborative UAS based on combining ideas from forward-chaining planning with partial-order planning, leading to a new hybrid partial-order forward-chaining (POFC) framework that meets the requirements on centralization, abstraction, and distribution found in realistic emergency services settings. A prototype implementation of POFC is in the process of being integrated with the UASTech RUAS architecture.

7. DISCUSSION AND CONCLUSION

In the last 20 years, significant progress toward the development of autonomous RUAS has been made. This survey has reviewed the current state of the art in guidance, navigation, and control systems for RUAS and provided a detailed overview of published papers, with a particular focus on recent and practical GNC algorithms that have been implemented and flight-tested. The key conclusions of this paper are the following:

- Many papers have been published in the GNC areas, but very few of them have reported convincing experimental results.
- Flight control is a well understood and well developed area that can offer immediate solutions to RUAS technology needs.
- LIDAR-based perception and obstacle avoidance or path planning systems are less mature, but some of the developed systems can be deployed for some particular missions.
- Vision-based navigation, including state estimation and perception, is an active research topic, but the technologies developed so far are not reliable enough for real deployments.
- There is a gap between theoretical work and reported experimental results. Indeed, a significant amount of theoretical work has been done in the design of control, navigation, and guidance algorithms for RUAS. However, most of these systems have been validated in simulations, offline processing, or very basic flight tests without convincing results. We believe that significant engineering work is also needed to implement the developed systems properly and to evaluate their performance in extensive flight-test programs.
- It was difficult to compare the reviewed algorithms, especially in the navigation and guidance areas. More work needs to take place on the development of benchmarks and metrics to compare algorithms and to provide some design standards for navigation and guidance technologies.
- From this review, it was clear that most milestones in GNC for RUAS have been achieved using robotic helicopters. This is mainly because of their increased payload capabilities, which allow them to carry different reliable sensors and powerful embedded computers. However, research using other small rotorcraft such as quadrotors is growing fast and many interesting results have been published in recent years.

7.1. Control

Early research on RUAS focused on modeling their dynamics and designing control algorithms for automated flight. In recent years, there have been fewer published papers on the control of RUAS, except a few papers on the improvement of existing controllers. In fact, many researchers have shifted from control-related research to the areas of vision-based navigation and guidance. Although significant work has been done in the design of advanced and nonlinear flight controllers, reported experimental

results did not show significant progress in flying capabilities when compared to standard linear controllers. Indeed, most research platforms and commercial RUAS still use autopilots based on standard linear controllers such as PID, LQR, and H_∞. Successful implementations of these linear flight controllers and some nonlinear controllers such as the Georgia Tech. adaptive flight controller have contributed to making control technologies for RUAS mature enough to be deployed for real world applications. We believe that there are three aspects of RUAS control that have not really been investigated and that are very important for real-world applications:

- Development of self-tunable and flexible controllers that can be integrated into different platforms in a short time. This also includes adaptation to platform changes such as payload and sensors.
- Design of robust controllers that guarantee good flight performance under windy and severe weather conditions.
- Design of reconfigurable control systems that have the ability to change or switch between different control strategies based on flight and mission conditions. This also includes compensation for most failures and mission and environment changes.

7.2. Navigation

UAS navigation in obstacle-free environments using conventional avionics and sensors (GPS, IMU, and altimeter) is a solved problem. Most current research activities focus on non-GPS state estimation and perception using active LIDAR units or passive imaging systems. Although there have been some interesting results on autonomous navigation using LIDAR or vision, current navigation technologies have not yet reached the level of robustness and reliability required for real-world applications. The most successful implementations of LIDAR-based perception and obstacle avoidance are probably the ones by NASA (Goerzen & Whalley, 2011; Tsenkov et al., 2008), ARCAA/CSIRO (Merz & Kendoul, 2011), and CMU (Scherer et al., 2008) for outdoor navigation and MIT (Bachrach et al., 2009) for indoor navigation. 3D navigation in unknown, cluttered, and complex environments is still challenging and requires adequate 3D sensing technologies and efficient onboard algorithms for data processing. The technologies of compact 3D LIDAR and 3D time-to-flight cameras are progressing quickly and have been identified as potential candidates for active sensing and perception onboard RUAS.

Computer vision is the most relevant localization and perception technology for RUAS navigation. It has been used for motion estimation and localization, target/obstacle detection, and environment mapping. We found that research on vision-based navigation (state estimation and perception) for RUAS is still in its early stages, especially from implementation and flight-testing points of view. Indeed, many papers have been published on vision-based navigation, but the conducted experiments are limited to special and simple cases. Furthermore, most algorithms developed for obstacle detection and mapping have not been validated and demonstrated in closed-loop autonomous flights. There is thus a real need to developing embedded real-time and robust algorithms for vision-based localization and perception given real-world conditions. We also recommend more efforts in rigorously implementing and extensively flight-testing developed algorithms. We believe that bio-inspired visual navigation approaches are very promising, and we recommend further research in this direction. A new generation of bio-inspired vision sensors (such as artificial compound eyes and frameless event-based cameras) has started to emerge, providing a new paradigm and framework for developing very efficient and reliable vision algorithms.

7.3. Guidance

Despite active research on UAS guidance, most current unmanned rotorcraft still use rudimentary guidance systems that are limited to waypoint navigation. From the reviewed literature, we found that the majority of existing algorithms do not include much discussion of practical implementation, and very few experimental results have been reported. Often, only simulation results are provided, based on idealized vehicle and operational conditions. The main problem of implementing and demonstrating effective guidance systems may be attributed to two main factors. First, a guidance system generally relies heavily on the perception system for obstacle detection and avoidance. Because perception technologies are not yet mature, as discussed before, it limits practical implementation and demonstration of guidance systems in realistic scenarios. Second, 3D path planning algorithms that are complete, optimal, and sound and that consider UAS dynamics are computationally expensive and their implementation onboard RUAS is very challenging. Therefore, among the papers surveyed, most of the practical implementation (Bachrach et al., 2009; Merz & Kendoul, 2011; Scherer et al., 2008; Tsenkov et al., 2008), of UAS guidance have been of the hierarchical decoupled control type, or in some cases focus only on the local planning needed to avoid obstacles. From review of the different papers, most work on UAS guidance has focused on path planning for simple goal-directed or goal-oriented tasks, without considering the different sources of uncertainty. Therefore, there is still much work to be done on developing and demonstrating efficient and robust path planning algorithms, mission planning and management systems, and higher-level guidance modules for a single RUAS or a team of multiple vehicles.

7.4. Operational and Certification Issues

In addition to technological factors such as GNC technologies, other important aspects of RUAS development and deployment are regulations and operational procedures. UAS operations and use have remained almost exclusively military because of the lack of a regulatory framework that allows UAS integration into the civilian national airspace system (NAS). Indeed, the development of such as framework is a prerequisite for greater UAS access to civil airspace and, subsequently, the continued growth of the UAS industry and research. Most academic research has focussed on the embedded autonomous capabilities (GNC challenges) of UAS. Issues related to operation and certification of UAS are completely missed by the research community. Although these factors have clear implications for future UAS research and deployment, they are beyond the scope of this paper. The main objective of this section is to encourage UAS developers, especially researchers from academia, to participate actively in the process of developing a regulatory framework and operational procedures. We also encourage them to be aware of the current operational restrictions, as well as making informed decisions on their research and development efforts so that their designs will be airworthy when the regulatory framework is in place. At ARCAA in Australia, we are closely collaborating with CASA (the Australian Civil Aviation Safety Authority) in developing rules, regulations, procedures and standards for integration of UAS in civil airspace (Clothier, Palmer, Walker, & Fulton, 2011). Currently, there are no specific standards and regulations for the type certification of civil UAS. In their absence, the risks to people and property on the ground are insured through substantial restrictions on where UAS operations can take place. However, there are a lot of efforts by national/international organizations and industry to produce and develop a regulation framework for integration of UAS in to the NAS. Dalamagkidis, Valavanis, & Piegl (2009) did a very good job of reviewing current manned aviation regulations and available UAS regulations worldwide. It also presented UAS safety assessment and functional requirements, as well as recommendations on a UAS integration roadmap. Another recent paper by Guglieri, Mariano, Quagliotti, and Scola (2011) also outlined the international initiatives that deal with the development of the regulatory framework on airworthiness and certification of UAS.

Some interesting academic works have addressed the problem of UAS operations and regulations. The study by Pratt, Murphy, Stover, and Griffin (2009) is a good example of addressing issues related to RUAS operations. The aim of the research was to identify concept of operations (CONOPS) and autonomy needs for RUAS operating in cluttered urban environments. Results from a small teleoperated helicopter performing a structural inspection task following Hurricane Katrina in 2005 provided a basis for extracting recommendations and key findings for structural inspections CONOPS (crewing organization and flight operations protocol), but also needed directions in research for developing autonomous capabilities for inspection-like missions.

The development of autonomy technologies for RUAS follows a bottom-up approach, because most GNC systems for UAS typically have a hierarchical structure. This is because recent advances in UAS autonomy have been largely driven by practitioners in the field of systems control and robotics, and not computer science and artificial intelligence. This mode of incremental technology development in the field of UAS autonomy resulted in a gradual increase of UAS autonomy and the complexity of missions that can be executed. Indeed, most research activities in the last decade have focused on flight control, i.e., addressing issues related to autonomy level 1. However, in recent years, there has been more research on higher autonomy levels incorporating new navigation and guidance capabilities in the UAS. For example, much work and many results have been published recently on non-GPS navigation (autonomy level 2) and obstacle detection and path planning (autonomy level 4). There is also some work on fault detection (level 3), cooperative navigation and path planning (autonomy level 5), and mission planning (autonomy level 6), and very little work on cooperative mission planning (autonomy level 7). In our review, we have not found any work that has addressed the autonomy problems related to level 8 and above.

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